12 Lipid Mediators in Acute Inflammation and Resolution: Eicosanoids, PAF, Resolvins, and Protectins

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INTRODUCTION

Autacoids are locally acting substances that are rapidly biosynthesized in response to specific stimuli, act quickly, and are usually deactivated by metabolism. Eicosanoids are a chemically diverse family of arachidonic acid–derived autacoids that have critical roles in cardiovascular, inflammatory, and reproductive physiology. Pharmacologic interventions in eicosanoid pathways – including the nonsteroidal anti-inflammatory drugs (NSAIDs), COX-2 inhibitors, and leukotriene inhibitors – are useful in the clinical management of inflammation, pain, and fever. Given the many important bioactivities of lipid mediator eicosanoids, resolvins, and protectins, future research may lead to the development of new therapeutics for the treatment of inflammatory conditions, autoimmune diseases, asthma, glomerulonephritis, cancer, sleep disorders, and Alzheimer's disease.

BIOSYNTHESIS OF EICOSANOIDS

Eicosanoids are crucially involved in a number of metabolic pathways that have diverse roles in inflammation and cellular signaling. These pathways center on reactions involving the metabolism of arachidonic acid (Figure 12.1). The following considers the biochemical steps leading to arachidonic acid synthesis, then discusses the cyclooxygenase (COX), lipoxygenase (LOX), epoxygenase, and isoprostane pathways of arachidonic acid metabolism. Arachidonic acid (*cis*-,*cis*-,*cis*-,*cis*-5,8,11,14-eicosatetraenoic acid), the common precursor to eicosanoids, must be synthesized from the essential fatty acid precursor linoleic acid (*cis*,*cis*-9,12-octadecadienoic acid), which can be obtained only from dietary sources. In cells, arachidonic acid is esterified to the *sn*2 position of the membrane phospholipids predominantly in phosphatidylcholine phosphatidylethanolamine, and phosphatidyl inositol. Arachidonic acid is released from cellular phospholipids by the enzyme phospholipase A_2 (Figure 12.1), which hydrolyzes the acyl ester bond. This important reaction, which represents the first step in the arachidonic acid cascade, is the overall rate-determining step in the generation of eicosanoids produced by most effector and immune cells.

Membrane-bound and soluble isoforms of phospholipase A_2 are classified as secretory ($sPLA_2$) and cytoplasmic ($cPLA_2$), which are differentiated based on molecular weight, pH sensitivity, regulation and inhibition characteristics, calcium requirements, and substrate specificity. The existence of multiple isoforms allows for tight control and regulation of the enzyme in different tissues to achieve selective biologic responses. Phospholipases relevant in inflammation are stimulated by such cytokines as TNF-α, GM-CSF, and IFN-γ, such growth factors as EGF, and the MAPK–PKC cascade. Glucocorticoids were once thought to act via the direct inhibition of phospholipase A_2 activity. More recent findings indicate that glucocorticoids act by inducing the biosynthesis of lipocortins, a family of phospholipase A_2–regulatory proteins that comprise Annexin 1, which mediates part of the anti-inflammatory actions of glucocorticoids (see later and Chapter 10).

Cyclooxygenases: Biosynthesis of Prostanoids

The unbound form of intracellular arachidonic acid (unesterified: not bound to phospholipids) is rapidly converted in a cell type–specific manner by COX, LOX, or epoxygenase (cytP450) enzymes that dictate the class of local eicosanoids generated. The COX pathway leads to the formation of prostaglandins, prostacyclin, and thromboxanes (Figure 12.2); the LOX pathways lead to leukotrienes and lipoxins; and the epoxygenase pathways lead to epoxyeicosatetraenoic acids (Figure 12.1). Cyclooxygenases (also

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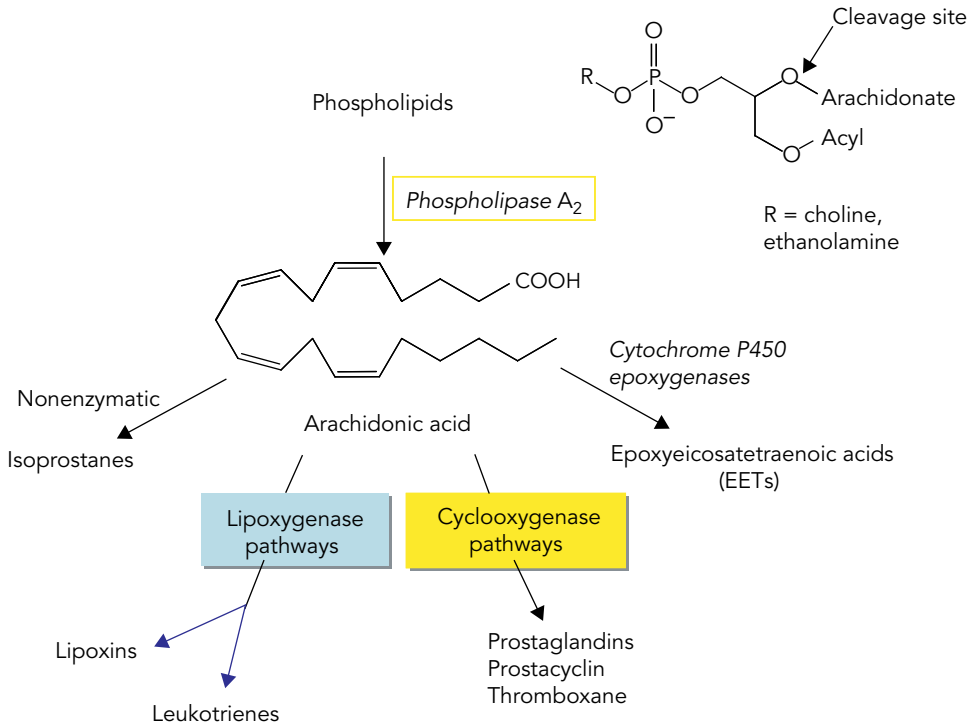


Figure 12.1. Overview of arachidonic acid cascade. Phospholipase A_2 acts on the phospholipids phosphatidylcholine (PC), phosphatidylethanolamine (PE), and phosphatidylinositol (PI) to release arachidonic acid. Phospholipase A_2 cleaves the ester bond marked by the arrow to release arachidonic acid. Unesterified arachidonic acid is substrate for the cyclooxygenase, lipoxygenase, and epoxygenase pathways. The cyclooxygenase pathways produce prostaglandins, prostacyclin, and thromboxane. The lipoxygenase pathways produce leukotrienes and lipoxins. The epoxygenase pathway produces epoxyeicosatetraenoic acids (EETs) and nonenzymatic oxidation of arachidonic acid can produce isoprostanes.

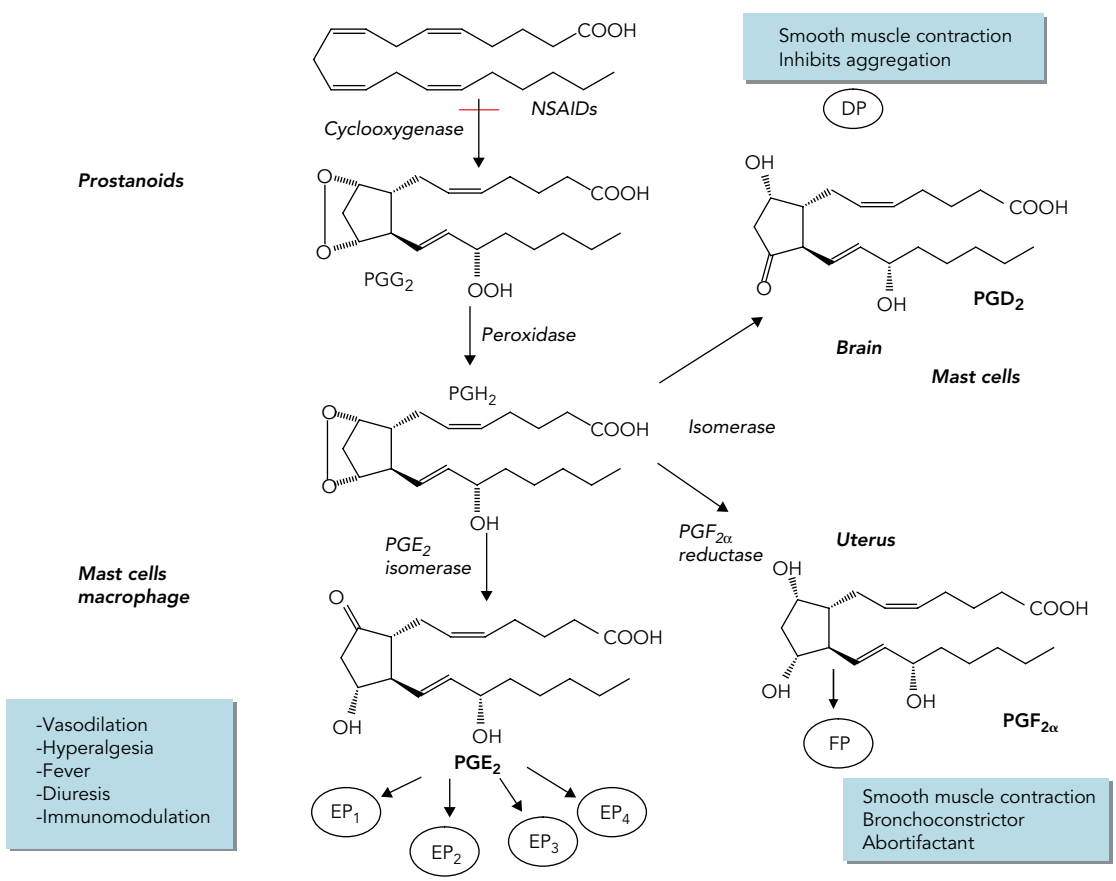


Figure 12.2. (*continued*).

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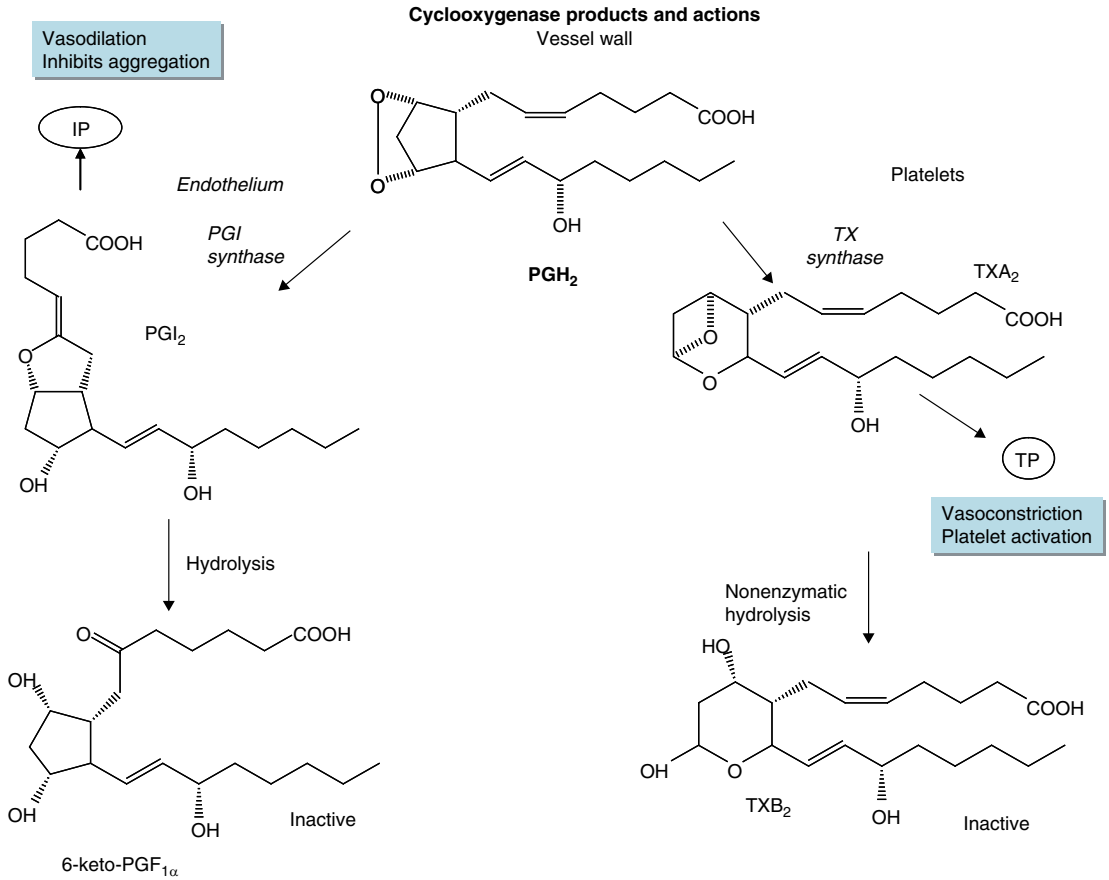


Figure 12.2. Prostaglandin and thromboxane biosynthesis, and major functions. The biosynthetic pathways from arachidonic acid to prostaglandins, prostacyclin, and thromboxane are shown. Note that tissue-specific enzyme expression determines the tissues in which the various PGH_2 products are produced. NSAIDs and COX-2 inhibitors are the most important classes of drugs that modulate prostaglandin and TXA_2 production.

Thromboxane antagonists and PGE_2 synthase inhibitors are each potential new pharmacologic strategies currently in development. COX, cyclooxygenase; PG, prostaglandin; TX, thromboxane.

GPCR specific for eicosanoids: denoted DP, PGD_2 receptor; EP, PGE_2 receptor; FP, $PGF_{2\alpha}$ receptor; IP, PGI_2 receptor; TP, TXA_2 receptor. Each of these receptors specifically and stereoselectively signals with its cognate ligand eicosanoid. NSAID, nonsteroidal antiinflammatory drug.

(A) Prostanoid biosynthesis: cell type specific formation of PGE_2, $PGF_{2\alpha}$, and PGD_2.

(B) Prostacyclin and thromboxane biosynthesis in the vessel wall and platelets.

known as prostaglandin H or endoperoxide synthases) are glycosylated, homodimeric, membrane-bound, heme-containing enzymes that are virtually ubiquitous in animal cells from invertebrates to humans. Two COX isoforms, denoted COX-1 and COX-2, are found in humans. Although COX-1 and COX-2 share 60% sequence homology and near superimposable three-dimensional structures, the genes are located on different chromosomes, and the enzymes differ in cellular, genetic, physiologic, pathologic, and pharmacologic profiles (Table 12.1). Each COX catalyzes two sequential reactions, that is, an oxygen-dependent cyclization of arachidonic acid into prostaglandin G_2 (PGG_2) followed by a peroxidase reaction that reduces PGG_2 to PGH_2.

As a result of differences in cellular localization, regulatory profile, tissue expression, and substrate requirement, COX-1 and COX-2 ultimately produce different sets of eicosanoid products. Constitutively expressed COX-1 is believed to function in physiologic, or "housekeeping," roles such as vascular homeostasis, maintenance of renal and gastrointestinal blood flow, renal function, intestinal mucosal proliferation, platelet function, and antithrombogenesis. A number of "as-needed," or specialized, functions are attributed to the inducible COX-2 enzyme, including roles in inflammation, fever, pain, transduction of pain stimuli in the spinal cord, mitogenesis (particularly in the gastrointestinal epithelium), vascular hemodynamics, deposition of trabecular bone, ovulation, placentation, and uterine contractions of labor. The role of constitutive COX-2 expression in areas of the nervous system as the hippocampus, hypothalamus, and amygdala remains to be elucidated.

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TABLE 12.1. COX-1 and COX-2

General properties	COX-1	COX-2
Expression	Constitutive	Inducible Constitutive in parts of nervous and vascular systems
Tissue location	Ubiquitous expression	Inflamed and activated tissues, tumor cells
Subcellular localization	Endoplasmic reticulum	ER and nuclear membrane
Substrate selectivity	Arachidonic acid, eicosapentaenoic acid	Arachidonic acid, γ-linolenate, α-linolenate, linoleate, eicosapentaenoic acid
Role	Gastroprotection and renal maintenance functions	Proinflammatory, mitogenic, and growth factors
Inducers	Generally not induced	LPS, TNF-α, IL-1, IL-2, EGF, IFN-γ mRNA rises 20- to 80-fold upon induction in many tissues regulated within 1–3 hours
Inhibitors	NSAIDs low-dose aspirin	Endogenous glucocorticoids, IL-1β, IL-4, IL-10, IL-13 NSAIDs, COX-2 inhibitors

Prostanoids: The Prostaglandins

Prostaglandins are a relatively large class of structurally similar compounds that each carries potent and stereospecific biological actions that are important in host mechanisms in acute inflammation. The name derives from their initial identification in the genitourinary system of male sheep. Prostaglandins all share a chemical structure, called a prostanoid, consisting of a 20-carbon carboxylic acid containing a cyclopentane ring and a 15-hydroxyl group, and are divided into three major subseries – PG_1, PG_2, and PG_3. The subscript numeral indicates the number of double bonds present in each molecule. The PG_2 series is the most prevalent because these are direct products of arachidonic acid. The PG_1 series derive from the arachidonic acid precursor dihomo-γ-linolenic acid (DHGLA), an eicosatrienoic acid, while the PG_3 series derive from eicosapentaenoic acid (EPA).

Prostaglandin $(PG)H_2$ is central to the COX pathway (see Figure 12.2) because it is the unstable transient precursor to PGD_2, PGE_2, $PGF_{2\alpha}$, and thromboxane A_2 (TxA_2), as well as prostacyclin (PGI_2). The distribution of these eicosanoids in tissues is determined by the expression pattern of the different enzymes of prostaglandin synthesis (i.e., PG synthases) present in specific cells in the tissues (see Figure 12.2), which gives rise to a cell type–specific biosynthesis profile of eicosanoids.

The prostaglandins are important in many physiologic processes; most are not directly related to inflammation (highlighted in Table 12.2). Note especially the important housekeeping functions of PGE_2, broadly referred to as cytoprotective roles, in which organs such as gastric mucosa, myocardium, and renal parenchyma are shielded from the effects of ischemia by PGE_2-mediated vasodilation and regulation of blood flow. PGE_2 is also involved in inflammatory cell activation, and PGE_2 that is biosynthesized by COX-2 and PGE_2 synthase in endothelial cells at the blood-brain barrier appears to have a role in fever.

Thromboxane and Prostacyclin

Platelets express high levels of the enzyme thromboxane synthase, but do not contain prostacyclin synthase. TxA_2 is the chief eicosanoid product of platelets. TxA_2 has a half-life of only seconds (~10–20 seconds) before it is nonenzymatically hydrolyzed to the inactive form TxB_2. TxA_2, which signals via a 7-transmembrane G-protein–coupled G_q mechanism, is both a strong vasoconstrictor and an agonist of platelet adhesion and aggregation. In contrast, the vascular endothelium lacks thromboxane synthase but expresses prostacyclin synthase. PGI_2 is, therefore, the primary eicosanoid product of the vascular endothelium. PGI_2, which signals via G_s, functions as a vasodilator and inhibitor of platelet aggregation. In other words, PGI_2 is a physiologic antagonist of TxA_2. The vasodilatory actions of PGI_2, as with those of PGE_2, also confer cytoprotective properties (Figure 12.2A and 12.2B).

The local TxA_2 and PGI_2 levels are critical in the regulation of systemic blood pressure and thrombogenesis. Imbalances can lead to hypertension, ischemia, thrombosis, coagulopathy, myocardial infarction, and stroke. In certain populations of the Northern latitudes (including Inuit, Greenlander, Irish, and Danish populations), the incidence of heart disease, stroke, and thromboembolic disorders is less than in Western populations. The diet of these Northern people is richer in whale and fish oils as compared to Western diets and, as a result, it contains relatively smaller amounts of arachidonic acid precursors but relatively

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TABLE 12.2. Prostanoid specific enzymes, receptors, and key immune bioactions

Prostaglandin	Synthetic enzyme	Tissues expressing synthetic enzyme	Receptor type and signaling mechanism		Bioactions
PGD_2	PGD_2 isomerase (a) Lipocalin type (b) Hematopoietic	Mast cells Neurons	DP1 DP2	G_s	Bronchoconstriction (asthma), resolution Sleep control functions
PGE_2	PGE_2 isomerase (a) Cytosolic (b) Microsomal 1 (c) Microsomal 2	Many tissues, including macrophages and mast cells	EP1 EP2 EP3 EP4 Other	G_q G_s G_i G_s	Potentiation of responses to painful stimuli Vasodilation Bronchoconstriction Cytoprotective: modulates gastric mucosal acid secretion, mucus, and blood flow Vasodilation Bronchoconstriction Inflammatory cell activation Pyrexia Mucus production
$PGF_{2\alpha}$	$PGF_{2\alpha}$ reductase	Vascular smooth muscle Uterine smooth muscle	FP	G_q	Vascular tone Reproductive physiology (abortifacient) Bronchoconstriction

The prostanoid receptors are G-protein–coupled receptors (see Brink et al. for further details).

larger amounts of EPAs. As a result, the thromboxane–prostacyclin levels tip toward vasodilation, platelet inhibition, and antithrombogenesis. These combined effects and change in balance of TX-PGI represent one possible explanation for the observation that these populations have a lower incidence of cardiovascular diseases and is one rationale for increasing dietary fish consumption in Western diets.

Lipoxygenase Pathways

The LOX pathways (Figure 12.3A and 12.3B) are a major route for converting arachidonic acid to bioactive mediators, namely formation of both leukotrienes and lipoxins. Lipoxygenases are enzymes that catalyze insertion of molecular oxygen into arachidonic acid using nonheme iron generating specific hydroperoxides. Three lipoxygenases, 5-, 12-, and 15-lipoxygenases (5-LOX, etc.), are the major LOX found in humans. The lipoxygenases are named for the position of the inserted molecular O_2 in arachidonic acid. The immediate products of LOX reactions are hydroperoxyeicosatetraenoic acids (HPETEs). HPETEs can be reduced to the corresponding hydroxyeicosatetraenoic acids (HETEs) by enzymes using glutathione peroxidase (GPx). The 5-LOX produces 5-HPETE, the direct precursor to LTA_4, which in turn is the precursor intermediate for the potent bioactive leukotrienes (Figure 12.3). Lipoxygenases are also involved in converting 15-HETE and LTA_4 to lipoxins (see later). 5-LOX requires translocation to the nuclear membrane for activity. The 5-LOX-activating protein (FLAP) helps 5-LOX translocate to the nuclear membrane, form an active enzyme complex, and accept the arachidonic acid substrate.

Leukotriene Biosynthesis

Leukotriene biosynthesis is initiated by the 5-LOX-mediated conversion of arachidonic acid into 5-HPETE and further into leukotriene A_4 (LTA_4). Note that 5-LOX catalyzes both the steps in leukotriene biosynthesis in a concerted manner (Figure 12.3). Depending on the cell type, LTA_4 is next converted to either LTB_4 or LTC_4. The enzyme LTA_4 hydrolase converts LTA_4 to LTB_4 in neutrophils as well as erythrocytes, macrophages, and monocytes. LTA_4 conversion to LTC_4 occurs in mast cells, eosinophils, basophils, and macrophages by the addition of a γ-glutamylcysteinylglycine tripeptide (glutathione), which may be cleaved by peptidases to generate LTD_4 and LTE_4. Together, LTC_4, LTD_4, and LTE_4 represent the cysteinyl leukotrienes (Figure 12.3).

LTB_4 acts via two G-protein–coupled receptors, BLT1 and BLT2. Binding of LTB_4 to BLT1, which is expressed on tissues involved in host-defense and inflammation (leukocytes, thymus, spleen), leads to proinflammatory sequelae, most importantly neutrophil chemotaxis, aggregation, and transmigration across epithelium and endothelium. LTB_4 upregulates neutrophil lysosomal function and free radical production, enhances cytokine production, and potentiates the actions of natural killer cells. The role of the second LTB_4 receptor denoted BLT2 remains unknown.

The cysteinyl leukotrienes (LTC_4, LTD_4, and LTE_4) bind to CysLT1 receptors to cause vasoconstriction,

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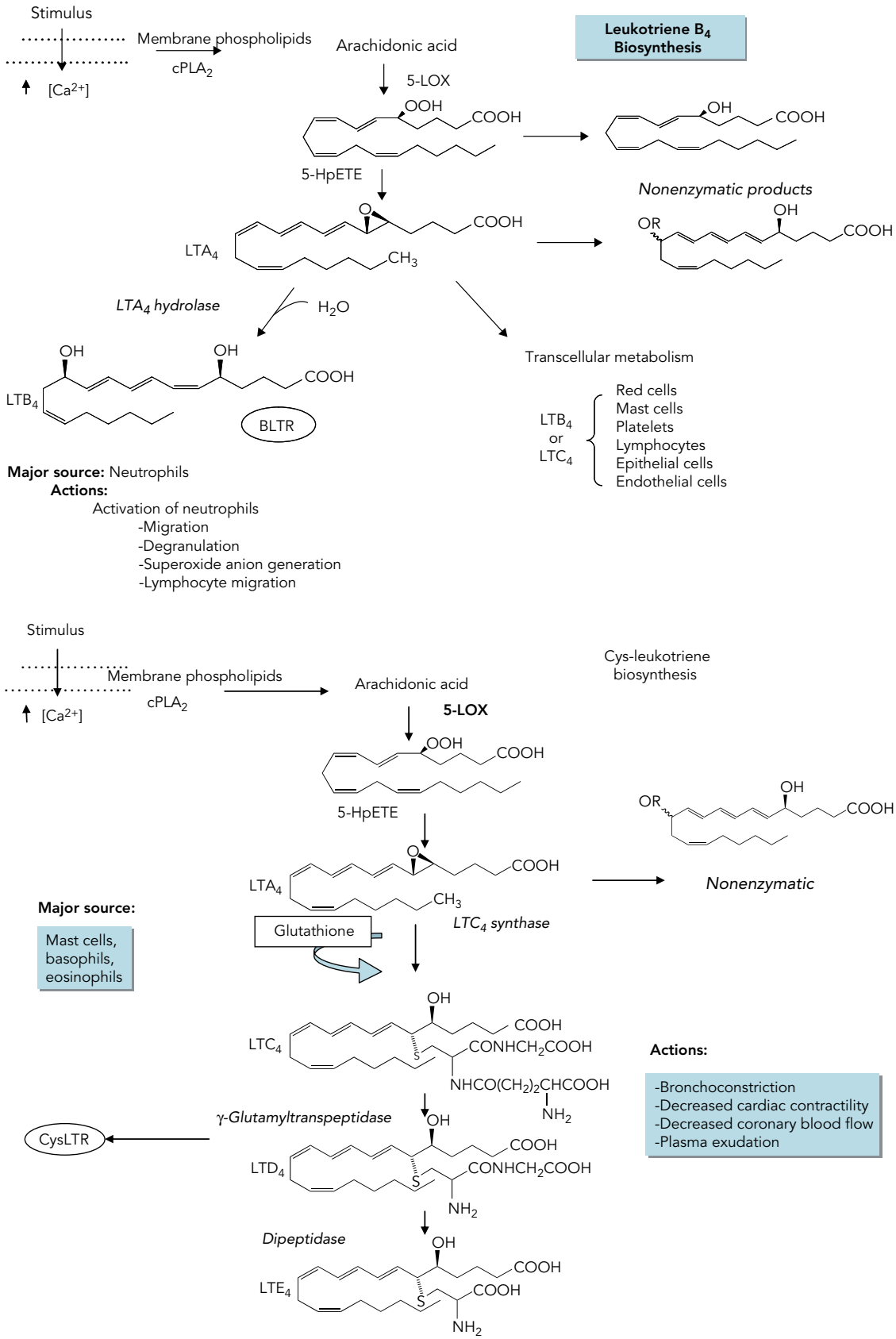


Figure 12.3. Leukotriene biosynthesis and function. The biosynthetic pathways from arachidonic acid to the leukotrienes are shown. (A) Leukotriene B_4 biosynthesis. Zileuton and 5-lipoxygenase-activating protein (FLAP) inhibitors prevent the conversion of arachidonic acid to 5-HPETE and LTA_4; zileuton is used in the chronic management of asthma. BLTR = BLT1 (and BLT2), LTB_4 receptor(s).
(B) Cys-Leukotriene C_4, D_4. Receptors are denoted as CysLTR compromising CysLT1, CysLT2, and GPR17, receptors for cysteinyl-leukotrienes. Zafirlukast and montelukast are antagonists at CysLT1, a receptor for all cysteinyl leukotrienes; these drugs are used in the chronic management of asthma.

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bronchospasm, and increased vascular permeability. Cysteinyl leukotrienes are responsible for airway and vascular smooth muscle contraction that occurs in asthma, allergic, and hypersensitivity processes. Leukotrienes are powerful lipid mediators that exert their bioactions already in low nM concentrations, and together, both arms of the leukotriene pathways, that is, LTB_4, and the cysteinyl leukotrienes, are held to play key roles in psoriasis, arthritis, and various inflammatory responses. More recent findings also place them as local mediators in vascular disease as well as atherosclerosis.

Lipoxin Biosynthesis

Lipoxins (LOX interaction products) are derived from arachidonic acid and contain four conjugated double bonds and three hydroxyl groups. The two main lipoxins are LXA_4 and LXB_4 (Figure 12.4). The lipoxins modulate the actions of leukotrienes and cytokines. LX are important in resolution, serving as agonists and counter-regulating mediators.

At sites of inflammation, there is typically an inverse relationship between the amounts of lipoxin and leukotriene present. LXA_4 receptors are present in the lung, spleen, and blood vessels, and on neutrophils. Lipoxins stop neutrophil chemotaxis, adhesion, and transmigration through endothelium (by decreasing P-selectin expression), inhibit eosinophil recruitment, stimulate vasodilation (by inducing synthesis of PGI_2 and PGE_2), inhibit LTC_4- and LTD_4-stimulated vasoconstriction, inhibit LTB_4 inflammatory effects, and inhibit the function of NK cells. Lipoxins are potent (active at similar concentrations as leukotrienes) agonists of resolution mechanisms, which include stimulating the uptake and clearance of apoptotic neutrophils by macrophages. Lipoxin production may, therefore, be important in the resolution of inflammation, and an

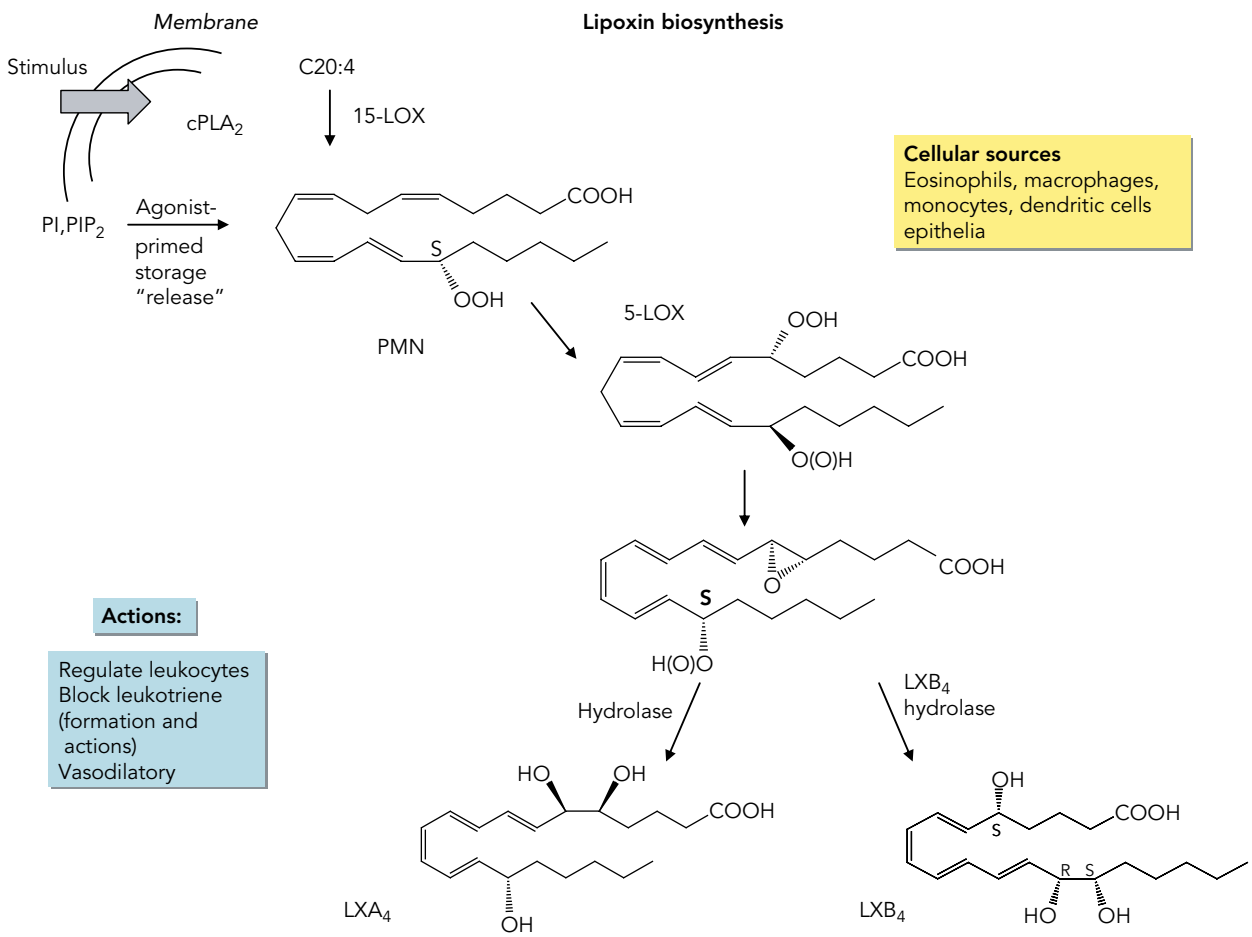


Figure 12.4. Lipoxin biosynthesis. Two main routes lead to biosynthesis of the lipoxins. In each pathway, sequential lipoxygenase reactions are required, followed by hydrolysis. The immediate precursor of the lipoxins is epoxytetraene; hydrolysis of epoxytetraene yields the lipoxins. Left pathway: Arachidonic acid is converted to 15-HETE by sequential activity of 15-LOX and peroxidase. 15-HETE is converted by 5-LOX to the chemical intermediate 5-hydroperoxy, 15-hydroxyeicosatetraenoic acid, and 5-LOX acts on this intermediate to form epoxytetraene. LTA_4 route: Arachidonic acid is converted to 5-HPETE by 5-LOX, and 5-HPETE is converted to LTA_4 by further action of 5-LOX. LTA_4 is converted to epoxytetraene by 15-LOX (see text for further details).

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imbalance in lipoxin–leukotriene homeostasis may be a key factor in the pathogenesis of inflammatory disease.

Epoxygenase Products

Microsomal cytochrome P450 epoxygenases are able to oxygenate arachidonic acid, resulting in the formation of epoxyeicosatetraenoic acid (EET) and hydroxyacid derivatives (Figure 12.1). These are major pathways in tissues that do not express COX or LOX, for example, certain cells of the kidney. The epoxygenation of arachidonic acid results in four different EETs, depending on which double bond in arachidonic acid is modified. Dihydroxy derivatives of EETs, formed by hydrolysis, may have roles in the regulation of smooth muscle cells and vascular tone by inhibiting the Na^+/K^+-ATPase, as well as in renal function by regulating ion absorption and secretion. With respect to inflammation, dihydroxy EETs inhibit platelet COX and expression of intercellular adhesion molecules (ICAMs). Downregulation of ICAMs inhibits platelet and inflammatory cell aggregation. Thus, specific EETs (i.e., 11,12-EET at micromolar levels) may play a role in regulating inflammation at certain sites and within select tissues. Future research may reveal more definitive functions for EETs in inflammatory cells and immune system.

Isoprostanes: Markers of Oxidative Stress

Phospholipid-esterified arachidonic acid is susceptible to free-radical–mediated peroxidation; cleavage of these modified lipids from the phospholipid by phospholipase A_2 gives rise to the isoprostanes (Figure 12.1). During oxidative stress, isoprostanes are found in the blood at levels much higher than those of COX products. Two isoprostanes in particular, 8-epi-$PGF_2\alpha$ and 8-epi-PGE_2, are potent vasoconstrictors. Isoprostanes may function in activating NF-κB, phospholipase Cγ, protein kinase C, and calcium flux. Because the formation of isoprostanes depends on cellular oxidation conditions, isoprostane levels may be indicative of oxidative stress and a wide range of pathologies. Urinary isoprostane levels are used as a marker of oxidative stress in ischemic syndromes, reperfusion injury, atherosclerosis, and hepatic diseases. Isoprostanes are not, however, known to have direct roles in host defense.

Metabolic Inactivation of Eicosanoids

Following their local actions, the prostaglandins, leukotrienes, thromboxanes, and lipoxins are inactivated locally by hydroxylation, β-oxidation (resulting in a loss of two carbons), or ω-oxidation (to dicarboxylic acid derivatives). These degradation processes render the molecules more hydrophilic and, therefore, excretable in the urine.

TABLE 12.3. Pathophysiology: human diseases with dysregulated eicosanoids

Asthma	Atherosclerosis
Inflammatory bowel disease	Cardiovascular diseases
Arthritis	Skin disorders
Glomerulonephritis	Reperfusion organ injury
Alzheimer's disease	Periodontal disease
Cancer	

Pathophysiology

The immune response and subsequent inflammation are the body's mechanisms for combating foreign invaders. This overall response is designed to remove the inciting stimulus and resolve tissue damage. In some cases, the response mechanism itself causes local tissue damage, such as when activated neutrophils inadvertently release proteases and reactive oxygen species (ROS) to the local milieu. In other settings, if the inflammatory reactions persist for too long, each counter-regulation, or the immune system misidentifies a part of self as foreign; these abnormal, misdirected immune responses can cause significant and chronic tissue injury. Eicosanoids are implicated in selected inflammatory diseases, including asthma, inflammatory bowel disease, rheumatoid arthritis, atherosclerosis, and glomerulonephritis (see Table 12.3).

MODULATORS AND INHIBITORS

Given the extensive and diverse roles of the many eicosanoids, the complexity of the pathways offers a variety of targets for controlling inflammation and aberrant immune responses. Strategies considered include altering the expression of key enzymes, competitively and noncompetitively inhibiting the activity of specific enzymes (i.e., PGE_2 synthases), activating receptors with exogenous receptor agonists, and preventing receptor activation with exogenous receptor antagonists. As with all other approaches, the therapeutic benefits must be weighed against the potential unwanted side effects.

Phospholipase Inhibitors: Inhibition of phospholipase A_2 prevents the generation of arachidonic acid, the rate-limiting step in eicosanoid biosynthesis. In the absence of proinflammatory mediators derived from arachidonic acid, inflammation is limited. Glucocorticoids (also known as corticosteroids, of which prednisone is a member) are a mainstay of

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therapy in a multitude of autoimmune and inflammatory diseases. Glucocorticoids induce a family of secreted calcium- and phospholipid-dependent proteins called lipocortins. Lipocortin interferes with the action of phospholipase A_2, and thus limits the release of arachidonic acid. Annexins, such as Annexin 1- and Annexin 1–derived peptides, are also induced by glucocorticoids that, in turn, act at GPCR on leukocytes to block proinflammatory responses and enhance endogenous anti-inflammatory mechanisms including activation of the lipoxin A_4 receptor. Small molecule inhibitors of specific phospholipases are in development; they may offer the potential for reducing some adverse effects associated with glucocorticoid use.

Cyclooxygenase Inhibitors: A number of COX pathway inhibitors are currently in wide use. These are some of the most frequently prescribed over-the-counter drugs. NSAIDs are important because of their combined anti-inflammatory, antipyretic, and analgesic properties. The ultimate goal of most NSAID therapies is to inhibit COX generation of proinflammatory eicosanoids and to limit the extent of inflammation, fever, and pain. The drugs' antipyretic activity is likely related to their decreasing the levels of PGE_2, particularly in the region of the brain surrounding the hypothalamus. Despite the benefits of current NSAIDs, these drugs only suppress the signs of the underlying inflammatory response.

A multitude of NSAIDs have been developed over the past century. All NSAIDs, except aspirin, act as reversible, competitive inhibitors of COX. Generally, these drugs block the hydrophobic channel in COX in which the substrate arachidonic acid binds, thereby preventing access of arachidonic acid to the active site of the enzyme. Traditional NSAIDs inhibit both COX-1 and COX-2 to different degrees. Because of inhibition of COX-1, long-term NSAID therapy has many deleterious effects. The cytoprotective roles of the COX-1 eicosanoid products are eliminated, leading to a spectrum of NSAID-induced gastropathy including dyspepsia, gastrotoxicity, subepithelial damage and hemorrhage, gastric mucosal erosion, frank ulceration, and gastric mucosal necrosis. Regulation of blood flow to the kidney is likewise perturbed, potentially causing renal failure.

Aspirin is the first and the oldest of the NSAIDs and is widely used to treat mild to moderate pain, headache, myalgia, and arthralgia (see Figure 12.5). In contrast to other NSAIDs, aspirin acts in an irreversible manner by acetylating the active site serine residue in both COX-1 and COX-2. Acetylation of COX-1 destroys the enzyme's COX activity and thereby prevents the formation of COX-1–derived prostaglandins, thromboxanes, and prostacyclin.

Daily low-dose aspirin is used as an antithrombotic agent for prophylaxis and postevent management of myocardial infarction and stroke. Recall that aspirin is antithrombotic because of its irreversible inhibition of COX that prevents platelets from biosynthesizing TxA_2. Within an hour of oral aspirin administration, the COX-1 activity in existing platelets is irreversibly destroyed (*vide supra*). The importance of this lies in the fact that platelets circulate in the bloodstream for ~10 days but cannot synthesize new protein. Therefore, irreversibly acetylated COX-1 is not replaced by freshly synthesized proteins. These platelets are irreversibly inhibited for their lifetime. Although aspirin also irreversibly inhibits vascular endothelial cell COX-1, the endothelial cells can synthesize new COX-1 protein, as well as possess constitutively expressed COX-2, and thus can rapidly resume synthesis of PGI_2. A single administration of aspirin decreases the amount of thromboxane that can be generated for several days, shifting the vascular local TxA_2–PGI_2 levels toward PGI_2-mediated vasodilation, platelet inhibition, and antithrombogenesis.

Aspirin-mediated inhibition of COX-2 prevents the generation of prostaglandins. Unlike COX-1 which is totally inactivated, aspirin-modified COX-2 retains a part of its catalytic activity that forms a new product, 15*R*-HETE, from arachidonic acid. By analogy to lipoxin biosynthesis (Figure 12.4), 5-LOX then converts 15*R*-HETE to 15-epi-lipoxins, which are stable epimers (carbon 15 position epimers) of native lipoxins. These epimers are collectively called aspirin-triggered lipoxins (ATLs). 15-Epi-lipoxins mimic the functions of lipoxins as anti-inflammatory and proresolving mediators. 15-Epi-lipoxins or 15*R*-epi-lipoxins may represent an endogenous mechanism of anti-inflammation, and their production is believed to mediate at least part of the anti-inflammatory effects of aspirin. Development of 15-epi-lipoxin analogues could lead to new anti-inflammatory drugs that do not have the unwanted side effects associated with COX-1 inhibition. There are several additional classes of NSAIDs that include ibuprofen, naproxen, ketoprofen, and flurbiprofen. The acetic acid NSAIDs include indole acetic acids – indomethacin, sulindac, and etodolac – and the phenylacetic acids diclofenac and ketorolac (a substituted phenylacetic acid derivative).

Acetaminophen, although sometimes classified along with NSAIDs, is by definition not an NSAID. Although acetaminophen has analgesic and antipyretic effects similar to aspirin, the anti-inflammatory effect of acetaminophen is insignificant because of its weak inhibition of cyclooxygenases. Acetaminophen therapy can, however, be valuable in patients, for example children, who are sensitive to the side effects of aspirin. The key side effect of acetaminophen is hepatotoxicity. Modification of acetaminophen by hepatic cytochrome P450 enzymes produces a highly reactive molecule that is detoxified by conjugation with

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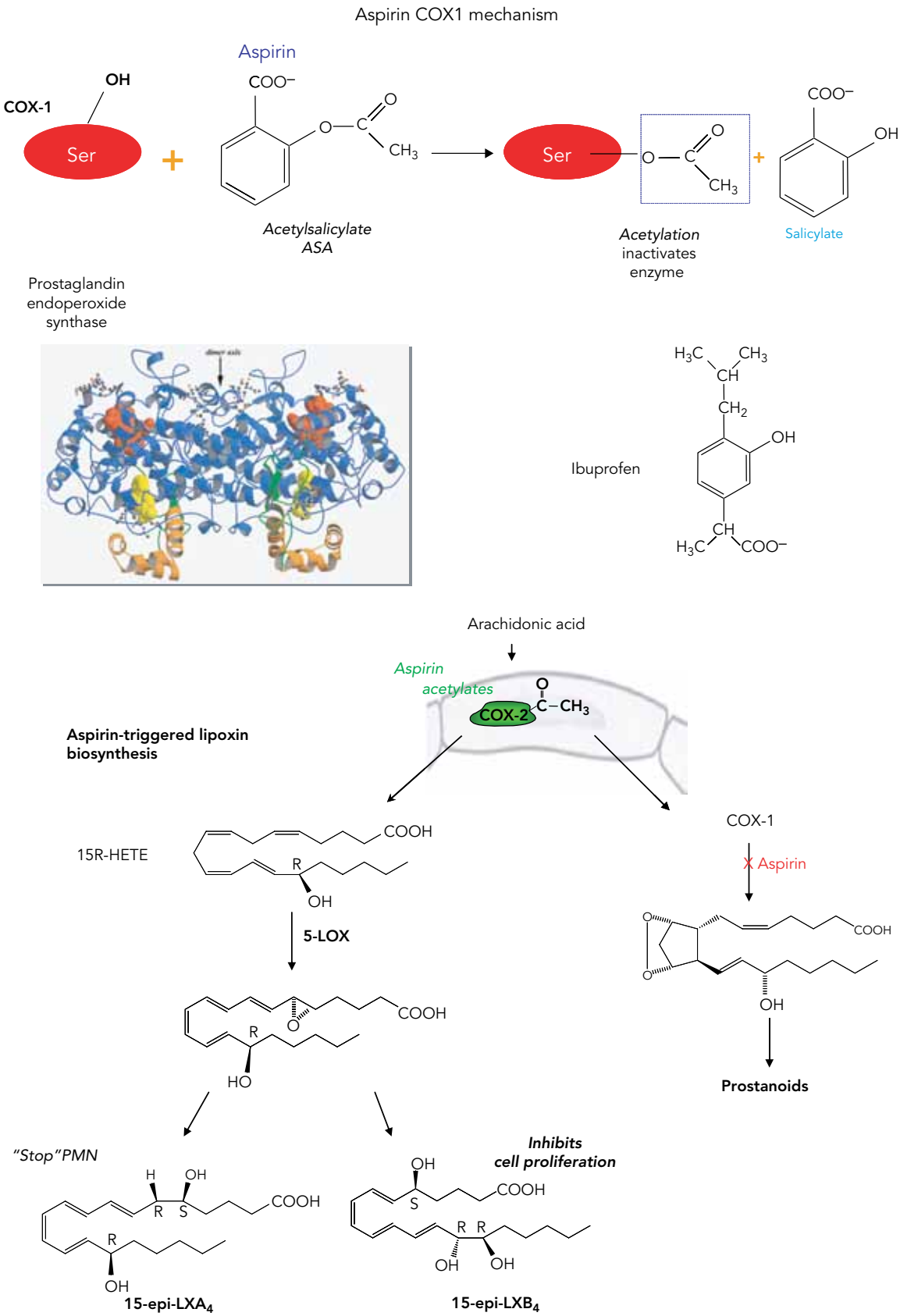


Figure 12.5. Aspirin mechanism of action. (A) Aspirin mechanism with COX-1: inhibition of prostanoid intermediates. (B) Aspirin and the biosynthesis of the aspirin-triggered Lipoxins (ATL) via COX-2.

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glutathione. An overdose of acetaminophen can overwhelm glutathione stores. This can lead to cellular and oxidative damage and, in some settings, to acute hepatic necrosis.

The anti-inflammatory, analgesic, and antipyretic effects of the NSAIDs do vary somewhat among the many drugs in this group. Hence, successful NSAID therapy is still considered more of an art than a science for each patient, and therapy should be directed toward achieving the desired anti-inflammatory, analgesic, and antipyretic effects while minimizing the unwanted side effects from relative inhibition of COX-1 and COX-2.

COX-2 Selective Inhibitors: COX-2 was identified in the early 1990s, and intense research swiftly led to the development of COX-2 selective inhibitors for clinical use. Compared with COX-1, COX-2 has a larger hydrophobic channel through which substrate enters the active site. Subtle structural differences between COX-2 and COX-1 topography allowed development of drugs that act preferentially on COX-2 rather than COX-1. The COX-2 selective inhibitors including celecoxib, rofecoxib, valdecoxib, and meloxicam are sulfonic acid derivatives that display 100-fold selectivity for COX-2 compared to COX-l. The relative inhibition of the two COX isozymes by these drugs in any given tissue is also a function of drug metabolism, pharmacokinetics, and possibly enzyme polymorphisms. The COX-2 selective inhibitors have similar anti-inflammatory, antipyretic, and analgesic properties as the traditional NSAIDs, but they do not share the antiplatelet actions of the COX-1 inhibitors.

Currently, it is not clear if they are less safe than other NSAIDs, and only celecoxib remains an approved drug. The long-term safety profiles of COX-2 inhibitors are in question. There is concern that COX-2 inhibitors – in particular, rofecoxib, a widely marketed COX-2 inhibitor – have deleterious effects on the cardiovascular and renal systems by inducing hypertension, renal failure, and cardiac failure. Prolonged inhibition of vascular COX-2 within endothelial cells reducing PGI_2 formation may account for the increased thrombogenesis uncovered in clinical trials. In addition, inhibition of COX-2 may generate potential problems in resolution, wound healing, and angiogenesis. Also, COX-2 selective inhibitors are much more expensive than equivalent doses of many NSAIDs, especially aspirin.

Lipoxygenase 5-LOX Inhibitors: Selective inhibition of the 5-LOX has the potential to represent a major therapeutic modality in diseases involving leukotriene-mediated pathophysiology, including asthma, inflammatory bowel disease, and possibly atherosclerosis and rheumatoid arthritis. Drugs that impair or alter the ability of LOX to utilize its nonheme iron properly would be expected to inhibit the activity of the enzyme. The only iron chelator drug in clinical use is zileuton, a benzothiophene derivative of N-hydroxyurea that inhibits 5-LOX. In asthma, zileuton induces bronchodilation, improves symptoms, and generates long-lasting improvement in pulmonary function tests. Interfering with the role of FLAP could represent an approach to the selective inhibition of 5-LOX activity and leukotriene function. Recall that 5-LOX is activated after the enzyme localizes to the nuclear membrane and docks with FLAP; FLAP also binds arachidonic acid released by phospholipase A_2 and shuttles it to the 5-LOX active site. FLAP inhibitors that both prevent and reverse LOX binding to FLAP are in clinical development.

CysLT Receptor Antagonists: Leukotriene receptor antagonism represents the receptor-based mechanism for inhibiting leukotriene-mediated bronchoconstriction and smooth muscle actions. Cysteinyl leukotriene receptor (CysLT1) antagonists are effective against antigen-, exercise-, cold-, and aspirin-induced asthma. These agents significantly improve bronchial tone, pulmonary function tests, and asthma symptoms. Montelukast and zafirlukast are the currently available cysteinyl leukotriene receptor antagonists; the main clinical application is in asthma treatment.

Crystal Structures in the Eicosanoid Cascade – Basis for Rational Drug Design

Over the past decade, crystal structures have been determined for several important enzymes in the eicosanoid cascade. In many instances, these structures have been used to perform rational, structure-based drug design, a process in which lead compounds are developed and optimized through repeated cycles of structure determinations of enzyme–inhibitor complexes and chemical refinement.

Starting from the top of the cascade where arachidonic acid is released from membrane phospholipids by the high-molecular weight cPLA2, a "master" enzyme which funnels substrate to both the COX and LOX pathways. The structure of cPLA2 has been solved and revealed an N-terminal calcium-dependent lipid-binding/C2 domain and a catalytic unit with a Ser-Asp dyad. Interestingly, there is a lid that must move to allow substrate access to the active site, explaining the interfacial activation of the enzyme.

As mentioned, crystal structures of the membrane associated COX-1 and COX-2 are available and have been instrumental for development of the second generation COX-2 inhibitors that exploit the small structural differences of a side pocket in the two isoenzymes. In addition, several downstream prostaglandin synthases have been structurally characterized, including PGD, PGI, and PGF synthases, as well as cytosolic and microsomal PGE synthase type 2.

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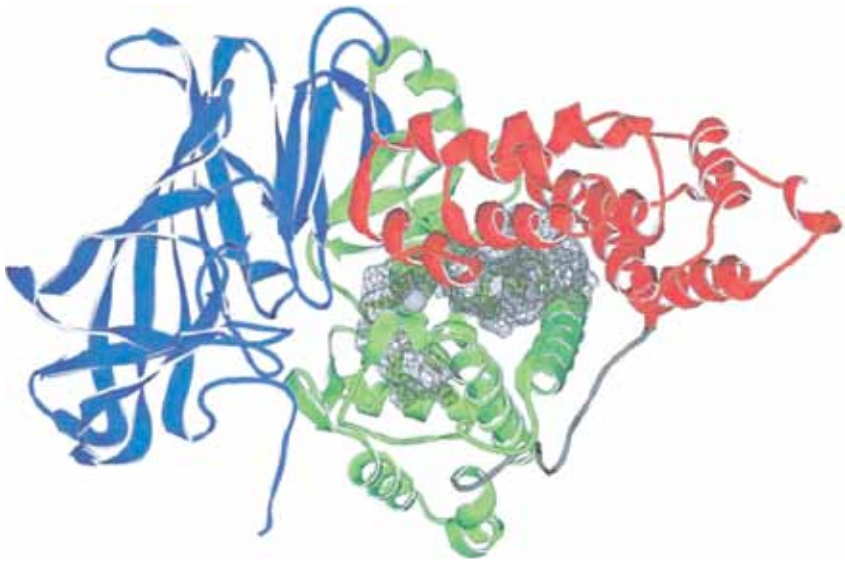


Figure 12.6. Overall structure of LTA_4 hydrolase. The N-terminal, catalytic, and C-terminal domains are colored in blue, green, and red, respectively. The active site is indicated as a gray volume in the central parts of the protein. Figure created by Fredrik Tholander.

To date, only a single mammalian LOX has been structurally characterized, *viz*. rabbit 15-LOX. This enzyme has an N-terminal β-barrel domain resembling the C-terminal domain of certain lipases, and a catalytic domain with an active site pocket starting at the protein surface that puts the fatty acid substrate in an optimal position relative to the catalytic iron. This structure has been used to create model structures of 5-LOX. However, the inherent limitations of these models preclude their use in rational drug design.

Continuing along the 5-LOX/leukotriene pathway, the soluble, 70 kDa, LTA_4 hydrolase has also been crystallized. The protein is folded into three domains and the L-shaped active site goes all the way from the surface deep into the protein (Figure 12.6). The structure reveals the molecular background of the enzyme's ability to convert LTA_4 into LTB_4 and also to cleave peptide bonds. Moreover, the 3-D structure offers a chemical terrain that is almost ideal for structure-based drug design.

The other branch of the leukotriene pathway leading to the cysteinyl-leukotrienes is governed by LTC_4 synthase, an integral membrane protein. This enzyme is a member of a larger protein family denoted MAPEG, which also includes FLAP (five-lipoxygenase activating protein) and mPGES-1 (microsomal PGE synthase type 1), believed to synthesize proinflammatory PGE_2. Recently, high-resolution crystal structures of LTC_4 synthase were presented revealing a trimeric enzyme, with a GSH binding site composed of residues from two neighboring monomers and a hydrophobic, superficial cleft, presumably accommodating the substrate LTA_4 (Figure 12.7). These structures will certainly enable the molecular characterization of the catalytic machineries as well as the design of potent and specific inhibitors of LTC_4 synthase, a hitherto unexploited drug target. Furthermore, crystal structures of FLAP with bound inhibitors have been presented, demonstrating the protein sites at which this class of antileukotrienes interact. Also, another 5-LOX supporting protein, the 16 kDa CLP (coactosin-like protein), was structurally determined by both NMR and crystallography. The next step in the structure biology of the eicosanoid cascade will be the structure determinations of eicosanoid receptors, a huge challenge that requires enormous human and financial efforts. If successful, it will certainly have a profound impact on drug development.

PLATELET-ACTIVATING FACTOR

Biosynthesis and Actions

In the early 1970s, the name *platelet-activating factor* (PAF) was coined to describe a bioactivity or substance(s) that, when released from basophils during IgE-induced anaphylaxis, activated platelets. This bioactive substance was subsequently isolated and its structure elucidated as a group of molecules characterized by the structure 1-*O*-alkyl-2-acyl-*sn*-glycero-3-phosphocholine (Figure 12.8). Individual molecules of PAF differ only within the number of carbons in the alkyl group. Like the eicosanoids, PAFs are derived from membrane phospholipids by phospholipase A_2. Upon cellular activation, phospholipase A_2 also affects the deacylation of 1-*O*-alkyl-2-acyl-glycero-3-phosphocholine yielding lyso-PAF plus unesterified fatty acid. Lyso-PAF can then be acetylated by

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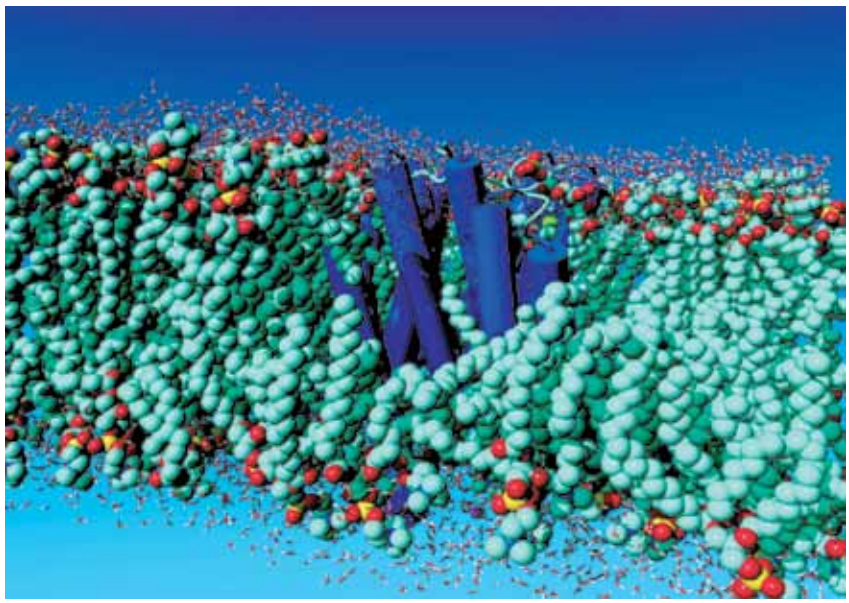


Figure 12.7. Structure of the integral membrane protein LTC_4 synthase. Transmembrane α-helices are shown as dark blue cylinders. The architecture of the lipid bilayer is computer designed. Figure created by Fredrik Tholander.

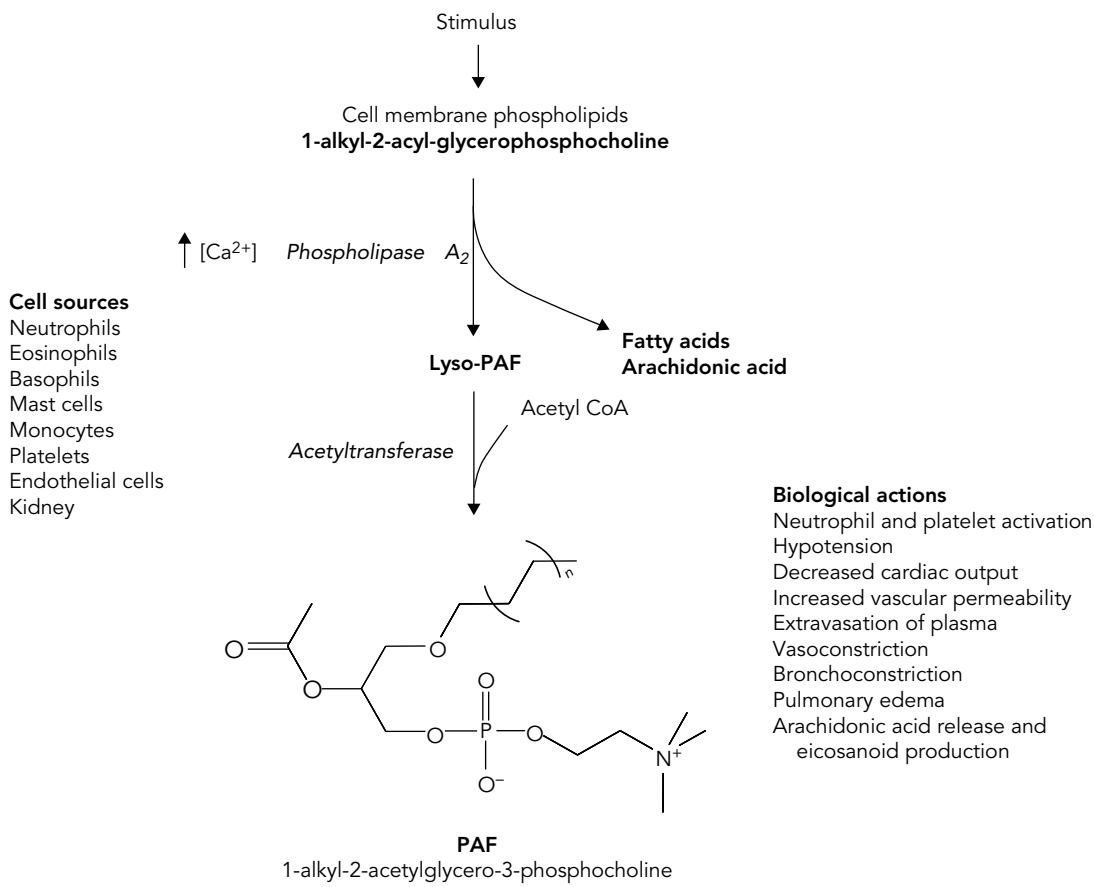


Figure 12.8. Biosynthesis of PAF and its major actions.

acetyl-coenzyme A (acetyl-CoA) transferase to give PAF. As much as 40% of the fatty acid released from the sn-2 position during PAF biosynthesis by human neutrophils is arachidonic acid. Hence, PAF biosynthesis by activated inflammatory cells is often accompanied by the generation of eicosanoids.

A number of different cell types generate PAF, including leukocytes (neutrophils, basophils, and

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eosinophils), macrophages, and mast cells, as well as platelets. Of interest, PAF biosynthesis can also occur in some noninflammatory and resident cells, such as endothelial and epithelial cells, suggesting that PAF may serve a physiologic function in addition to its role as a mediator of inflammation. A wide range of stimuli can activate PAF biosynthesis. For example, PAF biosynthesis by neutrophils is stimulated by phagocytosis of serum-treated zymosan, C5a fraction of complement and formyl-methionyl-leucyl-phenylalanine (a synthetic peptide that mimics the action of the formylated peptides contained in bacterial cell walls), as well as the divalent cation ionophore A23187. These stimuli increase the levels of intracellular calcium and lead to activation of phospholipase A_2. The stimuli for PAF biosynthesis are cell-type specific. For example, thrombin, angiotensin II, and vasopressin do not trigger the generation of PAF in inflammatory cells, but stimulate PAF formation in vascular endothelial cells.

PAF Actions *in vitro* and *in vivo*: As a mediator, PAF is implicated in the pathogenesis of both acute inflammation and hypersensitivity disorders. PAF is also likely to be an important mediator in several other conditions, including endotoxic shock, vasculitis, and arterial thrombosis. As the name indicates PAF is a potent activator of platelets (activation and aggregation) and of human neutrophils. PAF stimulates neutrophil adhesion, lysosomal enzyme release, the generation of ROS and eicosanoids. PAF causes rapid margination of neutrophils onto endothelial cell walls *in vivo*, and promotes neutrophil migration to the extracellular space. PAF may also be a regulator of lymphocyte function, either directly or via indirect routes that can involve the generation of specific eicosanoids, for example, prostaglandins and leukotrienes. PAF has been shown to inhibit lymphocyte proliferation in response to various mitogens. In turn, lymphocytes can generate PAF upon stimulation, particularly by agonists that increase intracellular calcium.

Local administration of PAF by intradermal injection in laboratory animals or human subjects is associated with neutrophil margination and intravascular thrombosis. There is an associated increase in vascular permeability, edema, and in hyperalgesia. Hence, PAF causes many of the cardinal features of inflammation.

Intravenous injection of PAF has profound effects on bronchial smooth muscle tone and the cardiovascular function of laboratory animals. Intravenous PAF causes a syndrome that is very similar to that of acute anaphylaxis (i.e., severe acute hypersensitivity reaction). PAF induces contraction of bronchial and vascular smooth muscle, including coronary artery smooth muscle. In addition, there is a marked increase in vascular permeability and exudation of plasma into the extravascular space, including the lungs (pulmonary edema). As a result, the animals develop respiratory insufficiency, decreased cardiac output, and hypotension.

Integrated Inflammation Schema: Lipid Mediators in Resolution

As described earlier, eicosanoids are generated locally in numerous complex reactions. It is not necessary to remember every mediator, but rather to understand the general scheme of these synthetic pathways. This section, along with Table 12.4, provides a concise overview of the functions of eicosanoids relevant to host defense, inflammation, and resolution of inflammation.

Acute inflammation is the result of an intricate network of molecular and cellular interactions induced by responses to a variety of stimuli, such as trauma, ischemia, infectious agents, or antibody reactions. Acute superficial inflammation generates local pain, edema, erythema, and heat; inflammation in visceral organs can have similar symptoms and result in severe impairment of organ function.

Leukotrienes and lipoxins, as well as thromboxanes, prostaglandins, and prostacyclins, are critical for generating, maintaining, and mediating inflammatory responses. The inflammatory cascade is initiated when cells in a particular region are exposed to a foreign substance or are damaged. That insult stimulates a local cytokine cascade (including interleukins or TNF) that raises the expression of COX-2 mRNA and enzyme levels. COX-2 then facilitates production of the proinflammatory and vasoactive eicosanoids (see additional illustrations in Chapter 2).

TABLE 12.4. Roles of lipid mediators in the steps of inflammation

Action	Lipid mediators involved
Vasoconstriction	$PGF_2\alpha$, TxA_2, LTC_4, LTD_4, LTE_4
Vasodilation (erythema)	PGI_2, PGE_1, PGE_2, PGD_2, LXA_4, LXB_4, LTB_4
Edema (swelling)	PGE_2, LTB_4, LTC_4, LTD_4, LTE_4
Chemotaxis, leukocyte adhesion	LTB_4, HETE, LXA_4, LXB_4
Increased vascular permeability	LTC_4, LTD_4, LTE_4
Pain and hyperalgesia	PGE_2, PGI_2, LTB_4
Local heat and systemic fever	PGE_2, PGI_2, LXA_4
Eicosanoid class switching	PGE_2, PGD_2
Stimulate resolution	Lipoxins, resolvins, and protectins (see Table 12.6 for specific steps and actions in resolution)

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Locally high concentrations of PGE_2, LTB_4, and cysteinyl leukotrienes promote the accumulation and infiltration of inflammatory cells by increasing blood flow and vascular permeability. LTB_4 and 5-HETE are also important in attracting and activating neutrophils. LTB_4, formed by activated neutrophils at the site of inflammation, recruits, and activates additional neutrophils and lymphocytes so that these cells adhere to endothelial surfaces and transmigrate into the interstitial spaces. Increased vascular permeability also results in fluid leakage and cellular infiltration, causing edema.

The aggregation of a multitude of inflammatory cells initiates the cell–cell interactions (as in pus or exudates) that foster transcellular biosynthetic routes that are exploited to generate eicosanoids (Figure 12.9). In transcellular biosynthesis, eicosanoid intermediates are donated from one cell type to another to generate a greater diversity of eicosanoids. This demonstrates the importance of cellular adhesion and cell–cell interaction in inflammatory and immune responses. The body tries to ensure that the inflammatory response cannot proceed unchecked.

Lipoxins help resolve inflammation and promote the return of the tissue to homeostasis. COX-2-derived eicosanoids may also function in wound healing and resolution. Hence, the temporal sequence of events is important in an organized inflammatory response (see Chapter 10). PGE_2 inhibits the functions of B and T lymphocytes and NK cells, while LTB_4 and the cysteinyl leukotrienes regulate T lymphocyte proliferation. PGE_2 and PGI_2 are potent sensitizers in pain and turn on lipoxin production that in turn reduces nociception. These many local factors can coordinate mediating and regulating the transition from acute to chronic inflammation versus the temporal events to resolution and homeostasis.

Lipoxins and ATLs offer the potential to counter-regulate/antagonize the inflammatory actions of leukotrienes and other inflammatory mediators, and as agonists to promote resolution of inflammation. Analogs of these compounds represent new approaches to treatment because they are agonists of endogenous anti-inflammation and proresolving mechanisms rather than direct inhibitors of biochemical pathways or receptor antagonists. Because lipoxins are endogenous regulators, they are expected to have selective actions with few adverse effects. Stable analogues of lipoxins and ATLs are currently being developed, and second-generation LX stable analogues have shown

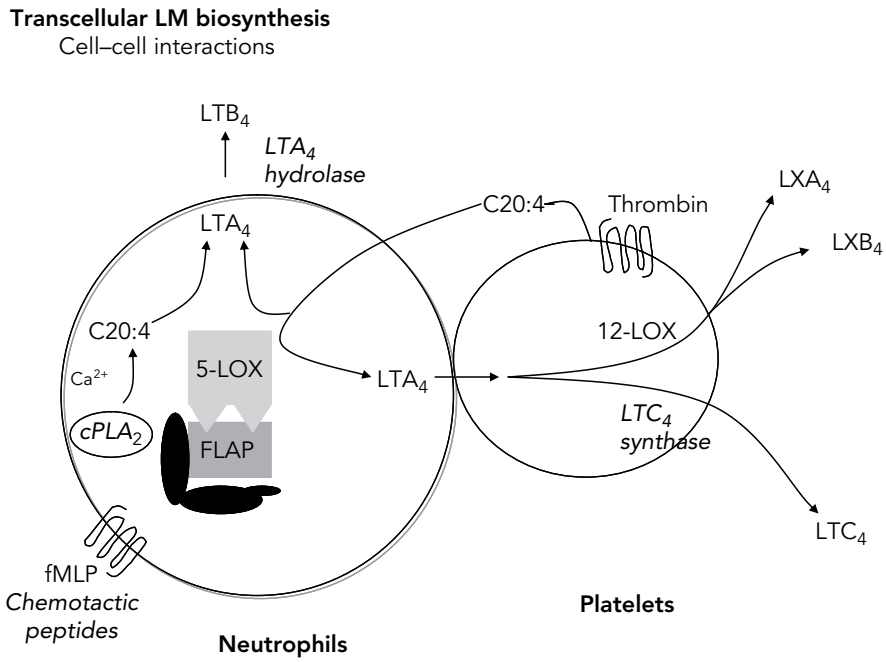


Figure 12.9. Transcellular biosynthesis is used to locally generate lipoxins and cysteinyl leukotrienes. In this example, the leukocyte (neutrophil) obtains arachidonic acid (AA) internally via activation of $cPLA_2$ and also externally from platelets, and uses this AA to biosynthesize LTA_4 and LTB_4. LTA_4 is transferred from the leukocyte to platelets and endothelial cells, which synthesize and secrete LTC_4. Platelets also biosynthesize lipoxins (LXA_4, LXB_4) from LTA_4. Note that the eicosanoids biosynthesized within each cell type are determined by the enzymatic repertoire of that cell type: for example, neutrophils biosynthesize primarily LTA_4 and LTB_4 because they express 5-LOX and LTA_4 hydrolase, whereas platelets biosynthesize LTC_4 and lipoxins because they express LTC_4 synthase and 12-LOX.

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efficacy in enhancing the resolution of recurring bouts of acute inflammation in skin inflammation and gastrointestinal inflammation models. This new approach to the treatment of inflammation remains to be established in human trials.

SPECIALIZED LIPID MEDIATORS IN RESOLUTION

Resolvins and Protectins: New Families of Mediators in Resolution

The literature reports that essential omega-3 PUFAs given in high doses (milligrams to grams daily) have beneficial actions in many inflammatory diseases, cancer, and human health in general. The molecular basis of omega-3 fatty acid action in inflammation was not established until recently. To identify potential mechanisms that are actively involved in resolution of inflammation, a systems approach was taken using a new lipid mediator lipidomics and informatics approach, with liquid chromatography–ultraviolet–tandem mass spectrometry-based analyses, to map and profile the appearance and/or loss of mediators in resolving inflammatory exudates. When novel bioactive compounds were encountered, their structures were elucidated and their bioactivity and role(s) confirmed *in vivo*. These studies uncovered two new families of bioactive mediators, termed resolvins and protectins that are biosynthesized from omega-3 essential PUFAs.

Resolvins: The first resolvin was identified in exudates collected from inflamed murine dorsal air pouches in the spontaneous resolution phase and so-named because it proved to be a potent regulator of resolution. Resolvins are derived from EPA and DHA with two chemically unique structural forms, the E-series and D-series of resolvins, respectively. E-series member resolvin E1 reduces inflammation *in vivo* and blocks human neutrophil transendothelial migration. Resolvin E1 can be produced *in vitro*, recapitulating events *in vivo*, by treating human vascular endothelial cells in a hypoxic environment with aspirin. These cells convert EPA to 18*R*-hydroperoxyeicosapentaenoic acid (18*R*-HPEPE) and release 18*R*-hydroxyeicosapentaenoic acid (18*R*-HEPE) that is rapidly transformed by activated human neutrophil 5-LOX. Resolvin E1 is produced in healthy individuals and is increased in plasma of individuals taking aspirin and/or EPA (Figure 12.10).

Resolvin E2 is the second member of the E-series that reduces zymosan-initiated neutrophil infiltration, thereby displaying potent anti-inflammatory actions. These EPA-derived E-series resolvins may contribute to beneficial actions attributed earlier to omega-3 PUFA in human diseases, such as skin inflammation, peritonitis, periodontal disease, and colitis. These results illustrate that 5-LOX in human leukocytes is pivotal for these beneficial effects and is controlled in part by temporal and spatial events *in vivo* to signal production of either leukotrienes or anti-inflammatory mediators, such as lipoxins and resolvins. Of interest, microbial and mammalian cytochrome P450 enzymes convert EPA into 18-HEPE, which can be transformed by human neutrophils to resolvin E1 and resolvin E2. Hence, it is likely that microorganisms at inflamed sites or in the gastrointestinal tract can contribute to production of E-series resolvins in humans.

At least two receptors involved in the actions of resolvin E1 are defined (Figure 12.3B). The GPCR

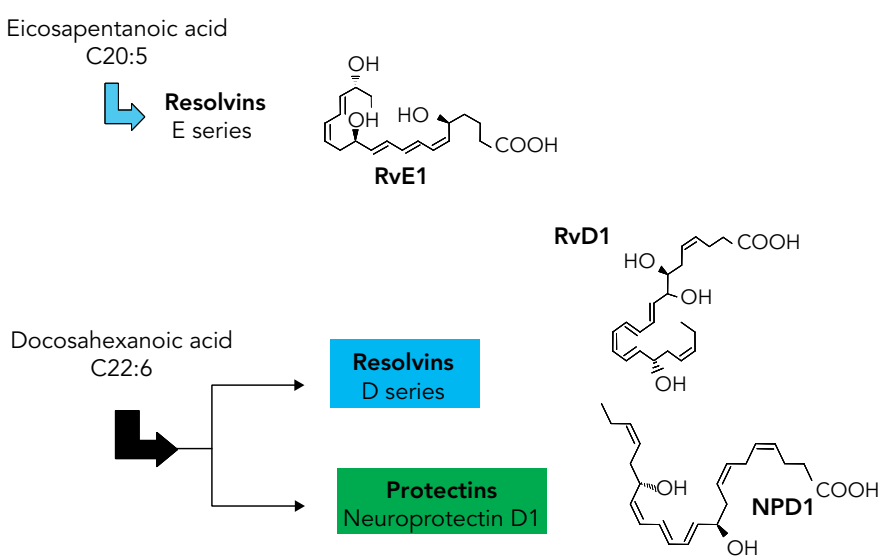


Figure 12.10. Resolvins and protectins: biosynthesis from omega-3 EPA and DHA precursors.

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chemokine-like receptor 1 (CMKLR1, a.k.a. ChemR23) attenuates TNF-stimulated nuclear factor-κB activation in response to resolvin E1 binding, indicating a counter-regulatory action of this ligand–receptor pair. Counter-regulation of TNF signaling was used to identify this GPCR because TNF is a key mediator in the early steps of acute inflammation. Recently, a second GPCR that interacts with resolvin E1 was identified, the leukotriene B_4 receptor, denoted BLT1. Resolvin E1 interacts in a stereospecific manner with BLT1 on human neutrophils as a receptor antagonist, attenuating leukotriene-B_4-dependent proinflammatory signals via BLT1. DHA is substrate for two groups of resolvins of interest during resolution in inflammatory exudates produced by different biosynthetic routes, denoted the 17*S* and 17*R* D-series resolvins (Figure 12.10). D-series resolvins display potent anti-inflammatory actions and are particularly interesting because the brain, synapses, and retina are highly enriched in DHA. Endogenous DHA is converted *in vivo* via LOX-initiated mechanisms to the 17*S*-hydroxy-containing series of four resolvins, denoted D1–D4. Each of these potent bioactive resolvins was first isolated in exudates from mice given aspirin and DHA that led to the identification of several new 17*R*-hydroxy-containing products isolated from exudates in the resolution phase. Their ability to stop neutrophil infiltration was used to assess biological function for structural elucidation studies and their biosynthesis was reconstructed to establish their potential origins. Results from these studies showed that human recombinant COX2 converts DHA into a 13-hydro(peroxy)-containing product. In the presence of aspirin, oxygenation at carbon-13 switches to the carbon-17 position with an *R* configuration that is a precursor for potent bioactive aspirin-triggered 17*R* D-series resolvins, denoted aspirin-triggered (AT)-RvD1-D4. These are produced in exudates and in the brain in response to aspirin treatment. Resolvin D1 and aspirin-triggered resolvin D1 have both been shown to be potent regulators of human and mouse neutrophils (Table 12.5). In microglial cells, both 17*S* and 17*R* D-series resolvins block TNF-induced transcripts for proinflammatory cytokine interleukin-1β (IL-1β), which is expressed rapidly in response to neuronal injury. Resolvins control inflammation at multiple levels, reducing peritonitis and skin inflammation, protecting organs from reperfusion injury and neovascularization.

Protectins. DHA is converted in resolving exudates to another new family of mediators named protectins. Mediators of this family are distinguished by the presence of a conjugated triene double bond system and their potent bioactivity. They are biosynthesized via a LOX mechanism that converts DHA to a 17*S*-hydroperoxide-containing intermediate that is converted by human leukocytes into a 16(17)-epoxide intermediate. This epoxide is enzymatically opened in these cells into a 10,17-dihydroxy-containing anti-inflammatory molecule. This bioactive compound, initially coined 10,17*S*-docosatriene, is now known as protectin D1 (10R,17S-dihydroxy-docosahexaenoic acid), owing to its potent protective activity in inflammatory and neural systems. When produced by neural tissues the prefix *neuro* is added to signify its biosynthetic origin, hence the term neuroprotectin D1.

Protectin D1 is stereo-selective and log-orders of magnitude more potent *in vivo* than its precursor DHA. Protectin D1 is also produced by human peripheral blood mononuclear cells in T helper-2-type conditions. Protectin D1 blocks T-cell migration *in vivo*, reduces TNF and interferon-γ secretion, and promotes T-cell apoptosis.

Agonists of Resolution: Specific resolvins, protectins, and lipoxins stereoselectively stimulate resolution and reduce the magnitude of the inflammatory response *in vivo* (Table 12.5). The clearance of apoptotic neutrophils by professional phagocytes, such as macrophages, is a cellular hallmark of tissue resolution, and can be used to quantify resolution using specific indices. Resolvins and protectins initiate resolution, decreasing neutrophils in exudates at earlier times than spontaneous resolution. Disruption of biosynthesis of these proresolution mediators by either COX-2 or LOX inhibitors gives a "resolution deficit" phenotype, characterized by impaired phagocytic removal, delayed resolution, and prolonged inflammation. These findings emphasize a pivotal homeostatic function for lipoxygenase(s) and COX-2 pathways in the timely resolution of acute inflammation. More importantly, proresolution mediators can rescue the deficit in resolution caused by these interventions.

Resolution of Inflammation in Disease Models

Uncontrolled inflammation is now appreciated in the pathogenesis of many diseases that were not previously considered classic inflammatory diseases. These include atherosclerosis, cancer, asthma, and several neurological disorders, such as Alzheimer's disease and Parkinson's disease. Natural proresolution mechanisms involving lipoxins, resolvins and protectins were tested for their ability to promote resolution and control inflammation (Table 12.6). It is now clear that endogenous anti-inflammation alone is not an identical mechanism of action compared to mediators that possess dual anti-inflammatory and proresolution actions. In this regard, lipoxins, resolvins, and protectins have potent multilevel mechanisms of action in disease models and promote resolution in animal models of oral, lung, ocular, kidney, skin, and gastrointestinal inflammation as well as in ischemia-reperfusion injury and angiogenesis.

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TABLE 12.5. Key cellular actions of lipoxins, resolvins, and protectins in the innate immune system and resolution*

Mediators	Cell type	Action(s)
Lipoxin A_4 (or aspirin-triggered lipoxin A_4)	Whole blood leukocyte	Downregulates CD11b/CD18 expression, prevents shedding of L-selectin, and reduces peroxynitrite generation on neutrophils, monocytes and lymphocytes
	Neutrophil	Stops chemotaxis, adherence, and transmigration Stops neutrophil-epithelial and neutrophil–endothelial cell interactions Blocks superoxide anion generation Reduces CD11b/CD18 expression and IP_3 formation Inhibits peroxynitrite generation Attenuates AP-1, NF-kB accumulation, inhibits IL-8 gene expression
	Eosinophil	Stops migration and chemotaxis in vivo, and inhibits eotaxin and IL-5 generation
	Monocyte	Stimulates chemotaxis and adhesion to laminin without increase in cytotoxicity Inhibits peroxynitrite generation Reduces IL-8 release by cells from asthma patients
	Macrophage	Stimulates nonphlogistic phagocytosis of apoptotic neutrophils
	T-cell	Inhibits TNF secretion by blocking ERK activation Upregulates CCR5 expression
	Dendritic cell	Blocks IL-12 production
	Epithelia	Inhibits TNF-induced IL-8 expression and release in enterocytes Inhibits *Salmonella typhimurium*–induced IL-8 in enterocytes
	Endothelia	Stimulates protein-kinase-C–dependent prostacyclin formation Blocks reactive oxygen species generation Inhibits VEGF-induced endothelial cell migration
	Fibroblast	Inhibits IL-1β-induced IL-6, IL-8, and MMP3 production Inhibits CTGF-induced proliferation
	Hepatocyte	Reduces PPARa and CINC1 expression levels
	Mesangial cell	Inhibits leukotriene-D_4–induced proliferation Inhibits CTGF-induced chemokine production
	Neural stem cell	Attenuates cell growth
	Astrocyte	Inhibits ERK and JNK activation
Resolvin E1	Neutrophil	Stops transepithelial and transendothelial migration
	Macrophage	Stimulates nonphlogistic phagocytosis of apoptotic neutrophils
	Dendritic cell	Blocks IL-12 production
	T cell	Upregulates CCR5 expression
Resolvin D1	Microglia	Inhibits IL-1β expression
Aspirin-triggered resolvin D1	Neutrophil	Stops transmigration
Protectin D1	Neutrophil	Upregulates CCR5 expression
	Macrophage	Stimulates nonphlogistic phagocytosis of apoptotic neutrophils
	T cell	Inhibits TNF and IFNγ secretion, promotes apoptosis Upregulates CCR5 expression
	Microglia	Inhibits IL-1β expression
	Epithelia	Protects from oxidative stress–induced apoptosis in retinal pigment epithelia

*See Serhan and Chiang (*Br J Pharmacol.* 2007) and Yacoubian and Serhan (*Nat Clin Pract Rheumatol.* 2007) for further details and the review of original literature.

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Periodontal diseases such as gingivitis and periodontitis are leukocyte-driven inflammatory diseases characterized by soft tissue and osteoclast-mediated bone loss. As a model of inflammatory diseases, periodontitis has several advantages in that many, if not all, of the tissues involved in inflammatory processes of other organ systems are affected in periodontitis, including the epithelium, connective tissue, and bone. There are many noteworthy similarities in the pathogenesis of periodontitis and arthritis. Results from rabbit periodontitis demonstrate an important role for resolution in disease prevention. Overexpression of 15-lipoxygenase-type 1 in transgenic rabbits increases the levels of endogenous lipoxin A_4 and protects against periodontitis, as well as reduces atherosclerosis. In prevention studies using this model, topical treatment with resolvin E1 prevents >95% of alveolar bone destruction. Histological analysis of resolvin-E1–treated rabbits revealed few, if any, neutrophils and little tissue damage. In addition, osteoclasts responsible for bone resorption are reduced in resolvin-E1–treated rabbits. In established disease, resolvin E1 prevents periodontitis tissue destruction; both soft tissue and bone that were lost during disease were regenerated.

In humans, the differential actions of resolvin E1 were studied using neutrophils from patients with localized aggressive periodontitis (LAP) and healthy individuals. Resolvin E1 reduces neutrophil superoxide generation in response to TNF or the bacterial surrogate peptide *N*-formyl-methionyl-leucyl-phenylalanine. Neutrophils from both healthy subjects and LAP patients produced ~80% less superoxide treated with resolvin E1. In comparison, neutrophils from LAP patients do not exhibit inhibition of superoxide production following treatment with lipoxin A_4, suggesting a molecular basis for excessive inflammation in these patients.

In murine air pouches, nanogram amounts of resolvin E1 (~100 nM) reduce leukocyte infiltration by 50%–70%, levels comparable to those achieved using microgram amounts of dexamethasone (~30 μM) or milligrams of aspirin. Similarly, in spontaneously resolving peritonitis induced by the yeast cell-wall component zymosan, both resolvin E1 and protectin D1 activate and accelerate resolution. Resolvins and protectins reduce neutrophil infiltration and increase nonphlogistic recruitment of monocytes. Notably, resolvin E1, protectin D1, and an ATL analogue each display different kinetics and molecular profiles of action (Tables 12.5 and 12.6).

Summation

Eicosanoids are critical mediators of homeostasis and of many pathophysiologic processes, especially host-defense and inflammation. Arachidonic acid, the primary substrate, is converted into prostaglandins, thromboxanes, prostacyclin, leukotrienes, lipoxins, isoprostanes, and epoxyeicosatetraenoic acids. Prostaglandins have diverse roles in vascular tone regulation, gastrointestinal regulation, uterine physiology, analgesia, and inflammation. Prostacyclin and thromboxane in coordination control vascular tone, platelet activation, and thrombogenesis. Leukotrienes (LTC_4, LTD_4) are the chief activators of bronchoconstriction and airway hyperactivity; LTB_4 is a major activator of mediators of leukocyte chemotaxis and infiltration. Lipoxins antagonize the effects of leukotrienes and reduce the extent of inflammation and active resolution pathways.

Pharmacologic interventions at many critical points in these pathways are useful in limiting inflammatory sequelae. Glucocorticoids downregulate several steps in eicosanoid generation, including the rate-determining step involving phospholipase A_2. However, chronic glucocorticoid use is associated with deleterious side effects, including osteoporosis, muscle wasting, and abnormal carbohydrate metabolism. COX inhibitors block the first step of prostanoid biosynthesis preventing the generation of prostanoid mediators of inflammation. LOX inhibitors, FLAP inhibitors, leukotriene synthesis inhibitors, and leukotriene receptor antagonists prevent leukotriene formation and signaling, thereby limiting inflammation and its deleterious effects. Future drug development efforts will allow selective targeting of eicosanoid pathways involved in many clinical conditions. The discovery of resolvins and protectins opens the possibility of treating inflammatory diseases with agonists of resolution, namely stable resolvins and protectins. Efforts are currently underway for the clinical development of this novel agonist of resolution. Results from clinical trials will, in the near future, determine whether treating inflammation via accelerated resolution is indeed a clinically useful approach to diseases characterized by uncontrolled inflammation.

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GLOSSARY

15-HETE

15-hydroxy-5*Z*,8*Z*,11*Z*,14*Z*-eicosatetraenoic acid

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TABLE 12.6. Dual anti-inflammatory and pro-resolution actions of lipoxins, resolvins, and protectins in complex disease models

Mediators	Disease model	Action(s)
Lipoxin A_4/ATL	Rabbit/Periodontitis	Reduce neutrophil infiltration; prevent connective tissue and bone loss
	Mouse/Peritonitis	Stop neutrophil recruitment and lymphatic removal of phagocytes
	Mouse/Dorsal air pouch	Stop neutrophil recruitment
	Mouse/Dermal inflammation	Stop neutrophil recruitment and vascular leakage
	Mouse/Colitis	Attenuate proinflammatory gene expression and reduce severity of colitis. Inhibit weight loss, inflammation, and immune dysfunction
	Mouse/Asthma	Inhibit airway hyperresponsiveness and pulmonary inflammation
	Mouse/Cystic fibrosis	Decrease neutrophilic inflammation, pulmonary bacterial burden, and disease severity
	Mouse/Ischemia-reperfusion	Attenuate hind-limb I/R-induced lung injury Detachment of adherent leukocytes in mesenteric I/R
	Mouse/Cornea	Accelerate cornea re-epithelialization, limit sequelae of thermal injury (i.e., neovascularization, opacity) and promote host defense
	Mouse/Angiogenesis	Reduce angiogenic phenotype:endothelial cell proliferation and migration
	Mouse/Bone-marrow transplant (BMT)	Protect against BMT-induced graft-versus-host diseases
	Murine/Glomerulonephritis	Reduce leukocyte rolling and adherence Decrease neutrophil recruitment
	Rat/Hyperalgesia	Prolong paw withdraw latency, reducing hyperalgesic index Reduce paw edema
	Rat/Pleuritis	Shorten the duration of pleural exudation
Resolvin E1	Rabbit/Periodontitis	Reduces neutrophil infiltration; prevents connective tissue and bone loss; promotes healing of diseased tissues; regeneration of lost soft tissue and bone
	Mouse/Peritonitis	Stops neutrophil recruitment; regulates chemokine/cytokine production Promotes lymphatic removal of phagocytes
	Mouse/Dorsal air pouch	Stops neutrophil recruitment
	Mouse/Retinopathy	Protects against neovascularization
	Mouse/Colitis	Decreases neutrophil recruitment and proinflammatory gene expression; improves survival; reduces weight loss
Resolvin D1	Mouse/Peritonitis	Stops neutrophil recruitment
	Mouse/Dorsal skin air pouch	Stops neutrophil recruitment
	Mouse/Kidney ischemia-reperfusion	Protects from ischemia-reperfusion–induced kidney damage and loss of function; regulates macrophages
	Mouse/Retinopathy	Protects against neovascularization
Protectin D1	Mouse/Peritonitis	Inhibit neutrophil recruitment; regulate chemokine/cytokine production Promote lymphatic removal of phagocytes Regulate T-cell migration
	Mouse/Asthma	Protect from lung damage, airway inflammation, and airway hyperresponsiveness
	Human/Asthma	Protectin D1 is generated in humans and appears to be diminished in asthmatics
	Mouse/Kidney ischemia-reperfusion	Protect from ischemia-reperfusion–induced kidney damage and loss of function; regulate macrophages
	Mouse/Retinopathy	Protect against neovascularization
	Rat/Ischemic stroke	Stop leukocyte infiltration, inhibit NF-κB and cyclooxygenase-2 induction
	Human/Alzheimer's disease	Diminished protectin D1 production in human Alzheimer's disease

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AA

Arachidonic acid, *cis*-,*cis*-,*cis*-,*cis*-5,8,11,14-eicosatetraenoic acid

Alzheimer's disease

A degenerative neurological disease that is characterized by progressive deterioration of the brain, dementia, and the presence of senile plaques, neurofibrillary tangles, and neuropil threads. Disease onset can occur at any age, and women seem to be affected more frequently than men.

BLT1

Leukotriene B_4 receptor 1

Cytochrome P450 enzymes

A large and diverse superfamily of hemoproteins. Cytochrome P450 enzymes use a plethora of both exogenous and endogenous compounds as substrates. The most common reaction catalyzed by cytochrome P450 is a monooxygenase reaction, that is, insertion of one atom of oxygen into an organic substrate while the other oxygen atom is reduced to water.

DHA

Docosahexaenoic acid, (4*Z*,7*Z*,10*Z*,13*Z*,16*Z*,19*Z*)-docosa-4,7,10,13,16,19-hexaenoic acid

Eicosanoids

A family of bioactive products that contain 20 (eicos in Greek) carbons. They are biosynthesized from arachidonic acid by the initial activities of either cyclooxygenases (isoforms COX1 or COX2) or lipoxygenases and downstream enzymatic reactions. There are several main classes of eicosanoids: prostaglandins, prostacyclins, thromboxanes, leukotrienes, and lipoxins.

EPA

Eicosapentaenoic acid, (5*Z*,8*Z*,11*Z*,14*Z*,17*Z*)-eicosa-5,8,11,14,17-pentaenoic acid

Exudate

Biological fluids that filter from the circulatory system into lesions or areas of inflammation. Exudate formation is caused by inflammation, and is characterized by a high content of plasma proteins, cells, and cellular debris. Pus is an example of an exudate found in infected wounds that contains bacteria and high concentrations of white blood cells.

G-protein–coupled receptor

GPCR. One of a large family of receptors that bind diverse molecules, including for example lipid mediators, eicosanoids, chemokines, complement components, and neurotransmitters. GPCRs are seven transmembrane–spanning receptors and are coupled to heterotrimeric, GTP-regulated signaling proteins composed of αβ and βγ subunits.

Ischemia-reperfusion injury

An injury in which the tissue first suffers from hypoxia as a result of severely decreased or completely arrested blood flow. Restoration of normal blood flow then triggers inflammation, which exacerbates the tissue damage.

Leukotrienes

A family of eicosanoids derived from the metabolism of arachidonic acid by the action of leukocyte 5-lipoxygenase and other enzymes. They have a conjugated triene double-bond structure and various proinflammatory activities, including leukocyte activation (by leukotriene B_4) and bronchoconstriction (by leukotriene C_4 and leukotriene D_4).

Lipoxins

A class of eicosanoids produced via lipoxygenase-mediated conversion of arachidonic acid. They are trihydroxytetraene-containing structures with potent biological activities in the resolution of inflammation.

LOX

Lipoxygenase

LTB_4

Leukotriene B_4, (5*S*,12*R*)-dihydroxy-(6*Z*,8*E*,10*E*,14*Z*)-eicosatetraenoic acid

Murine dorsal air pouch

A well-characterized system for studying inflammatory responses. Air pouches are formed by subcutaneous injection of air in the back of a mouse, into which potential inflammatory stimuli, such as tumor-necrosis factor, can be added. In importance when the stimulus is titrated, inflammatory reactions undergo spontaneous resolution. These structurally contained compartments have been likened to the inflamed synovium (see Chapter 27 from Peretti et al.).

Prostaglandins

Cyclopentane ring–containing lipids derived from the metabolism of arachidonic acid by the action of cyclooxygenases and downstream synthase enzymes. They have a wide range of biological activities and a well-recognized role in inflammation and pain.

Protectins

A family of docosahexaenoic acid (DHA)-derived mediators characterized by the presence of a conjugated

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Box 1. The stereochemistries of lipoxins, resolvins, and protectins

- Resolvin D1: 7*S*,8*R*,17*S*-trihydroxy-4*Z*,9*E*,11*E*,13*Z*,15*E*,19*Z*-docosahexaenoic acid
- Resolvin E1: 5*S*,12*R*,18*R*-trihydroxy-6*Z*,8*E*,10*E*,14*Z*,16*E*-eicosapentaenoic acid
- Neuroprotectin D1/Protectin D1: 10*R*,17*S*-dihydroxy-docosa-4*Z*,7*Z*,11*E*,13*E*,15*Z*,19*Z*-hexaenoic acid
- Lipoxin A_4: 5*S*,6*R*,15*S*-trihydroxy-7*E*,9*E*,11*Z*,13*E*-eicosatetraenoic acid
- Aspirin-triggered lipoxin A_4: 5*S*,6*R*,15*R*-trihydroxyl-7,9,13-*trans*-11-*cis*-eicosatetraenoic acid
- Lipoxin B_4: 5*S*,14*R*,15*S*-trihydroxy-6*E*,8*Z*,10*E*,12*E*-eicosatetraenoic acid.
- Aspirin-triggered Resolvin D1: 7*S*,8*R*,17*R*-trihydroxy-4*Z*,9*E*,11*E*,13*Z*,15*E*,19*Z*-docosahexaenoic acid

triene double-bond structure and a 22 carbon backbone structure with six double bonds.

PUFA

Polyunsaturated fatty acid

Rabbit model of periodontitis

A model of periodontitis in rabbits induced by application of *Porphyromonas gingivalis* to ligatures tied to second premolars.

Resolvins

Biosynthesized lipid mediators that are induced in the resolution-phase following acute inflammation. They are synthesized from the essential omega-3 fatty acids eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA).

Transcellular biosynthesis

The biosynthesis of biologically active mediators that involves two or more cell types, for example, when the necessary enzymes are differentially expressed in two or more cell types.

In this example, a donor cell converts a precursor compound (such as arachidonic acid, EPA and DHA) into an intermediate product. The acceptor cell then converts the intermediate product into the final active product. Transcellular biosynthesis therefore provides a means to produce mediators that neither cell type can generate alone.

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13 Cytokines and Chemokines in Inflammation

Dennis M. Lindell and Nicholas W. Lukacs

SUMMARY

Mediators produced during inflammatory/immune responses dictate the type of response, as well as its magnitude ("quality and quantity"). The profile of cytokines and chemokines produced are responsible for the cell-to-cell communication that facilitates initial recognition of infection or damage. These signals, in turn, communicate with primary lymphoid tissues (the thymus and bone marrow) to mobilize inflammatory cells to the bloodstream. At the tissue site, chemokines and cytokines orchestrate leukocyte adhesion to vascular endothelium, extravasation, and localization of leukocytes at the site of inflammation. Recruitment of leukocyte populations into inflamed tissues is initiated by cytokine-induced expression of adhesion molecules on vascular endothelium. These adhesion molecules play an essential role in capturing and tethering circulating leukocytes from the bloodstream. Chemokines promote the tight adherence of leukocytes to activated endothelium, as well as direct the extravasation of cells into inflamed tissue. Cytokines and chemokines further coordinate the response of inflammatory effector cells as they arrive in inflamed tissues. The sequential activation and cellular recruitment cascades mediated by cytokines and chemokines are essential for successful resolution of infection or other tissue damage. However, overproduction or dysregulation of these inflammatory mediators can also lead to destructive, pathological consequences. This chapter will examine the role played by cytokines and chemokines in the initiation, amplification, and shaping of inflammatory responses.

INTRODUCTION

The actions of cytokines are dictated by the expression of specific receptors. These actions may be on the producing cell itself (autocrine), on neighboring cells (paracrine), or on distant sites carried via circulation (endocrine) (Figure 13.1). Cytokines are very potent at low concentrations and their effects are targeted via directional release, meaning that the cell producing the cytokine often releases it predominantly in one direction, to target the effect. The majority of cytokines can be divided into several major families, including the tumor necrosis factor (TNF) family, hematopoietins, and interferons. Cytokines have a number of characteristics: they are typically pleiotropic and can act upon many different cell types. Second, they are redundant; a number of similar cytokines may produce similar effects. Third, cytokines from a profile may exhibit synergy – that is, the effects of two cytokines are greater than the sum of each individually. Conversely, cytokines from opposing functional categories tend to be antagonistic toward one another – blocking each other's production or activity.

A great deal of insight of the in vivo significance of cytokines has been gained through the use of genetically deficient mice (knockouts), conditional knockouts, transgenics, and neutralizing antibodies. Each of these approaches has its own limitations. In the case of knockout mice, animals tend to develop compensatory pathways, potentially leading to an underestimate of the true in vivo importance of a particular cytokine. Conversely, transgenically overexpressing a protein can lead to artifactual effects due to supraphysiologic concentrations. Neutralizing or depleting antibodies can produce nonspecific effects, or induce antiantibody responses (serum sickness) when given long term making it difficult to assess their effects under chronic disease conditions. Thus, any single approach to the understanding of cytokine function should be regarded with some skepticism. However, these approaches offer a great deal of value for understanding the impact of a particular cytokine in a complex in vivo disease response.

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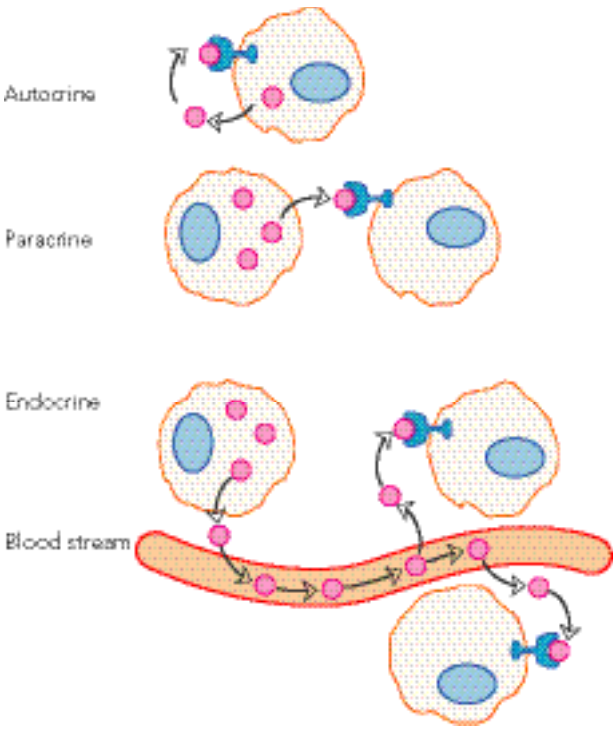


Figure 13.1. Types of cytokine signaling. Autocrine signaling refers to secreted agents that act through specific receptors on the secreting cell. In paracrine signaling, cytokines act on nearby cells. Endocrine signaling refers to action on distant cells following transport by the circulation.

NOMENCLATURE OF CYTOKINES

Many cytokines are given the designation interleukin (IL). Cytokines may be classified by their structural similarity. For instance, the IL-2 family, interferons, and the IL-10 family all share a similar four alpha-helix structure. Other families include the IL-1 family, and the IL-17 family of cytokines. Cytokines may also be classified based on their function. One of these classifications is whether a particular cytokine is Type 1 (T helper 1 associated) or Type 2 (T helper 2 associated). Usually, the classification of cytokines is structural or function based and can be divided depending on a particular disease or class of diseases. Many other functional categories are possible depending on context.

CYTOKINE RECEPTORS/SIGNAL TRANSDUCTION

The cytokine receptors can be classified into several major families, including the TNF receptor superfamily, hematopoietins, interferons, and immunoglobulin (Ig) superfamily. Cytokine receptors are typically composed of several subunits. Many exist as heterodimers or heterotrimers, with one or two subunits primarily responsible for binding and recognition, and the other subunit required for signal transduction through the cell membrane. Often, the signal transduction subunit is common to many different receptors. A prototypical example of this theme is the common gamma chain in the hematopoietin cytokine receptor family.

Hematopoietic cytokine receptors (class I) have similar extracellular domains, and often form multimeric structures with one another. For instance, the IL-2 receptor gamma subunit is shared by IL-4, IL-7, IL-9, and IL-15 receptors, and is thus referred to as the common γ chain (γ_c). A common β chain is shared by the IL-3, IL-5, and GM-CSF receptors. The interferons are also referred to as class II cytokine receptors, and include the receptors for IFN-α and IFN-β, IFN-γ, and the IL-10 receptor. The TNF ligand and receptor family (class III) includes ligands that can be cell associated or soluble. Many members of this family are homotrimers in both the ligand and ligand-bound receptor. Many members of the TNF family are not cytokines, but are membrane-bound signaling molecules important for costimulation of immune responses. In addition, many members of the TNF receptor family have a cytosolic tail that contains a number of death domains, which can activate the caspase cascade, resulting in apoptosis or cell suicide. However, signaling via the TNF receptor through TNF-receptor associated factors (TRAFs) can also promote the activation of NF-κB, resulting in prosurvival and growth promoting activities.

Cytokine receptor signaling results in complex sequential intracellular events [1]. A simplified representation of Class I/II cytokine receptor signaling is depicted in Figure 13.2. Initially, the intracellular domains of the receptor are openly spaced. Janus kinases (JAKs) are associated with the intracellular tail of the receptor. Ligand binding results in recruitment of the JAKs bringing them closer together where they can cross-phosphorylate one another, as well as other tyrosine residues in the cytosolic tail of the receptor. Phosphorylated JAKs in turn can phosphorylate a family of signal transduction and transcription factors (STATs) making them functionally active. The dimerization of STATs results in translocation to the nucleus, where phosphorylated, dimerized STATs bind to the promoters of target genes, activating transcription of the genes. Depending on the cytokine specific STAT pathways are initiated. Several of these pathways have been well-defined and include the IFN (α,β,γ) signaling that activate STAT1, IL-6 signaling initiates STAT3, IL-12 drives the STAT4 pathway, and IL-4 as well as IL-13 that drive the STAT6 pathways. Each of these signaling pathways has been linked to critical aspects of the innate and acquired immune responses via the use of genetically deleted animals as well as rather specific inhibitors. Using in vitro and in vivo experiments the STAT signaling molecules have been demonstrated to influence the direction of the immune responses. The STAT1 pathway initiates signaling that promotes antipathogen responses and without it the immune system cannot properly

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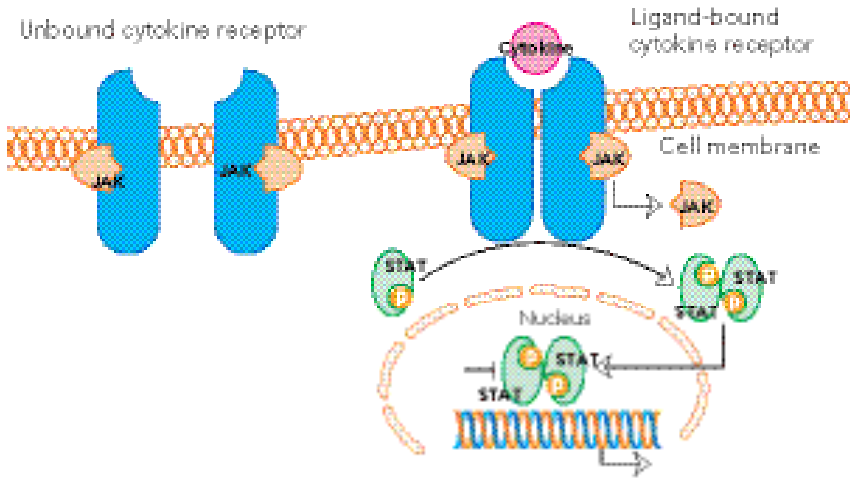


Figure 13.2. A simplified model of cytokine receptor signaling (Type I/II cytokine receptors). Initially, the unbound receptor is present as a dimeric structure, with the subunits spaced. Cytokine receptors are associated with Janus kinases (JAKs). Upon ligation of the receptor by ligand, the subunits are brought into apposition facilitating phosphorylation of Tyr residues on the cytosolic side of the receptor. These in turn are bound via SH2 containing signal transducers and activators of transcription (STATs). STAT phosphorylation leads to dimerization of STATs, which accumulate in the nucleus, activating transcription.

clear pathogens and further increases pathology due to skewed immune responses with increasing Th2 cytokines (IL-4, IL-5, and IL-13). In a similar fashion, the inhibition or deletion of the STAT4 pathway blocks production of IFN and the development of the Th1-mediated cellular immune responses, leading to increased Th2 cytokine production. As indicated earlier, the key Th2 cytokine (IL-4, IL-13) associated signaling protein, STAT6, is essential to initiate the Th2 response and without it there are deficiencies in proper B cell–mediated humoral responses related to isotype switching and extracellular pathogen immune responses. More recently, the development of Th17 subsets of lymphocytes has been shown to be dependent on IL-6-mediated STAT3 responses and links to the severity of autoimmune disease have been established. Thus, these critical signaling molecules have a central role in determining the direction and maintenance of acute and chronic immune responses.

CHEMOKINES (MORE THAN "CHEMOTACTIC CYTOKINES")

Initially, chemotactic cytokines or "chemokines" were viewed as leukocyte chemoattractants that regulate cellular movement from circulation into inflamed tissues (Figure 13.3). However, chemokines also have other functions, including cell activation, hematopoiesis, and angiogenesis. Chemokines have a number of characteristics: (1) Chemokines may be abundantly produced. For instance, mRNA for CCL3/MIP1-α accounts for greater than 20% of the total mRNA produced by LPS-stimulated macrophages. (2) Chemokines tend to be redundant, with many ligand receptor pairs producing seemingly identical effects. (3) Chemokines are (for the most part) promiscuous, meaning that a single ligand can bind multiple receptors, and multiple ligands are bound by a single receptor (Figure 13.4). These characteristics may be misleading, however, as apparent redundancy may be simply due to our lack of understanding of the complex signaling pathways induced by chemokines. Furthermore, specific profiles of chemokines will specifically initiate the migration of different leukocytes to the site of inflammation/immune responses. For example, during acute inflammatory responses early production of CxCR1/2 binding chemokines, such as IL-8 (CxCL8), initiate intense neutrophilic responses since these are the primary cell that expresses the receptors. Thus, the profile of chemokines produced at the site of inflammation dictate the type of leukocytes that migrate due to the profile of receptors that are expressed by those cells. Key findings regarding the diverse roles of chemokines in development, inflammation, as well as resolution and end-stage fibrotic disease has highlighted that chemokines are key players in the regulation of a diverse set of responses. This is likely the reason that there are such a large number of different chemokines that can be produced that allow distinct fidelity and control of a particular inflammatory response.

Nomenclature: Chemokines have been divided into four families based upon their sequence homology

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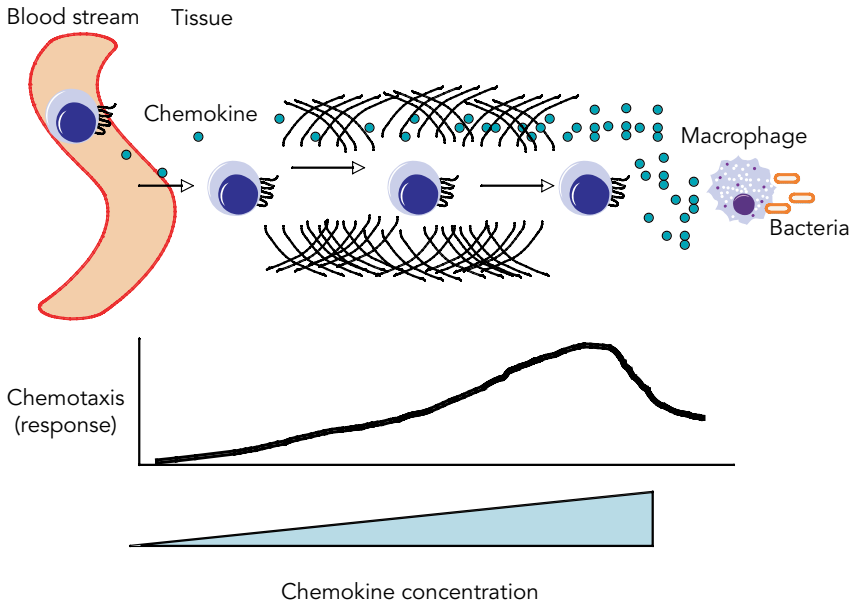


Figure 13.3. Inflammatory cells follow chemokine gradients. Chemokines mediated the homing and extravasation of inflammatory cells through specific receptors via the establishment of chemokine gradients. These gradients are facilitated by the binding of chemokines to extracellular matrix. Inflammatory cells follow the increasing concentration of chemokine through tissues until they reach a threshold concentration, whereupon the cell becomes desensitized.

and the position of the first two conserved cysteine residues, CXC (alpha) and CC (beta), C, and CX3C (Figure 13.4). Both of these latter families have only a single member thus far, XCL1 (lymphotactin) and CX3CL1 (fractalkine), respectively. The nomenclature of chemokines and receptors is, admittedly, confusing. Historically, chemokines were named by the laboratories that discovered them, usually based on the cell type or tissue from which they were isolated (e.g., macrophage inflammatory protein 1 alpha – MIP1-α). While this nomenclature had some advantages, cloning by multiple laboratories resulted in multiple names for a single chemokine. To resolve this issue, nomenclature for chemokine ligands has been replaced by the more systematic CCL, CXCL, XCL, or CX3CL designation (e.g., MIP1-α became CCL3). However, the original names are often still used in the literature, usually in conjunction with the newer, systematic one. The CXC or alpha chemokine family can be further subdivided depending upon whether the chemokines have an amino acid E-L-R motif within its sequence in the N-terminal region. This consensus sequence appears to determine whether the chemokines have the ability to induce chemotaxis of neutrophils and determines whether the CXC chemokine is angiogenic (ELR) or angiostatic (non-ELR). The ELR CXC chemokines are bound by CXCR1 and CXCR2, expressed predominantly by neutrophils. Non-ELR CXC chemokines bind CXCR3–5, and serve as chemoattractants for monocytes, dendritic cells, T, and B lymphocytes. The CC or beta chemokine family is a large group of chemokines with nearly 30 members, and at least 10 receptors. This family serves as recruitment molecules for monocytes and lymphocytes, but also basophils, eosinophils, NK cells, and mast cells. Different inflammatory stimuli result in the induction of different chemokine profiles. Additionally, as will be discussed in more detail, specific cell types typically express a limited array of chemokine receptors. The patterns of chemokines produced and the receptor expression responding cells (in conjunction with cytokines and other inflammatory signals) shapes the trafficking and activation of both resident and inflammatory cells.

CHEMOKINE RECEPTOR SIGNALING

Chemokine receptors are seven transmembrane spanning G-protein coupled receptors. Chemokines mediate their effects predominately via Gi-protein mediated pathways. The activation of chemokine receptors results in a number of cellular events, including Ca^{2+} flux, activation of phosphoinositides, various protein and lipid kinases, and small GTPases (Figure 13.5) [2]. The activated G protein alpha subunit results PLC-mediated cleavage of phosphotidylinositol (4,5)-bisphosphate (PIP_2) into inositol triphosphate (IP_3) and diacylglycerol (DAG), triggering Ca^{2+} flux and protein kinase C activation. Beta/gamma subunits

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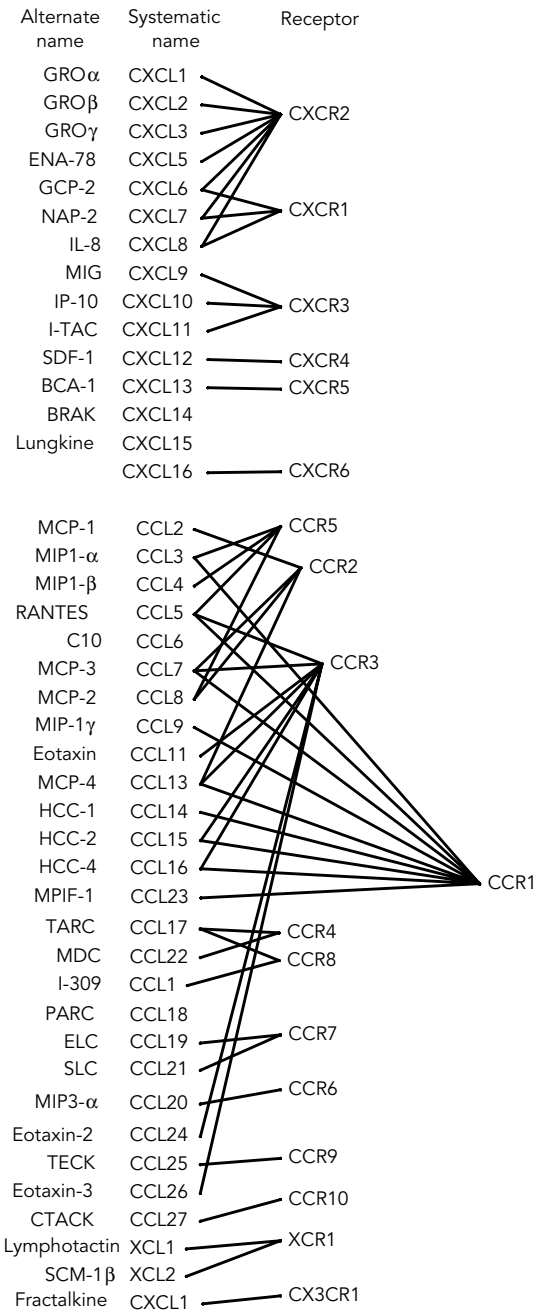


Figure 13.4. Chemokine and chemokine receptor specificity. Receptor–ligand specificity is depicted via connecting lines. Chemokines and chemokine receptor vary significantly in their specificity, with some receptor–ligand pairs being monogamous, whereas others are more promiscuous.

together are responsible for chemotaxis. Signaling through these subunits together is known to activate important pathways including PI3K, MAPK, and FAK. PI3K activation generates phosphatidylinositols (Ptdlns) species, especially Ptdlns (3,4,5)P_3 (PIP3), that are enzymatically modified by a series of kinases and phosphatases to regulate chemotaxis. PI3K isoforms appear to be differentially expressed and activated depending on the chemokine receptor, as well as the responding cell type. For instance, T-cell chemotaxis is dependent on P110γ, whereas B cells may rely upon P110δ. PI3K-dependent signaling can be negatively regulated by PTEN (phosphatase and tensin homology deleted on chromosome 10 protein), as well as by SH2-containing inositol 5-phosphate (SHIP) [3]. Small GTPases Rac, Rho, and Cdc42 play key roles in regulating the morphology of migrating cells via effects on the actin cytoskeleton. During chemotaxis, the cell exhibits polarization, localized to the leading edge during migration; PTEN is found in the trailing edge and is actively excluded from the leading edge. Thus, the polarized regulation of this system facilitates directional leukocyte movement. Thus, many intracellular signaling pathways are stimulated via chemokine receptor signaling. These have both direct chemokine-mediated effects, as well as opportunities for cross talk with other cellular signals.

REGULATION OF HEMATOPOIESIS BY CYTOKINES AND CHEMOKINES

One prominent function of cytokines is the regulation of hematopoiesis. Included in this family are the colony-stimulating factors (e.g., GM-CSF, G-CSF, M-CSF), and a number of interleukins, such as IL-3, summarized in Table 13.1. Many of these cytokines are produced by stromal cells, but can also be produced by hematopoietic cells themselves to provide autocrine signals that modify their own maturation. Hematopoietic cytokines may be lineage restricted, such as erythropoietin (EPO) which acts primarily on red blood cells (RBCs), those that act on multiple lineages (e.g., GM-CSF), or those that act sequentially, such as in the case of SCF and EPO on RBCs. Most hematopoietic cytokines can be made by many cell type and take the form as membrane-bound or cell-secreted factors. They are potent in their actions, functioning at low nM concentrations. They are typically present at low concentrations, but are highly inducible upon stimulation in specific diseases or when needed (e.g., 1,000-fold). They are polyfunctional and can promote survival, proliferation, differentiation, maturation, and cell activation along with other potential functions. Therapeutically, the hematopoietic growth factors have been very useful. For example, GM-CSF and G-CSF are commonly used therapeutically to treat neutropenia, whereas EPO is used to treat anemia.

Chemokines also have effects on the hematopoietic system. Many of these effects are indirect, but some chemokines have been demonstrated to have direct roles, as well. Among the best characterized of the hematopoietic chemokines is CXCL12/SDF-1. CXCL12 is the most abundantly produced chemokine in the

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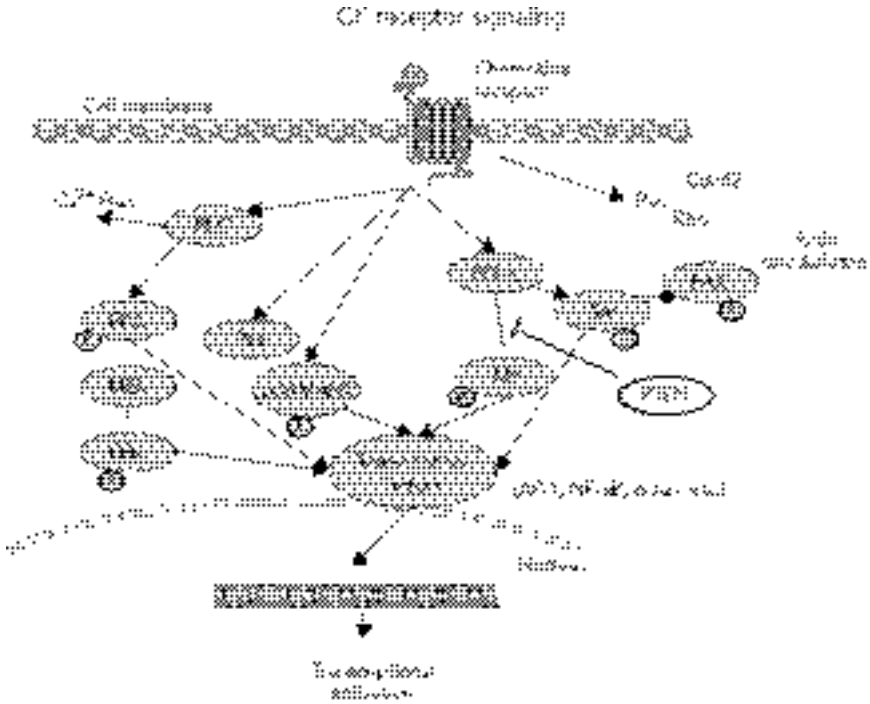


Figure 13.5. Signaling induced by chemokine receptors. Chemokines receptors are seven transmembrane spanning G-coupled receptors. A number of signaling pathways are induced by chemokine ligation, including PI-3 kinases, small GTPases, MAP kinases, and activation of phosphoinositides. These pathways mediate a wide variety of responses, including adhesion, locomotion, chemotaxis, calcium flux, as well as activation of gene transcription.

TABLE 13.1. Prominent/major hematopoietic cytokines

Hematopoietic cytokines
G-CSF (granulocyte-colony stimulating factor)
GM-CSF (granulocyte macrophage-colony stimulating factor)
M-CSF (macrophage-colony stimulating factor)
IL-3
SCF (stem cell factor)
LIF (leukemia inhibitory factor)
Flt3L (FMS-like tyrosine kinase 3 ligand)
IL-11
TPO (thrombopoietin)
IL-2
IL-6
IL-7
OM (oncostatin M)

bone marrow. Although CXCL12 promotes the survival of hematopoietic stem cells (HSCs) and hematopoietic precursor cells (HPCs) in vitro, a number of studies suggest that CXCL12 (through its receptor CXCR4) serves as a bone marrow retention signal in vivo. These functions have relevance to a number of immunologically mediated diseases. CXCR4 also serves as a coreceptor for the HIV-1 virus, which may allow the virus to spread to recent bone marrow emigrants or the bone marrow itself. In cancer, CXCL12 promotes chemotaxis and homing in many tumor cell lines. In addition, CXCL12 promotes the formation of new blood vessels (neoangiogenesis), which tumors require to continue to grow beyond self-limiting size. Other chemokines may promote the mobilization of HSCs or HPCs. One chemokine with this function is CXCL2, which enhances the homing and engrafting capabilities of HSCs. In contrast to CXCL12 and CXCL2, most chemokines are generally believed to have suppressive function on HPCs. CCL2, 3, 19, 20, CXCL4, 5, 8, and 9 all suppress the numbers and proliferation of HPCs when administered in vivo. It is important to note, however, that rarely (if ever) will the host experience a bolus of a single chemokine, in the absence of other signals. Therefore, the effects of chemokines on hematopoiesis during disease are still poorly understood.

INITIATION OF INNATE IMMUNE RESPONSES

Cytokines and chemokines are among the first signals induced by the immune system upon recognition of infection. Initial recognition of microbial infection occurs via pattern recognition receptors (PRRs) that distinguish pathogen-associated molecular patterns

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(PAMPs) from normal host associated proteins [4]. Viral infection is accompanied by the induction of IFN-β and IFN-α (type I interferons). This response depends on the recognition of viral nucleic acids and/or viral particles. Toll-like receptors (TLRs) 3, 7, and 8 play important roles in recognition of viral nucleic acids by their ability to recognize either double-stranded RNA (TLR3) or single-stranded RNA (TLR7, 8). In addition, cytosolic sensors of viral replication and double-stranded RNA (such as PKR and RIG-I) also play a role in recognition of cytosolic-replicating viruses. Type I interferons are produced by virus infected cells and facilitate the establishment of an antiviral state in infected (autocrine action) and nearby (paracrine action) cells. Other PRRs recognize other microbial products, such as bacterial lipopolysaccharide (via TLR4), flagellin (via TLR5), CpG DNA (via TLR9), and fungal products (via TLR2 and mannose receptor) [5]. Recognition of microbial products by TLRs and other PRRs initiates the production of proinflammatory cytokines including IL-1, TNF-α, IL-6, as well as a number of chemokines (including CXCL8, CXCL10, CCL2, and CCL3). This is regarded as the acute phase response, and results in fever, increased vascular permeability, edema, and increased expression of adhesion molecules on the vascular epithelium. These early cytokines form an amplification cascade, collectively referred to as the "cytokine storm." These mechanisms serve to clear, or at least limit, infections during the establishment of the more efficient adaptive immune response. This intense inflammatory activity is followed by a compensatory downregulation of inflammation. One anti-inflammatory cytokine, IL-10, plays a critical role in modulating the inflammatory state. Dysregulation of the highly inflammatory state can occur during systemic infection or injury, resulting in sepsis. The highly inflamed state (systemic inflammatory response syndrome, or SIRS) can be lethal. Potentially life-threatening, as well, is the downregulation of severe inflammation. During this time the immune system is reflexive to further challenges and this is referred to as the compensatory antiinflammatory response syndrome (CARS). At this time, the host exhibits increased susceptibility to infections. For example, patients who survive an initial septic episode often shortly succumb to pneumonia. Cytokines and chemokines play an essential role in the establishment of the inflammatory state of the host, which is essential to combat infection, but may be detrimental if not properly controlled. Thus, many types of insults, bacterial infection, head trauma, burns, and so on can result in a similar life-threatening condition due to the overproduction and/or dysregulation of cytokine responses, leading to multiorgan shutdown.

CHEMOKINES AND THE REGULATION OF LEUKOCYTE TRAFFICKING

The acute phase response establishes conditions that favor the rapid migration of leukocytes into affected tissues. The vascular endothelium upregulates a series of adhesion molecules ("activated endothelium"), which in turn, serve to trap circulating leukocytes. The initial adhesion is mediated by E and P selectins that facilitate rolling of the leukocytes on the activated endothelium. In response to chemokine signals, leukocytes firmly adhere to the activated endothelium via the upregulation of beta integrins. Subsequently, the firm adhesion allows the leukocytes to spread along the endothelium and begin extravasating into the inflamed tissues. Leukocytes are guided to the site of inflammation via a chemokine gradient (Figure 13.3). However, the nature of these chemokine gradients have been difficult to assess in an in vivo setting. Multiple studies have suggested that chemokines, in particular, have specialized binding attributes that allow them to be deposited into the tissue so that they are not simply diluted or "washed" away. Extracellular matrix proteins are important for the maintenance of chemokine gradients. Most members of the chemokine family have the ability to bind to glycosyaminoglycans (GAGs), a component of extracellular matrix. This function serves to immobilize chemokines, and provides a scaffolding for migrating leukocytes. In addition, the oligomerization of chemokines on GAGs may be important for migration as leukocytes or even hypertrophic structural cells allowing them to haplotax (crawl) along the solid phase chemokine gradients. At high concentrations of chemokines (signaling arrival at the target site), leukocytes no longer migrate. Instead, other cell activating functions, such as degranulation of mast cells and eosinophils, take place. Thus, chemokines play essential roles in the firm adherence, extravasation, and chemoattractant-direct trafficking of leukocytes to inflammatory sites, as well as the initiation of the effector function once at the site of the inciting agent. While most research has been performed targeting a single chemokine or receptor at a time, it is clear that an individual inflammatory response initiates a series of chemokines to be produced. Since leukocytes have a profile of receptors, there are coordinated mechanisms that allow cells to respond, traffic, and be activated by multiple chemokines during any one response. Thus, a chemokine that initiates the firm adhesion event at the vascular endothelial cell surface is different from the chemokine that eventually localizes the migrating cell at the site of inflammation. This complex system along with the redundancy of chemokine function enables more fidelity and control within the system as well as increased protection if a signal chemokine/receptor system is perturbed.

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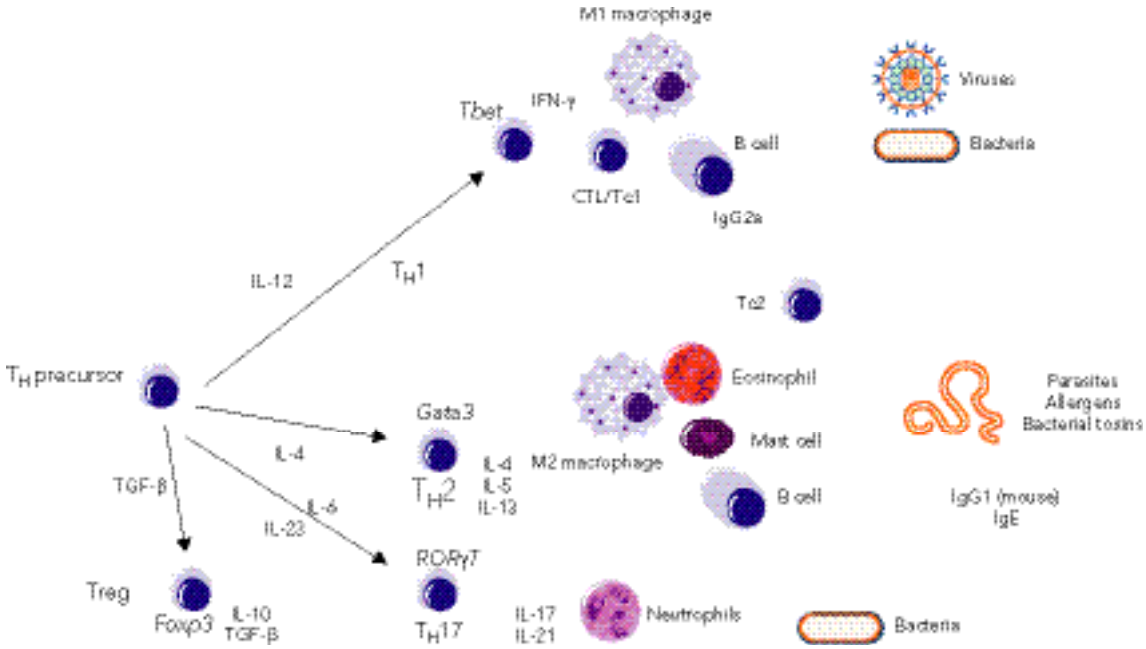


Figure 13.6. Adaptive immune responses are controlled by the actions of cytokines. The adaptive immune response is tailored to the type of pathogen via the actions of cytokines. Cytokine signals from innate immune cells and non-immune cells promote one of several pathways of CD4+ T cells (T_H) cell differentiation. IL-12 promotes the generation of IFNγ-producing T_H1 cells, important for the clearance of viruses and intracellular bacteria. IL-4 promotes the generation of T_H2 responses, which mediated antibody-dependent (or "humoral") immunity. T_H2 responses are generated in response to bacterial toxins, parasites, but also drive allergic responses. IL-6, TGF-β, and IL-23 favor the generation of T_H17 responses, important for the clearance of extracellular bacteria. T_H17 responses are also involved in autoimmunity. The immunosuppressive regulatory T-cell lineage (Treg) is promoted by TGF-β and IL-10 and serves to negatively regulate many types of immune responses.

CYTOKINES SHAPE THE ADAPTIVE IMMUNE RESPONSES

A critical function of cytokines is in shaping the development of adaptive immunity. Different types of microbial infection (i.e., viruses, bacteria, fungi, and parasites) are effectively controlled by different effector cell types. The cytokine profiles generated initially by resident cells in the infected tissue, and via antigen-presenting cells (APC) in the lymph nodes, dictate the type of immune response generated by T and B lymphocytes. Ideally, the immune response serves to activate the proper cell populations for antimicrobial effector functions. The role of cytokines in controlling adaptive immunity is exemplified in the differentiation and polarization CD4+ or helper T-cell populations (Figure 13.6) [6]:

- T helper type 1 (Th1) responses are generated in response to most viruses and intracellular bacteria. Early production of IL-12 from macrophages and dendritic cells in response to PRR engagement promotes the development of Th1 responses. IL-12 promotes induction of the transcription factor regulating Th1 development, T-bet. Th1 cells in turn produce large amounts of IFN-γ, which promotes the classical pathway of macrophage activation, inducing phagocytosis and intracellular killing via reactive oxygen intermediates.
- T helper type 1 (Th2) responses are generated primarily in response to extracellular pathogens, such as parasites, some fungi, as well as allergens. The development of Th2 responses is promoted by the cytokine IL-4, and Th2 lineage development is controlled via the transcription factor Gata-3. Prominent effector functions of Th2 responses include antibody production, basophil, mast cell, and eosinophil activation. Since most of these pathogens are too large to be engulfed by phagocytes, effector cells dump antimicrobial mediators on these organisms, or literally expel them from the host. Effector cytokines produced by Th2 cells include IL-4 IL-5, IL-13, as well as others.
- T helper 17 (Th17) cells, thus named for their production of IL-17, are involved in the elimination of extracellular bacteria (from mucosal surfaces especially), but are also thought to be involved in the development of chronic diseases, including autoimmune disease [7]. Th17 cells develop in the simultaneous presence of TGF-β and IL-6, and differentiation is mediated via the transcription factor

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RORγt. The major target cells of Th17 cells are thought to be neutrophils.

- Regulatory T cells (Tregs) are critical for controlling excessive inflammation, as well as maintaining tolerance to self-antigens [8,9]. Their development is dependent on the cytokine TGF-β, and controlled via the transcription factor Foxp3. In the absence of functional Tregs, lethal autoimmunity results. In addition to TGF-β, IL-10 is also a critical factor in Treg responses.

In addition to promoting the development of a particular lineage directly, cytokines also serve to antagonize the development of other lineages. For instance, in addition to promoting Th2 development through its actions on STAT 3, IL-4 also negatively regulates the development of Th1 cells in part via downregulation of the IL-12 receptor. By both promoting one response, and antagonizing others, cytokines shape the character of adaptive immunity.

The cytokine environment determined via inflamed tissues, as well as inflammatory cells, regulates B-cell activation, proliferation, and antibody production. IL-4 is secreted in a directional fashion by Th2 cell to target antigen-specific B cells. B-cell activation in the presence of IL-4, as well as other signals derived from direct T-cell contact, leads to B-cell proliferation. In addition, IL-4 promotes isotype switching in B cells to IgG1 and IgE, important for mast cell and eosinophil-mediated defense against parasites. IL-5 and TGF-β can both promote the switch to IgA secretion, critical for defense of the mucosa. IFN-γ from Th1 cells promotes isotype switching to IgG2a and IgG3, important for opsonic phagocytosis. Thus, each of these antibody subtypes is specialized for a different effector function. The antibody isotypes promoted via the cytokine environment serve to promote the antibody effector functions appropriate for the particular pathogen encountered.

The cytokine environment that predominates during an immune response affects many other cell types, including macrophages. Macrophages can serve as both initiators of adaptive immunity, as well as effector cells. In the presence of microbial products (such as LPS), IFN-γ is critical for the classical activation of macrophages (also referred to as M1 macrophages). This pathway serves to support Th1 responses by promoting IL-12, IL-18, and IL-15 secretion, as well as inducing the upregulation of phagocytosis and the respiratory burst. Activated M1 macrophages undergo efficient phagolysosomal fusion, exposing intracellular and/or recently ingested extracellular microbes to a variety of microbicidal lysosomal enzymes. In contrast, alternatively activated macrophages (or M2 macrophages) are activated under conditions of Th2 cytokines IL-4 and IL-13, as well as other conditions such as TGF-β. M2 macrophages are associated with Th2 responses to eukaryotic pathogens and allergens, resolution of inflammation, and tissue repair. M2 macrophages exhibit suppression of nitric oxide, and the enhancement of arginase, IL-10 production, and numerous gene products associated with tissue remodeling and repair.

DIRECTION OF INNATE AND ADAPTIVE IMMUNITY BY CHEMOKINES

Although the precise factors that govern the expression of chemokine receptors are incompletely understood, it is clear that certain chemokines and receptors direct the trafficking of specific cell types [10,11]. During the innate immune response, inflammatory infiltrate is typically composed of neutrophils, followed by macrophages. Among the first inflammatory signals induced, the ELR CXC chemokines, and their receptors, CXCR1 and CXCR2, play primary roles in the recruitment of neutrophils. The recruitment of macrophages tends to be more context-specific, but CCR1, CCR2, and CCR5 (and their ligands) play predominant roles in the recruitment of macrophages in most settings. Most leukocyte subtypes express CCR1, which binds CCL3, CCL4, and CCL5. CCR2 binds all of the CCL2 family members (CCL2, 7, 8, 12, and 13). Under both Type 1 and Type 2 conditions, CCR2 and CCR6 play important roles in the trafficking of monocytes and dendritic cells to inflamed tissues.

G-protein–coupled receptors have historically served as good targets for pharmaceuticals. Consequently, a significant amount of investigation has sought to identify the chemokine receptor (or receptors) required for Th1 versus Th2 inflammatory diseases. This effort has met with limited success. In general, differential chemokine receptor expression tends to be present under Th1 and Th2 inflammatory conditions. Under Th1 conditions, CXCR3 and, to a lesser extent, CCR5 predominate. Conversely, under Th2 conditions, CCR3, CCR4, and CCR8 are preferentially expressed. Thus, under Th1 promoting conditions, CXCR3 ligands including IP-10/CXCL9, MIG/CXCL10, and I-TAC/CXCL11 preferentially attract Th1 T cells. CCR3, CCR4, and CCR8 ligands, including eotaxins/CCL11, 24, 26, MDC/CCL22, TARC/CCL17, and TCA-3/CCL1 preferentially attract Th2 cells, eosinophils, mast cells, and basophils.

CXCR3 is expressed by a number of cell types, but preferentially on Th1 and Tc1 cells. CXCR3 ligands CXCL9–11 were all originally identified as IFN-γ inducible chemokines, so it comes as no surprise that CXCR3 ligands predominate under Th1 conditions. However, CXCR3 is expressed by dendritic cells, both myeloid dendritic cells and plasmacytoid dendritic cells (Type I IFN producing innate immune cells), so while CXCR3 may be preferentially expressed during Th1 conditions,

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its expression is not limited to Th1 cells. Perhaps the best-characterized receptor is CCR5 due to its use as a coreceptor for HIV infectivity. This receptor binds CCL3, CCL4, and CCL5 and when ligated with these chemokines can inhibit HIV infectivity in vitro. CCR5 antagonists have been tested as anti-HIV therapeutics with minimal success.

Although a number of studies demonstrated that Th2 cells preferentially express CCR3, CCR4, and CCR8, these chemokine receptors may not be specific for Th2-mediated diseases. Although CCR3 is highly expressed on Th2 cell and eosinophils (a hallmark of allergic disease), the biology of CCR4 appears to be more complex. CCR4 has been linked to both Tregs and modulation of M1/M2 phenotype in macrophages.

Clearly, the regulation of chemokines dictates the nature of cellular recruitment. On the basis of numerous signaling pathways induced by chemokine receptor ligation, it is likely that chemokines also play a role in the differentiation of responding cells. However, the precise relationship between chemokine production and cellular differentiation is still unclear. Chemokines can aid in directing the immune response either directly by activating APCs and T cells or indirectly by recruiting the proper cell populations.

DEVELOPMENT AND ORGANIZATION OF LYMPHOID TISSUES

Chemokines and cytokines mediate the development and organization of lymphoid tissues [12]. Lymphoid tissues such as thymus, spleen, and lymph nodes are distinct organized tissues. One cytokine in particular, lymphotoxin, is integrally important for the development of lymph nodes, and LT-deficient mice do not develop lymph nodes. B-cell–rich follicles require the chemokine receptor/ligand pair CXCL13/CXCR5. Conversely, T-cell zones develop in response to stromal and dendritic cell production of the chemokines CCL19 and CCL21, which both bind CCR7. The position of B cells and T cells within secondary lymphoid tissues depends on their differential responsiveness to CCR7 and CXCR5 to first enter the lymph node and subsequently localize to the follicle, respectively. This is also a dynamic process. For instance, upon antigen exposure, B cells decrease responsiveness to CXCR5 and upregulate responsiveness to CCR7, which promotes localization to the B-cell/T-cell margin. In general, cells in the lymph nodes migrate through these opposing chemokine gradients by modulation of receptors, facilitating distribution and redistribution of leukocytes within the node. While this process is not completely understood, the manipulation of these chemokine receptor systems are central to proper positioning of immune cells for activation during immune events.

When dendritic cells encounter pathogen-associated antigen in peripheral tissues, they upregulate CCR7 on their surface. This facilitates the trafficking and entry into draining lymph nodes. At the same time naive and central memory T cells express CCR7 allowing entry of the lymph node from peripheral circulation. The expression of CCR7 on dendritic cells and T cells allows the localization in the proper zone (T-cell zone) of the lymph node and therefore antigen presentation is optimal. When T cells are primed in the lymph nodes, downregulation of CCR7, and the upregulation of other chemokine receptors, such as CXCR3, CCR4, or CCR5 mediates their sensitivity to chemokines produced in the inflamed tissues. Subsequent egression from the lymph node and recirculation to the inflamed tissue allows extravasation to the site of immune response and allows the activated T cell to perform their effector function in the local tissue. Thus, the coordinated expression of chemokine receptors mediates the T cells to be properly activated and subsequently perform their effector function in the most appropriate manner to rid the host of the inciting pathogens.

CYTOKINES IN REPAIR AND REGENERATION OF TISSUE

Whether an acute or chronic inflammatory response occurs in tissue to pathogens or any other inciting agent the damage induced by the responses requires that tissue repair occur eventually allowing normal tissue function. A number of key cytokines are required for the repair process to proceed. Those cytokines studied in this process include TGF-β, TNF-α, platelet-derived growth factor (PDGF), basic fibroblasts growth factor (βFGF), monocyte chemoattractant protein-1 (MCP-1), macrophage inflammatory protein-1 α (MIP-1α) and IL-1, IL-13 and IL-8. Within hours of injury, reepithelialization is initiated and the release of numerous growth factors, such as EGF, TGF-α, and FGF stimulate epithelial cell migration and proliferation allowing the wound to be closed.

Perhaps the most defined repair cytokine has been TGF-β. The most prominent effect of TGF is its ability to induce fibroblast production of extracellular matrix (ECM), especially collagen. This process is most important during tissue repair and allows the activation local fibroblasts to produce ECM, which provides a foundation for epithelial cell proliferation to restructure the tissue. Once the wound is closed the re-epithelialization can occur. In an optimal situation TGF-β is produced for a defined period of time allowing both matrix deposition for basement membrane organization and epithelial cell growth and a stimulus for other important processes including angiogenesis as well as immune cell regulation [13]. Together, these important mechanisms that are induced by TGF-β

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signal the end of the immune response and the beginning of tissue reorganization and repair. However, the overproduction of repair cytokines, such as TGF-β, can have grave detrimental effects on the host system leading to end-stage disease [14]. The chronic overexpression of TGF-β has been linked to almost all fibrotic diseases in the kidney, lung, liver, as well as in cardiac infarct repair leading to increased lesion size. Thus, many therapeutic approaches have attempted to target this cytokine but it has been fraught with complications since it is central to the several important biologic processes.

CONCLUSION

The coordination of inflammatory and immune responses is a complex, ongoing balance between the clearance of pathogens and simultaneous damage control for the host. The production of cytokines and chemokines during these responses participate in all aspects of the regulation, including initiation and maintenance of inflammation while inciting agents are present, as well as resolution and repair of damaged tissue after the response has subsided. A better understanding of the differential regulation of immune responses by particular cytokines and chemokines facilitates the development of therapies to target immune/inflammatory mediators during chronic disease. Thus, our continued research and application in appropriate models of disease will enhance our ability to identify clinically relevant targets.

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14 Adenosine Receptors: Therapeutic Aspects for Inflammatory and Immune Diseases

György Haskó and Bruce Cronstein

Metabolic stress, hypoxia, and cell damage cause adenosine to accumulate in the extracellular space, and increases in extracellular adenosine levels are observed in hypoxia, ischemia, inflammation, and trauma [1,2]. Extracellular adenosine levels accumulate following the release of adenosine from cells or as a consequence of extracellular degradation of released ATP and ADP. Intracellular adenosine, which can originate from increased intracellular metabolism of ATP during cellular stress or *S*-adenosyl homocysteine, is released through equilibrative nucleoside transporters ENT1 and ENT2. Extracellular ATP and ADP are catabolized by a cascade of ectoenzymes which consists of CD39 (ENTPD1 [ectonucleoside triphosphate diphosphohydrolase-1]), an enzyme that hydrolyzes ATP and ADP to AMP, and CD73 (ecto-5'nucleotidase), which in turn, rapidly dephosphorylates AMP to adenosine [3]. Owing to the widespread expression of equilibrative nucleoside transporters, adenosine derived from extracellular ATP is rapidly recycled from the extracellular space by uptake into cells. Adenosine in the cytosol is then either phosphorylated by adenosine kinase to AMP or metabolized by adenosine deaminase to inosine [4,5]. As a net result of these various metabolic processes, adenosine levels in the extracellular space are maintained in a range of 10–200 nM in normal, healthy tissues. In contrast, under pathophysiological conditions adenosine is generated at a rate that is higher than the rate of degradation leading to markedly increased extracellular adenosine levels that can range between 10 and 100 μM.

Adenosine produces its biological effects by binding to and activating one or more of four membrane spanning adenosine receptors, designated A_1, A_{2A}, A_{2B}, and A_3. All four adenosine receptors contain seven transmembrane domains and couple to intracellular GTP binding proteins (G proteins). Adenosine elicits activation of A_1, A_{2A}, and A_3 receptors with EC_{50}s that range from 0.01 to 1 μM, and A_{2B} receptor activation occurs at adenosine levels that exceed 10 μM (EC_{50}: 24 μM) [6]. Since physiological adenosine concentrations are less than 1 μM, physiological levels of adenosine can activate only A_1, A_{2A}, and A_3 receptors, and A_{2B} receptor activation requires pathophysiological conditions [5]. In addition to adenosine concentrations at the cell surface, receptor density and the functionality of the intracellular signaling pathways coupled to adenosine receptors are also key factors in dictating the nature and magnitude of the effect of adenosine on the cell. For example, A_{2A} receptor activation inhibits production of the T helper (Th)1-inducing cytokine interleukin-12 (IL-12) more potently by human monocytes that are pretreated with the proinflammatory cytokine IL-1 or tumor necrosis factor-α (TNF-α), mediators that also increase A_{2A} receptor expression in these cells [7]. In addition, the effect of adenosine can also be affected by the polarized localization of adenosine receptors: A_3 adenosine receptors accrue at the leading edge of migrating neutrophils and are instrumental in directing the movement of cells in response to chemotactic mediators [8]. Finally, it is important to keep in mind that results regarding adenosine receptor function in one species cannot be readily extrapolated to another one, because sequence differences in cloned adenosine receptors have been shown to be associated with differential pharmacological responses to selective agonists and antagonists. In this regard, A_3 receptors were not even discovered until they were cloned [9], because the A_3 receptor, especially in rodents, is insensitive to xanthines such as caffeine and theophylline, antagonists, which had been pivotal in identifying adenosine receptor-mediated effects.

ADENOSINE RECEPTOR SIGNALING

Adenosine receptors in general dictate cell function through coupling to G proteins, but some G-protein–independent actions have also been reported [5]. Adenosine receptors were initially classified as A_1

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(cAMP-decreasing) or A_2 receptors (cAMP-increasing) [10]. Subsequently, the cAMP-increasing A_2 receptors were divided into two groups: high-affinity A_{2A} receptors and low-affinity A_{2B} receptors [11]. More recent studies have revealed that in addition to A_1 receptors, A_3 receptors also decrease intracellular cAMP concentrations [12]. In addition to signaling via the adenylyl cyclase-cAMP system, adenosine receptors can signal through a variety of other pathways. For example, A_1 receptor activation can be linked to various kinase pathways, including PI3-kinase, protein kinase C (PKC), and mitogen-activated protein (MAP) kinases [13]. In addition, A_1 receptor engagement can directly activate K^+ channels and inhibit Q-, P-, and N-type Ca^{2+} channels. A_{2A} receptors, as other Gs protein–coupled receptors, signal via chiefly the adenylate cyclase–cAMP–protein kinase A (PKA) canonical pathway, but they can also activate exchange factor directly activated by cAMP (Epac) [5]. Signaling downstream from PKA occurs through phosphorylation of the transcription factor CREB on serine residue 133 leading to direct CREB-mediated transcriptional activation [14]. Activated CREB can also modulate gene expression indirectly by competing with NF-κB or other transcription factors for an important cofactor, CBP [15]. In other cell types adenosine A_{2A} receptors stimulate collagen production via MAP kinases [16] and inhibit neutrophil superoxide production through activation of protein phosphatases [17]. Furthermore, recent results implicated c/EBPβ in the stimulatory effect of A_{2A} receptor agonists on IL-10 production by *Escherichia coli*–challenged macrophages [18]. A_{2B} receptor stimulation can induce both adenylyl cyclase activation via Gs and phospholipase C activation via Gq [19]. Interaction between these two pathways is important for upregulation of IL-4 production by mast cells upon A_{2B} receptor activation [20]. Specifically, Gq-mediated activation of phospholipase Cβ causes calcium mobilization and an increase in NFATc1-dependent IL-4 transcription, which response is further facilitated by Gs-mediated NFATc1 protein accumulation. Traditionally, A_3 receptor activation is linked to Gi-mediated inhibition of adenylyl cyclase and Gq-mediated stimulation of PLC [21] and A_3 receptors can activate PLD, RhoA, WNT, MAP kinase, and PI3 kinase pathways in governing cell function. For example, A_3 receptor–mediated augmentation of histamine released in sensitized murine mast cells was blocked by inactivating Gi proteins with pertussis toxin and by using pharmacological PI3 kinase inhibitors [22].

EFFECTS OF ADENOSINE ON THE CELLS OF INFLAMMATION

Neutrophils

Neutrophils are the most abundant white blood cell in the peripheral blood and are generally the earliest responders to trauma, infection, and bacterial invasion or other inflammatory stimuli. Adenosine is a potent regulator of neutrophil function and it has long been known that adenosine, acting at its receptors, regulates stimulated production of reactive oxygen species by these cells [23–26]. Adenosine, acting at A_{2A} receptors, also regulates phagocytosis [27,28]. In general neutrophils do not secrete large quantities of cytokines but because there are so many of these cells the cumulative contribution to proinflammatory cytokine levels at a given site are large and adenosine, acting at A_{2A} receptors regulate the production of a variety of cytokines including TNF-α, MIP-1α/CCL3, MIP-1β/CCL4, MIP-2α/CXCL2, and MIP-3α/CCL20 [29]. Neutrophils are recruited to an inflamed site by the postcapillary venular endothelium, which alters the expression of adhesive molecules on their surface in order to "capture" neutrophils from the circulation. Adenosine, acting at A_{2A} receptors, inhibits the adhesion of neutrophils to endothelium by diminishing the expression and "stickiness" of the neutrophil's adhesive molecules [30–38]. Interestingly, adenosine A_1 receptors increase neutrophil adhesion to different adhesive molecules on the endothelium and on other surfaces [30]. Once in the tissue neutrophils follow a gradient of chemoattractants to their source. There are many chemoattractants including activated complement components (C5a), bacterial products (formylated peptides), and chemokines and recent studies have indicated that adenosine receptors promote directed migration of neutrophils via A_1 and A_3 receptors [8,39,40]. More strikingly, neutrophils cluster their A_3 receptors at the leading edge of the cell and release ATP that is converted at the cell surface to adenosine, which then acts in an autocrine fashion to promote migration [8]. At inflamed sites, neutrophils undergo apoptosis and this process can be further stimulated by such cytokines as TNF-α. Adenosine, acting at A_{2A} receptors, prevents neutrophils from undergoing apoptosis [41–43]. Thus, nearly every function carried out by neutrophils is regulated by adenosine and its receptors.

Macrophages

The effect of adenosine on cytokine production by macrophages has been extensively investigated, for macrophage-derived cytokines are pivotal initiators and orchestrators of immune responses. The earliest studies concentrated on the ability of adenosine receptor ligands to suppress TNF-α production by activated macrophages. These early results have recently been complemented with data obtained by utilizing adenosine receptor KO mice, allowing a better elucidation of the receptor subtypes involved. On the basis of these results, it appears that the A_{2A} receptor is the dominant adenosine receptor subtype mediating the inhibition of TNF-αI secretion by adenosine [44–46]. In

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addition, since adenosine, NECA, and IB-MECA can each inhibit, albeit to a lesser extent, TNF-α production even in A_{2A} receptor KO mice [44,46], other adenosine receptors can also contribute to the inhibition of TNF-α production. By employing a combination approach consisting of A_{2A} receptor KO mice and the A_{2B} receptor antagonist MRS 1754, a recent study demonstrated a role for A_{2B} receptors as the other inhibitory receptor [44]. Nevertheless, A_{2B} receptors become functional only when their effect is not overshadowed by A_{2A} receptors, for both MRS 1754 and knockout of A_{2B} receptors in the presence of A_{2A} receptors fails to affect the inhibition of TNF-α production [44,45].

The regulation of IL-10 production also appears to be dependent primarily on A_{2A} receptors, with A_{2B} receptors having a minor role. In this context, adenosine failed to upregulate *E. coli*–induced IL-10 release by macrophages lacking A_{2A} receptors but not WT macrophages [18]. In RAW264.7 macrophages, which express low copy numbers of A_{2A} receptors, adenosine upregulates IL-10 production through an A_{2B} receptor-mediated posttranscriptional mechanism [47]. While A_{2B} receptor activation has a marginal effect on TNF-α and IL-10 production, it is central to the stimulatory effect of adenosine on IL-6 production, as NECA fails to induce the release of IL-6 in A_{2B} receptor KO but not WT macrophages [45].

Finally, it is noteworthy that the question of which adenosine receptors regulate cytokine production by human monocytes/macrophages is even less clear, with data supporting a role for all four receptors [48]. However, because these human results are derived from pharmacological approaches, they must be interpreted with caution, as many of the ligands used are not particularly selective. However, the recent demonstration of elevated concentrations of adenosine contained in human newborn plasma when compared to adult plasma coupled with the observation that degrading adenosine with adenosine deaminase or interrupting adenosine signaling by adenosine receptor blockade augments TNF-α production by neonatal but not adult blood, indicate the clinical importance of the suppressive effect of adenosine on TNF-α production in human immunity [49].

Dendritic Cells

The only available human studies demonstrate a relatively consistent role for adenosine receptors in regulating dendritic cell function. Gi-coupled (A_1, A_3, or both depending on the experimental system) adenosine receptors are present on both immature myeloid [50] and plasmocytoid dendritic cells [51], and their activation leads to the liberation of intracellular calcium from its stores and reorganization of the actin cytoskeleton. This actin reorganization is instrumental in causing immature dendritic cells to migrate along different concentration gradients of adenosine, indicating that adenosine receptor activation triggers chemotaxis in immature dendritic cells [50,51]. This chemotactic effect can be seen only in immature dendritic cells, as the Gi-coupled adenosine receptors undergo downregulation during dendritic cell maturation [50,51].

Although immature dendritic cells express A_{2A} receptors, they appear to be nonfunctional, as their activation is unable to trigger intracellular signaling events such as increase of intracellular cAMP [50]. Dendritic cell maturation, however, is associated with the appearance of A_{2A} receptor-mediated signaling responses, due to both increased expression and coupling of A_{2A} receptors [50,51]. A_{2A} receptor activation on mature dendritic cells results in a shift in the pattern of cytokine release from a proinflammatory to an anti-inflammatory profile, characterized by decreased IL-12, IL-6, and IFN-α production and increased IL-10 production [50–52]. Owing to this adenosine-induced shift in dendritic cell cytokine production away from the Th1-inducing IL-12 toward the Th2-inducing IL-10, dendritic cells in the presence of adenosine have a decreased ability to elicit T helper (Th)1 versus Th2 polarization of naive CD4+ cells [52].

Overall, the available data demonstrate a dual role for adenosine in regulating dendritic cell function: adenosine promotes the migration of immature dendritic cells to inflammatory sites through A_1 or A_3 receptors, where adenosine triggers, via A_{2A} receptors, an anti-inflammatory dendritic cell phenotype shifting T-cell responses toward a Th2 direction.

Mast Cells

As a result of the demonstration that inhaled adenosine elicits bronchoconstriction in patients afflicted with asthma but not healthy volunteers, adenosine, and adenosine receptors have emerged as potential therapeutic targets in asthma [53]. Although the bronchospastic mechanisms of adenosine are incompletely understood, there is an increasing body of evidence incriminating mast cell–derived mediators such as cytokines and histamine as having a central role in provoking airway constriction following adenosine inhalation. Although A_{2A}, A_{2B}, and A_3 receptors are expressed on the cell membrane of mast cells [54], it is unclear which receptor(s) account for the augmented mast cell activation in response to adenosine.

Adenosine augments release of the pivotal asthma-inducing cytokine IL-13 by murine mast cells from WT but not A_{2B} receptor KO mice indicating that A_{2B} receptors are responsible for the proinflammatory effects of adenosine [55]. Furthermore, studies with adenosine receptor KO mice demonstrated that A_3 but not A_{2B}

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receptor activation can trigger the release of histamine from naive lung mast cells [22]. Thus, A_{2B} and A_3 receptors subserve different proinflammatory roles in murine mast cells in the absence of specific antigenic stimulation. The role of A_{2B} receptors in mediating antigen-stimulated histamine release by murine mast cells is subject to controversy. Although mast cells isolated from WT animals exhibit decreased histamine release in response to antigenic stimulation when compared to mast cells from A_{2B} receptor KO mice [56], exogenous adenosine retains its stimulatory effect in A_{2B} receptor KO mast cells [20]. Therefore, it is plausible that the decreased release of histamine from mast cells obtained from A_{2B} receptor WT versus KO mice indicates an impaired responsiveness of A_{2B} receptor WT versus KO cells to antigen that may originate from altered development of mast cells in A_{2B} receptor KO mice. In contrast to A_{2B} receptors, A_3 receptor activation directly augments histamine release in antigen-stimulated mouse mast cells, as the stimulatory effect of added adenosine that can be observed in WT mast cells is not seen in A_3 receptor KO mast cells [57].

Unlike results noted with rodent cells, in human and canine mast cells, adenosine-induced degranulation or cytokine release is mediated predominantly by A_{2B} receptor activation [58,59]. Stimulation of A_{2B} receptors on human mastocytoma HMC-1 cells elicits production of the T helper 2 cytokines IL-4 and IL-13, in addition to a several other proinflammatory cytokines including IL-1β and IL-8. Although adenosine receptor activation is incapable of triggering histamine release from human mast cells [56], the demonstration of augmented IL-4 and IL-13 production in response to adenosine receptor stimulation indicates that human mast cell adenosine receptors are important factors in the pathophysiology of asthma.

Lymphocytes

Several recent studies utilizing adenosine receptor KO mice have examined the effect of adenosine receptor activation on various lymphocyte functions, and dominant view based on these studies is that A_{2A} receptors are the primary adenosine receptors in shaping lymphocyte responses. Studies employing A_{2A} KO mice have shown that A_{2A} receptor activation limits IL-2 secretion by naive CD4+ T-cells [60] leading to their reduced proliferation [61] following T-cell receptor engagement. In addition, A_{2A} receptor activation decreases the production of both IFN-γ and IL-4 by both naive CD4+ T cells [60,62] and polarized Th1 and Th2 cells [63], casting doubt on the hypothesis that A_{2A} receptor stimulation in lymphocytes might switch Th cell responses toward a Th2 profile. A_{2A} receptor stimulation also upregulates the expression of negative costimulatory molecules including cytotoxic T-lymphocyte antigen-4 (CTLA-4) and programmed cell death-1 (PD-1), and downregulates expression of the positive costimulatory molecule CD-40L [61].

As with CD4+ cells, adenosine limits IL-2 production by both polarized Tc1 and Tc2 CD8+ cytotoxic cells, which effect is mediated through A_{2A} receptors [64]. However, the production of neither Tc1 (IFN-γ) nor Tc2 (IL-4 and IL-5) cytokines was affected by A_{2A} receptor stimulation. Moreover, A_{2A} receptor stimulation failed to inhibit Tc1 or Tc2 cell cytotoxicity [64], which indicates that another subtype, possibly the A_3 receptor may dictate the anticytotoxic effect of adenosine observed in earlier studies [65,66]. Recent analysis of A_1, A_{2A}, and A_3 receptor KO mice, however, indicate a key role for A_{2A} receptors in decreasing the cytolytic activity of IL-2-activated natural killer (NK) cells [67].

Adenosine has emerged as a major mediator of the immune suppressive effects of regulatory T (Treg) cells, which have a key role in harnessing the immune system, thereby preventing excessive immune-mediated tissue injury. Studies show that Tregs, as defined by expression of CD4+/CD25+/Foxp3+ and a capacity to limit the proliferation of CD4+/CD25– cells, express high levels of both CD39 [68] and CD73 [68,69]. On the basis of these results CD39 and CD73 were proposed to serve as novel-specific markers of Treg cells. Recent studies confirmed the intimate relationship between Foxp3+ expression and CD39 expression by demonstrating that Foxp3+ triggers the expression of CD39 [70].

Evidence indicates that CD73 and CD39 have important roles in mediating the immune suppressive properties of Tregs by hydrolyzing exogenous ATP/ADP to generate adenosine [68]. Specifically, Tregs obtained from CD39 KO mice lost their ability to suppress proliferation of CD4+/CD25– cells, which effect was reinstituted by introducing soluble exogenous NTPDases [68]. In addition, a role for adenosine was further highlighted by studies demonstrating that A_{2A} receptor KO target cells (CD4+/CD25–) proliferated more than WT cells when cultured with WT Tregs. The intimate relationship between A_{2A} receptors and Tregs is also indicated by the observation that A_{2A} receptor activation augments Foxp3+ expression in T cells [71].

A subset of T cells known as invariant NKT cells (iNKT) are important for the innate immune response that induces very rapid host responses to infection [72,73]. A_{2A} receptors are expressed on iNKT cells and markedly suppress the production of proinflammatory cytokines such as INF-γ [74]. Since NKT cell activation has been implicated in a wide range of disease processes including atherosclerosis, type 1 diabetes, arthritis, and various allergic diseases [75], the presence of A_{2A} receptors on NKT cells indicates novel therapeutic applications of A_{2A} receptor ligands.

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Endothelial Cells

Endothelial cells express predominantly A_{2A} and A_{2B} adenosine receptors although A_{2B} receptors appear to play a larger role in regulation of the microvasculature [76]. At inflamed sites the vascular endothelium expresses or upregulates the expression of adhesive molecules, including E-selectin, VCAM-1, and ICAM-1, on their surface, which mediates recruitment of leukocytes. Adenosine A_{2A} receptors inhibit the expression of E-selectin and VCAM-1 but do not affect ICAM-1 upregulation [77], effects apparently mediated by inhibition of NF-κB activation in the endothelium [78]. More recently, similar effects have been ascribed to the A_{2B} receptor as well [79]. In addition, the vascular endothelium secretes a variety of chemokines and cytokines at inflamed sites, including IL-8 and IL-6, and adenosine, acting at A_{2A} and A_{2B} receptors, diminishes their expression [77]. Thus, adenosine diminishes the endothelial contribution to inflammation and leukocyte recruitment to inflamed sites.

The vascular endothelium is also an important source of adenosine [80–82], particularly during inflammation, by extracellularly converting adenine nucleotides to adenosine via CD39/CD73 expressed on their surface [83–86]. Inflammatory cytokines such as IFN-α, promote the expression of CD73 in a form of feedback regulation of inflammation as well [87]. The adenosine released by the endothelium not only protects the endothelium and the underlying tissue from attack by stimulated leukocytes [85,88] but inhibits the vascular leakage (edema) that is so prominent at inflamed sites via activation of adenosine A_{2B} receptors [84,89–91].

As discussed later, adenosine and its receptors play an important role in granulation tissue formation in wounded tissues. The central event in granulation tissue formation is angiogenesis and adenosine, acting at both A_{2A} and A_{2B} receptors, stimulates angiogenesis. Application of an adenosine A_{2A} receptor agonist to wounds increases blood vessel formation [92–95] and enhances wound healing. This effect is mediated, in part, by promotion of angiogenic factors by endothelial cells and macrophages and downregulation of antiangiogenic factors [96–98]. In other tissues, such as the retina, adenosine A_{2B} receptors mediate this function [99–101].

OVERALL EFFECTS OF ADENOSINE AND ITS RECEPTORS ON INFLAMMATION

Asthma and Chronic Obstructive Pulmonary Disease

Inhaled adenosine can trigger bronchospasm in patients with asthma and chronic obstructive pulmonary disease (COPD), but not healthy patients, and adenosine receptor antagonists can block this bronchospastic response [54]. These observations coupled with the findings that both adenosine concentrations and adenosine receptor abundance on immune cells in the lung are increased in individuals with asthma and COPD link endogenous adenosine to pathophysiological mechanisms in lung diseases. Because the bronchospastic response to adenosine can be prevented by agents that stabilize the mast cell membrane and prevent the release of injurious mediators, and because histamine receptor blockers can also prevent adenosine-mediated bronchoconstriction, mast cells have been proposed as major targets of the bronchoconstrictive effects of adenosine [54]. Rodent models of asthma have confirmed this, and have also incriminated A_{2B} [55] and A_3 [22,57] receptors as contributing to the stimulatory effect of adenosine on mast cell activation. In contrast, studies employing human mast cells have demonstrated that A_{2B} receptors mediate the proinflammatory effects of adenosine in humans. A_{2B} stimulation is proinflammatory not only in mast cells, but also in human bronchial smooth muscle cells [102], human bronchial epithelial cells [103], and human lung fibroblasts [104], which produce increased amounts of IL-6 [102,104] and IL-19 [103] in response to adenosine. A_{2B} receptors promote the differentiation of human lung fibroblasts into myofibroblasts, a major cell type that produces extracellular matrix [22]. This suggests that adenosine may participate in the fibrosis and remodeling of the lung during asthma and COPD.

Preclinical data showing the effectiveness of A_{2B} receptor antagonism in preventing disease progression in rodent animal models [105,106] suggest that selective A_{2B} antagonists may be a feasible option for the treatment of human patients suffering from asthma and COPD. This notion is underlined by evidence that enprofylline, an antiasthmatic agent, is a relatively selective, albeit not potent A_{2B} receptor antagonist [107]. In this regard it is noteworthy that CVT-6883 [105,106], a selective A_{2B} receptor antagonist that has demonstrated efficacy in preventing disease in rodent models of asthma and COPD appeared to be safe and well tolerated in a recent human Phase 1 study.

A_{2A} agonists have been found to have widespread anti-inflammatory effects in pulmonary inflammation [108]. Bone marrow transfer experiments demonstrated that A_{2A} receptor activation only decreased lung inflammation when the A_{2A} receptor was present on bone marrow–derived cells. In addition, utilizing A_{2A} receptor KO mice, Nadeem et al. showed that the adenosine–A_{2A} receptor–cAMP axis is a potent endogenous anti-inflammatory signaling mechanism that prevents airway hyperreactivity and inflammatory cell sequestration following ragweed sensitization [109]. Thus, A_{2A} agonists appear to be effective at

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harnessing inflammatory lung tissue injury. However, in clinical trials, the utility of the A_{2A} receptor agonist GW328267X was limited by cardiovascular side effects [110].

Reperfusion Injury

A_{2A} receptors on bone marrow–derived cells can protect organs from reperfusion injury following an ischemic episode. The organs that have been shown to be protected in this way include liver [111], kidney [112], heart [113], skin [114], spinal cord [115,116], and lung [117]. In liver [74] and kidney [118], the proinflammatory activities of iNKT cells have been found to be inhibited by A_{2A} receptor activation, and to be critical targets of the organ protective effects of A_{2A} adenosine receptor activation.

Inflammatory Arthritis

Methotrexate is the mainstay of therapies for inflammatory arthritis. At low doses methotrexate enters cells and becomes polyglutamated to a long-lasting metabolite [119]. AICAR transformylase, an enzyme which is part of the de novo purine synthetic pathway, is inhibited by methotrexate polyglutamates [120] causing, the intracellular accumulation of AICAR [121,122]. AICAR competitively inhibits AMP deaminase causing, ultimately enhanced release of adenosine from cells [121]. Although it is difficult to measure adenosine levels in blood and other fluids because of the short half-life (2–8 seconds) of adenosine [123], in patients suffering from rheumatoid arthritis treated with methotrexate there is strong evidence that methotrexate therapy increases adenosine release [124,125]. Experiments employing rodents show that the anti-inflammatory effects of methotrexate are mediated by adenosine and that the anti-inflammatory effect of methotrexate is absent if animals are administered adenosine receptor antagonists or if their adenosine A_{2A} or A_3 receptors have been knocked out [122,126–132]. Similarly, methotrexate loses its efficacy in patients with rheumatoid arthritis who consume significant quantities of caffeine, an adenosine receptor antagonist [133]. Thus, by increasing adenosine release at inflamed sites methotrexate attenuates inflammation in patients with rheumatoid arthritis. These observations that methotrexate exerts some of its anti-inflammatory effects through A_3 receptors together with the demonstration that the selective A_3 receptor agonist IB-MECA suppresses the course of arthritis in collagen-induced arthritis in mice [134], led to clinical trials with IB-MECA (CFA101) in patients with rheumatoid arthritis. IB-MECA was safe and well tolerated in this Phase II trial, and patients that were given this drug attained a moderate attenuation of symptoms [135].

Sepsis

Mouse models employing both knockout and pharmacological approaches have delineated the role of the different adenosine receptors in governing the physiological response of the host to sepsis [136]. Blockade of A_1 receptors augmented mortality in intraperitoneal sepsis induced by cecal ligation and puncture in mice, which was associated with increased inflammation-induced hepatic and renal injury [137]. Similar to these results, both knockout and pharmacological inactivation of A_3 receptors enhanced mortality, in mice challenged with cecal ligation and puncture [138]. In contrast to the deleterious effects of blockade of A_1 and A_3 receptors, A_{2A} receptor blockade by either gene deletion or administration of ZM241385 diminished cecal ligation and puncture-induced lethality. This protective effect occurred by a mechanism that involved decreased bacterial growth, which was due to preserved immune system function [139]. Surprisingly, A_{2A} receptor activation, when combined with antibiotics, has been shown to decrease lethality from sepsis induced by injecting with live *E. coli*, possibly by suppressing an exaggerated inflammatory response that can be caused by the rapid drug-induced killing of large number of bacteria [140]. Finally, it appears that A_{2B} receptor inactivation in cecal ligation and puncture results in an augmented inflammatory response (Csoka, Hasko, Nemeth, and Pacher, unpublished observation).

Inflammatory Bowel Disease

Ulcerative colitis and Crohn's disease are chronic, relapsing diseases characterized by dysfunctional mucosal T lymphocytes, imbalances in cytokine production and inflammation leading to damage of the intestinal mucosa. Although the etiology of inflammatory bowel disease (IBD) is elusive, the consensus is that that disease is the result of a dysregulated immune response to luminal antigens in a genetically susceptible host [141,142]. A_{2A} receptor activation has been shown to be protective in several animal models of IBD, and these protective effects can be attributed to two main mechanisms: attenuation of inflammatory cell sequestration and function in the mucosa, and enhanced activity of Tregs. Contribution of the former mechanism is based on results that A_{2A} receptor stimulation with ATL146e diminishes both leukocyte infiltration and the release of inflammatory cytokines by disease-promoting effector T cells [143]. The latter mechanism was predicated on data that co-transfer of CD25+ CD4+ T regulatory cells obtained from A_{2A} receptor WT mice could prevent disease induction SCID mice that were transferred adoptively with disease-inducing $CD45RB^{high}$ CD4+ T cells, while

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co-administration of CD25+ CD4+ Tregs obtained from A_{2A} receptor KO mice together with disease-inducing $CD45RB^{high}$ CD4+ T cells was inefficient in preventing the development of disease [60].

Stimulation of A_{2B} receptors on intestinal epithelial cells increases IL-6 production by these cells leading to increased neutrophil activation [144]. In addition, A_{2B} receptors expression increases in epithelial cells isolated from murine or human colitis [145]. On the basis of these observations, recent studies have assessed the effect of A_{2B} receptor blockade in IBD. Treating mice with a selective A_{2B} antagonist, ATL-801, strongly decreased IL-6 production and neutrophil infiltration, reduced mucosal damage, and ameliorated the course of disease in a model of IBD [146]. These are encouraging results, because blockade of A_{2B} receptor might be a potentially advantageous treatment option that selectively targets the gut of patients suffering from IBD. This idea is based on results that A_{2B} receptor expression is highest in the gut as compared to other organs and that concentrations of endogenous adenosine that seem to propagate the disease process through A_{2B} receptors are elevated exceedingly in the gut [147].

Finally, it is noteworthy that in addition to endogenous adenosine, its metabolite inosine can also ameliorate the course of experimental IBD [148]. Since inosine can potently deactivate macrophages, in part, through A_{2A} receptors [149], one possibility is that inosine exerts its beneficial effect by preventing production of macrophage-derived inflammatory cytokines in the gut [150,151].

Wound Healing

Inflammation in the skin is part of a continuum in which damaged tissue is repaired and replaced, leading to wound healing. Montesinos et al. first showed that applying adenosine A_{2A} receptor agonists topically increases the rate of wound healing in normal and diabetic mice and rats [92]. Adenosine receptor agonists increased both matrix and vessel formation and their effects were superior to that induced by recombinant platelet-derived growth factor [92–94,152]. Activation of A_{2A} and A_{2B} receptors enhances angiogenesis both directly and indirectly: adenosine receptors directly promote proliferation of microvascular endothelial cells in part through the autocrine production of VEGF, a central stimulus for endothelial proliferation and angiogenesis [7,99–101,153–159] and suppress the production of thrombospondin 1, an antiangiogenic matrix protein [97]. Furthermore, adenosine receptor stimulation upregulates VEGF production by macrophages [96,152,160]. In addition, A_{2A} and A_{2B} receptor stimulation enhances matrix production by fibroblasts, a central step in tissue repair [104,161–163]. In this regard, it is worth mentioning that A_{2A} receptor KO mice fail to generate granulation tissue confirming that endogenous adenosine has a central role in wound healing [93]. Sonedenoson (MRE0094, King Pharmaceuticals), an A_{2A} agonist that regulates the inflammatory response and enhances tissue regeneration, is currently undergoing trials for the treatment of diabetic foot ulcers.

CONCLUSIONS

A recent increase in our understanding of the role of the various adenosine receptors in regulating tissue injury have led to the identification of novel pharmacological targets to restore tissue function in various diseases. Targeting adenosine receptors with agonists, antagonists, and interfering with the function of enzymes and transporters that are responsible for the accumulation of extracellular adenosine represent a wide range of approaches to alter adenosine signaling at inflammatory sites.

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15 Leukocyte Generation of Reactive Oxygen Species

William M. Nauseef

INTRODUCTION

The rapid response of the innate immune system of humans depends in large part on the behavior of a subset of leukocytes known as polymorphonuclear leukocytes or neutrophils. Constituting the majority of white blood cells in the circulation, neutrophils provide continuous surveillance for microbial and noninfectious threats to the integrity of the host. Once recruited to an area of infection, neutrophils ingest microbes and thereby confine them in the phagosome, a membrane-bound compartment where an array of antibacterial factors can be delivered to kill and degrade the trapped microorganism. In addition to proteins that exhibit direct antimicrobial action, cytoplasmic granules in neutrophils contain enzymes that utilize microbial constituents as substrate, thereby compromising the structural integrity of the organism and indirectly contributing to microbial killing. Under normal circumstances, these complex and synergistic events occur in the presence of reactive oxygen species (ROS) generated in situ by activation of a multicomponent enzyme, the phagocyte NADPH-dependent oxidase. Like the prefabricated granule constituents, ROS damage targets directly and act in concert with granule proteins to generate second derivative active agents. The concomitant release of granule proteins and activation of the NADPH oxidase thus provide phagocytic cells such as neutrophils with the capacity to respond rapidly to potentially life-threatening challenges and thus provide the initial wave of host defense.

Although monocytes, macrophages, and eosinophils share many of the fundamental features of the phagocytosis-coupled response exhibited by neutrophils, the discussion will use human neutrophils as the paradigm for presenting the biochemical principles of ROS generation by stimulated leukocytes.

CONTEXT OF THE PHAGOCYTE NADPH OXIDASE

Appreciation of the role of ROS in normal host defense against infection serves as another example in modern medicine of how basic research and clinical medicine reciprocally rely on each other for critical insights as well as for identifying the proper biological context. As investigators were probing the origins and significance of the "burst" in oxygen consumption that phagocytes exhibited when challenged with microbes, clinicians were characterizing the phenotype of a new clinical syndrome associated with a compromise in normal immune function.

Whereas resting neutrophils rely on glycolysis for energy and consume little to no O_2, stimulated neutrophils exhibit a rapid and robust burst of respiration, often referred to as the "respiratory burst," and thereby generate a large flux of ROS. Nearly all of the oxygen consumed by stimulated neutrophils can be recovered as H_2O_2 produced, demonstrating that the phagocyte oxidase represents the sole mechanism for oxygen utilization in phagocytes. The respiratory burst oxidase was not constitutively active but required stimulation to initiate activity, resisted inhibition by cyanide or azide, thus excluding a mitochondrial source, and required flavin adenine dinucleotide (FAD) as well as NADPH for activity. Over the same time span that investigators were characterizing the respiratory burst oxidase, clinicians described a new inherited disorder, chronic granulomatous disease (CGD), that was manifested as frequent and severe pyogenic infections. Patients with CGD were first recognized in the 1950s when four boys with recurrent infections, lymphadenopathy, and granulomatous reactions in liver and lymph nodes were reported. The concurrent pursuits to elucidate the enzymology and significance of the respiratory burst of phagocytes and to understand the basis of the increased susceptibility to infection seen in patients with CGD merged with recognition

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(1) that the biochemical phenotype of CGD reflected the absence of oxidase activity from neutrophils, monocytes, macrophages, and eosinophils and (2) that the failure of CGD phagocytes to generate ROS resulted in depressed antimicrobial activity and the resultant increased occurrence of pyogenic infection.

Collectively, ROS refers to molecules generated by the incomplete reduction of oxygen, which include, in the context of the phagocyte oxidase, superoxide anion (O_2^-), hydrogen peroxide (H_2O_2), and hydroxyl radical (HO).

O_2^- is relatively unstable and because of its negative charge does not diffuse efficiently through membranes. However, because of its negative charge, O_2^- avidly oxidizes charged redox centers such as iron–sulfur clusters in bacteria. In contrast, H_2O_2 is relatively stable and readily diffuses through membranes to oxidize susceptible targets, especially sulfur-containing amino acids such as cysteine and methionine residues. HO· ranks as the most toxic of these three ROS and reacts indiscriminately with lipids, proteins, DNA, and virtually all potential biologic substrates. As discussed in the following, the interactions among these ROS and between the ROS collectively and granule proteins create a hostile environment within the phagosome for the ingested microbe.

MEMBRANE COMPONENTS OF THE PHAGOCYTE OXIDASE

As noted earlier, the phagocyte NADPH-dependent oxidase is inactive in resting neutrophils but converted into an active form by exposure of the cells to a suitable agonist. The activation state of the phagocyte oxidase is regulated by the spatial disposition of its components. The essential elements of this multicomponent enzyme in resting cells are segregated either in membranes or as soluble proteins within the cytoplasm. Exposure to effective agonists triggers assembly of the oxidase at the plasma membrane or the phagosomal membrane. Once assembled, the enzyme becomes active, converting molecular oxygen to O_2^-.

The membrane component of the phagocyte NADPH oxidase is flavocytochrome b_{558}, a heterodimeric integral membrane protein composed of a glycosylated 91-kDa protein, gp91^{phox} (where *phox* indicates *ph*agocyte *ox*idase), and a nonglycosylated 22-kDa protein, p22^{phox}. Encoded by the *CYBB* gene on the X-chromosome and once thought to be uniquely expressed in phagocytes, gp91^{phox} is the electron transferase component of the phagocyte oxidase and has recently been recognized to be the patriarch of a family of NADPH oxidase proteins, referred to now as the NOX protein family (*vide infra*). Neutrophils, monocytes, eosinophils, and macrophages express relatively large amounts of gp91^{phox}, which is referred to as NOX2 in the nomenclature of the NOX protein family. Ongoing studies of the tissue distribution of NOX protein family members have revealed detectable amounts of both gp91^{phox} message and protein in a wide variety of cells, although in many cases the physiologic significance of the findings awaits further elucidation. The integral membrane protein p22^{phox} is encoded by the *CYBA* gene on chromosome 16q24 and associates with gp91^{phox} in 1:1 stoichiometry to create functional flavocytochrome b_{558}. During the biosynthesis of the individual subunits of flavocytochrome b_{558} in the endoplasmic reticulum (ER), the heme-containing 65-kDa precursor of gp91^{phox} associates with p22^{phox} to form the nascent heterodimer (Figure 15.1). In the absence of either subunit, as occurs in some types of CGD, the lone subunit undergoes degradation, mediated in part in the proteasome. Furthermore, heme acquisition in the ER is a prerequisite for maturation of the heterodimer in the secretory protein pathway and proper targeting to membranes at the neutrophil surface or in intracellular vesicles. In the absence of heme synthesis, heterodimers do not associate in the ER and flavocytochrome b_{558} biosynthesis is aborted, demonstrating the critical contribution of heme incorporation to the integrity and stability of the heterodimer.

In the absence of a solved crystal structure, the topography of flavocytochrome b_{558} and some of its structural features have been derived from sequence analysis, mapping by monoclonal antibodies, and the application of a host of analytical techniques (Figure 15.2). Data suggest that gp91^{phox} possesses six transmembrane helices and an extended C-terminus of ~300 amino acids in the cytoplasm. Several features of gp91^{phox} that are shared by all members of the NOX protein family are suited to its role as an electron transferase, shuttling electrons across the membrane from cytoplasmic NADPH to molecular oxygen. The C-terminal cytoplasmic tail of gp91^{phox} possesses binding sites for both NADPH and FAD and the transmembrane α-helices contain two nonequivalent heme groups, each ligated to histidine residues in parallel helices and stacked atop each other perpendicular to the plane of the membrane. This arrangement is reminiscent of that suggested for the iron reductase of *Saccharomyces cerevisiae*, where the individual ligating histidines appear 12–13 residues apart in the linear sequence. Direct data on neutrophils indicate that the midpoint potentials for the inner and outer heme groups are –225 and –265 mV, respectively, thus providing a favorable pathway for sequential electron movement from NADPH to FAD to the hemes and then to molecular oxygen, with the ultimate production of superoxide anion.

Whereas gp91^{phox} serves as the catalytic subunit of flavocytochrome b_{558}, supporting electron movement from NADPH to oxygen, p22^{phox} has not been implicated

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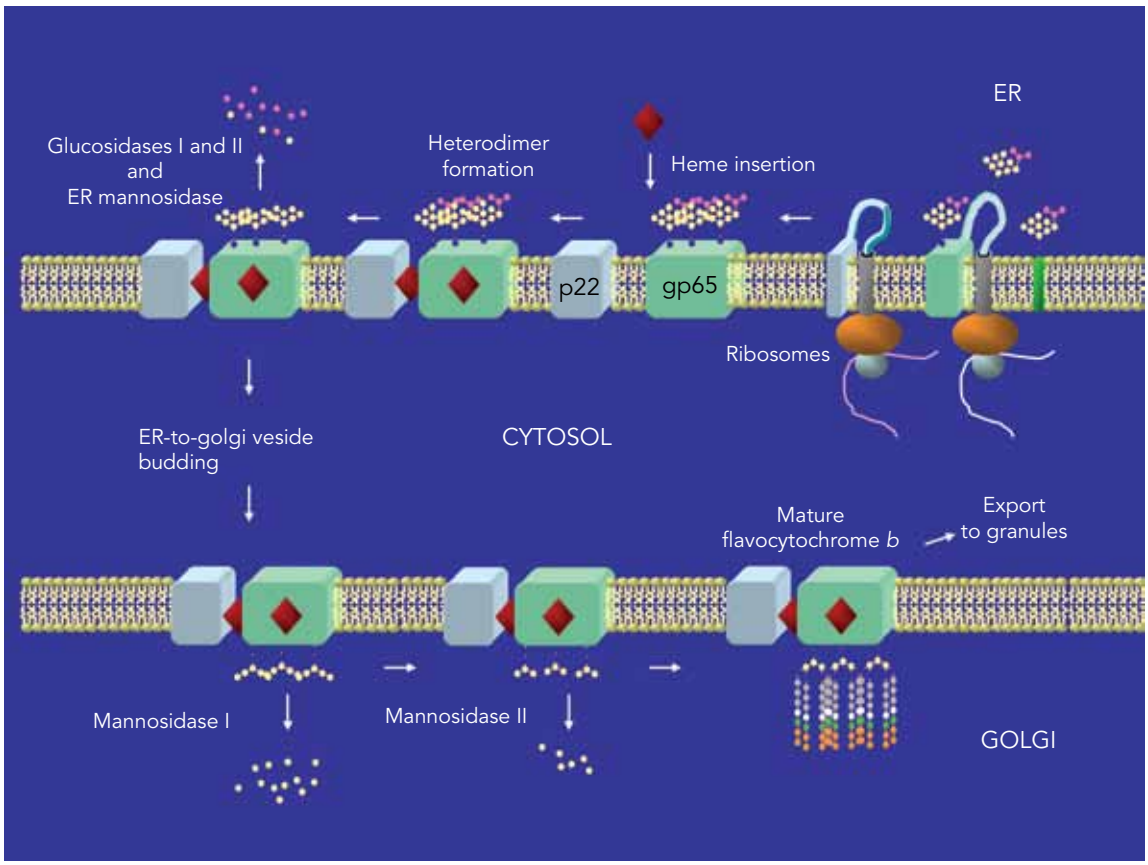


Figure 15.1. Biosynthesis of flavocytochrome b_{558}. Both subunits of flavocytochrome b_{558} are synthesized in the ER, with the primary translation product for gp91^{phox} undergoing cotranslational glycosylation to generate a 65-kDa precursor, gp65. The nascent subunits associate and acquire the heme groups in the ER, with any unassociated subunits undergoing degradation, in part mediated by the proteasome. The heme-containing gp65-p22^{phox} heterodimer has posttranslational modification of carbohydrates by ER glucosidases and mannosidases before exiting the ER and entering the Golgi, where significant modification of the oligosaccharides occurs. The net result of extensive glycosylation is the appearance of gp91^{phox}. The mature heterodimer exists the Golgi and transferred into the membranes of the secretory vesicles, secondary granules, and cell surface.

in directly contributing to enzymatic activity of the heterodimer. Rather, p22^{phox} is essential for the stability of gp91^{phox} during its biosynthesis and thus to the structural and functional integrity of flavocytochrome b_{558}; patients with absent or abnormal forms of p22^{phox} have the biochemical and clinical phenotype of CGD and suffer the same frequent and life-threatening infectious complications as do patients with CGD secondary to a lack of normal gp91^{phox}. In addition to the association with gp91^{phox} in the membrane, p22^{phox} supports interactions with cytosolic oxidase components that are essential to assembly of a functional oxidase (*vide infra*). It is noteworthy that the stabilizing effect of p22^{phox} on gp91^{phox} is not a feature shared by all NOX protein family members; neither NOX5 nor the two Duox proteins is associated with p22^{phox}. As discussed later, these same NOX isoforms also function independently of cytosolic cofactors, thus highlighting the specific contribution p22^{phox} makes to the infrastructure-supporting oxidase assembly.

CYTOPLASMIC COMPONENTS OF THE PHAGOCYTE OXIDASE

In addition to the membrane-associated flavocytochrome b_{558}, activity of the phagocyte NADPH oxidase requires three cytosolic proteins, p47^{phox}, p67^{phox}, and either Rac1 or Rac2 from the Rho family of low-molecular-weight GTPases. The absence of any of these proteins from phagocytes results in the biochemical and clinical phenotype of CGD. In addition to these three proteins, other cytosolic proteins have been implicated in modulating activity of the phagocyte oxidase. The essential proteins exist in the cytoplasm as multicomponent complexes; p47^{phox}, p67^{phox}, and p40^{phox} are associated in one cytosolic complex, and Rac is coupled with RhoGDI, the GDP dissociation inhibitor for Rho, in the other.

Serving as an adaptor protein that organizes assembly of the multiple oxidase components at the membrane, p47^{phox} contains several domains that have

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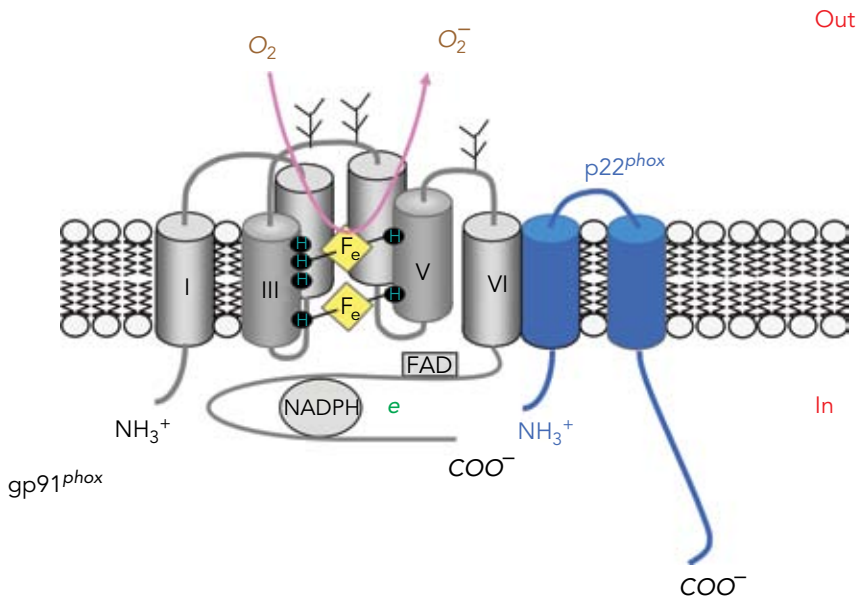


Figure 15.2 Electron transfer via flavocytochrome b_{558}. Substantial complementary data support a model for gp91^{phox} with six transmembrane α-helices, with both a short N-terminus and extended C-terminus extending into the cytoplasm. The cytoplasmic C-terminus possesses binding sites for NADPH and FAD. Stacked in the membrane, coupled in bis-histidine linkages between helices III and V, are two nonequivalent heme groups. When the oxidase is assembled and activated, electrons shuttle sequentially from NADPH to FAD, across the two hemes in the membrane, to reduce molecular oxygen to superoxide anion.

been implicated in mediating intermolecular interactions (Figure 15.3). Near the N-terminus of p47^{phox} is a PX domain, originally identified in Phox proteins and demonstrated to associate with specific phosphoinositides in or at the membrane. In addition, p47^{phox} possesses tandem Src homology 3 (SH3) domains and a proline-rich region (PRR), motifs known to support protein–protein interactions. However, the conformation of p47^{phox} in the cytoplasm of unstimulated phagocytes renders these domains cryptic and inaccessible for interactions with their potential binding partners. In addition to the protein domains that support intermolecular associations, p47^{phox} has a region rich in arginine and in serines that are susceptible to phosphorylation. Referred to as an autoinhibitory region, this cationic region of p47^{phox} includes 8–10 sites for agonist-dependent phosphorylation. During phagocyte stimulation, several kinases are activated and mediate phosphorylation of p47^{phox} in the autoinhibitory region, thereby releasing the conformational constraints on its potentially interactive domains. As a consequence of these posttranslational modifications in p47^{phox}, its SH3 domains become accessible to associate with the PRR in p22^{phox} and its PX motif can interact with targets in the stimulated plasma or phagosomal membrane. The translocation of p47^{phox} to the membrane is required to promote the movement of p67^{phox} and assembly of a functioning oxidase; in neutrophils of patients with CGD secondary to the absence of p47^{phox}, p67^{phox} remains in the cytoplasm after exposure to agonists. Thus, p47^{phox} contributes to oxidase activation in its role as an organizing element, providing a platform that is essential for the orderly translocation of p67^{phox} to the membrane and the stable association there with flavocytochrome b_{558}.

Whereas p47^{phox} provides critical organizational input into phagocyte oxidase assembly and action, p67^{phox} exhibits intrinsic catalytic activity, regulating the reduction of FAD by NADPH and interacting with Rac. In fact, in an experimental cell-free system using flavocytochrome b_{558}–enriched neutrophil membranes and high concentrations of recombinant p67^{phox}, superoxide anion can be generated in a Rac-dependent fashion in the absence of p47^{phox}. The catalytic activity of p67^{phox} depends as well on interactions with Rac, with an effector domain within the latter demonstrated to associate directly with p67^{phox}. In cytoplasm of resting phagocytes, isoprenylated Rac is in its GDP-bound inactive state, associated with RhoGDI. However, stimulation results in phosphorylation of RhoGDI, generation of the GTP bound form of Rac, and translocation of Rac-GTP to the membrane (Figure 15.4). The latter translocation is independent of concurrent translocation of the p47^{phox}–p67^{phox}–p40^{phox} complex, although

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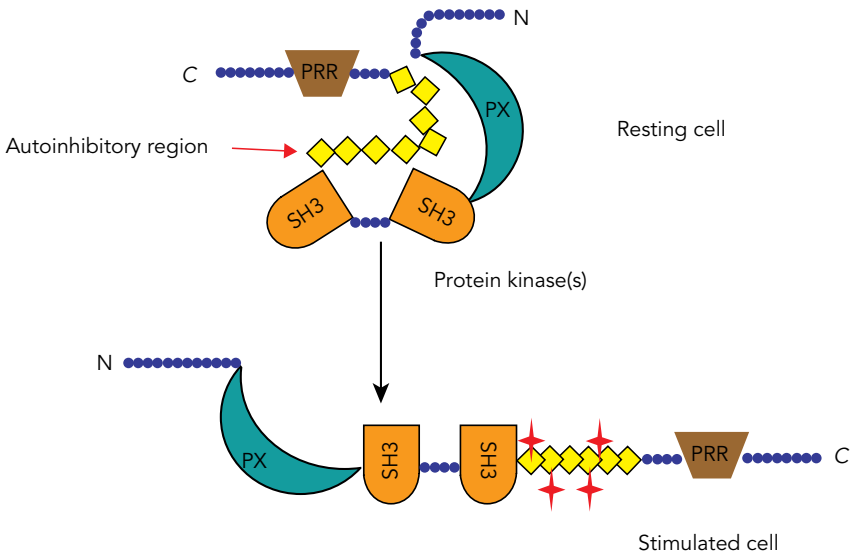


Figure 15.3. Agonist-triggered conformational changes in p47^{phox}. In the cytoplasm of resting phagocytes, p47^{phox} exists in a conformational state wherein domains that possess the capacity to support protein–protein and protein–lipid interactions are cryptic and inaccessible. There is a PX domain, which binds to specific phosphoinositides; tandem Src homology 3 (SH3) domains, which interact with proline-rich regions (PRR); and a C-terminal PRR. Very importantly, there is a cationic region between the second SH3 domain and the PRR known as the autoinhibitory region that is rich in arginine and serine residues. Stimulation of phagocytes activates several protein kinases that phosphorylate 8–10 serines in the autoinhibitory region. Consequently, there is a conformational change in p47^{phox}, rendering its interactive domains available for binding. SH3 domains interact with a PRR in p22^{phox}, PX domain binds phosphoinositides on the inner leaflet of the phagocyte membrane, and other interactions between cytosolic elements and flavocytochrome b_{558} stabilize the assembled complex.

evidence supports interactions of the membrane-bound Rac-GTP with translocated p67^{phox} as well as with flavocytochrome b_{558}. The efficient redistribution of the two cytosolic complexes upon phagocyte stimulation and the stable association of oxidase complexes to support a functional oxidase require multiple protein–protein and protein–lipid interactions, the precise details of which are still being elucidated.

In the most reductionist view, phagocyte oxidase activity can be fully reconstituted in vitro with lipidated flavocytochrome b_{558}, p47^{phox}, p67^{phox}, and Rac-GTP. However, additional proteins and lipids likely contribute to modulation or regulation of phagocyte oxidase activity in vivo. The lipid environment of flavocytochrome b_{558} influences its conformation and activity. Furthermore, the phospholipid composition of the membranes at which oxidase assembly occurs contributes to overall activity of the enzyme complex. Like p47^{phox}, p40^{phox} possesses a PX domain but may recognize phosphoinositide species that differ or overlap with those to which p47^{phox} binds. The PX domain of p40^{phox} binds phosphatidylinositol 3-phosphate [$PI_{(3)}P$], a species generated on the cytoplasm-oriented leaflet of nascent phagosome membranes. Experimental evidence indicates that robust oxidase activation during phagocytosis requires $PI_{(3)}P$-dependent binding of p40^{phox} at the phagosome, thereby providing another level of spatial control of the phagocyte oxidase. Several reports of flavocytochrome b_{558} purification have noted the co-isolation of Rap1A, a low-molecular-weight G protein, although a functional link with the phagocyte oxidase remains to be further defined. Of note, neutrophils from mice lacking Rap1A exhibit slightly reduced oxidase activity. There is recent evidence as well that the abundant cytosolic calcium-binding proteins MRP8 and MRP14 associate with flavocytochrome b_{558} in neutrophils and modulate oxidase activity in intact cells.

OXIDANT GENERATION WITHIN THE PHAGOSOME

The clinical consequences of CGD dramatically highlight the important contribution of oxidant-dependent antimicrobial activity to human defense against infectious agents. Discussions of neutrophil antimicrobial action often juxtapose such activity with microbial killing that occurs in the absence of O_2, the so-called O_2-independent system. The granule proteins delivered

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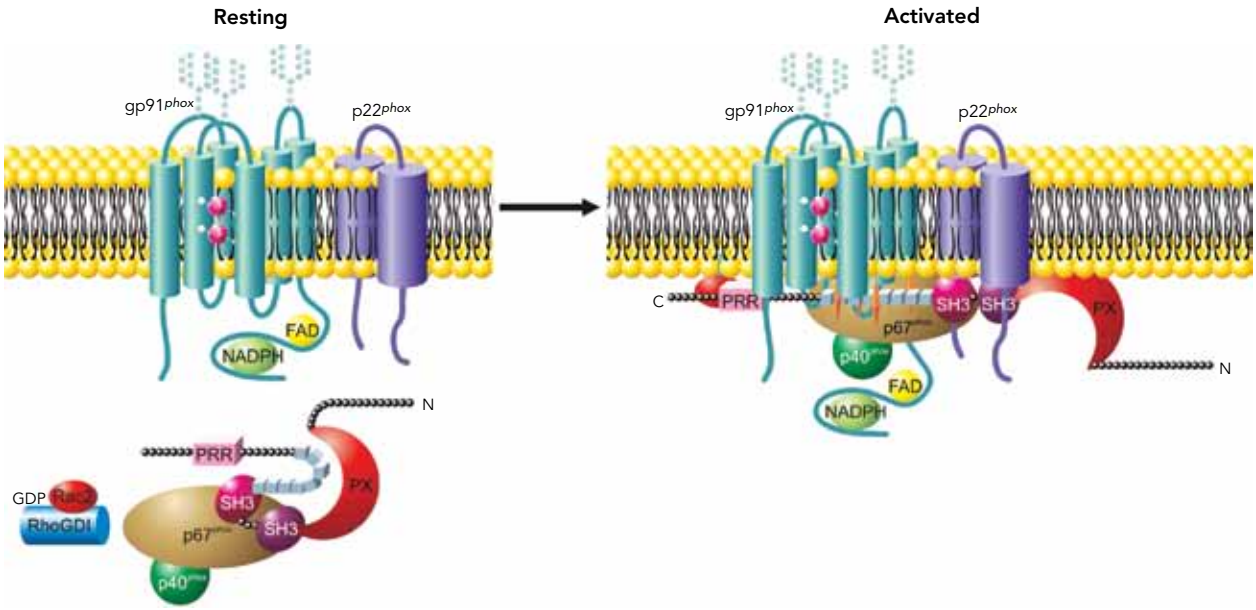


Figure 15.4. Assembly of the phagocyte NADPH oxidase. In resting phagocytes, the NADPH oxidase is unassembled and inactive. Upon stimulation, phosphorylation of p47^{phox} and RhoGDI relaxes conformational constraints in the complexes. The phosphorylation of p47^{phox} permits the p47^{phox}–p67^{phox} complex to associate with the membrane and with flavocytochrome b_{558} (Figure 15.3). Conversion of Rac-GDP to Rac-GTP permits the isoprenylated protein to translocate to the membrane, independently of the p47^{phox}–p67^{phox} complex, where interactions with gp91^{phox}, p67^{phox}, or both occur.

to attack the ingested microbe by granule–phagosome fusion include a variety of proteases with a broad array of substrate specificities that react with exposed or secreted microbial products. Some granule proteins are not enzymes but exert direct antimicrobial activity. For example, cationic antimicrobial peptides present in azurophilic granules of neutrophils, such as α-defensins or bactericidal-permeability increasing protein (BPI), lack protease activity but disrupt the integrity of the microbial surface upon binding and thereby compromise the viability of the organism. The intrinsic activity of both of these antimicrobial agents is independent of the presence of O_2, and their initial engagement with the target is at the microbial surface, dictated, in part, by electrostatic interactions between the cationic antimicrobial peptide and the anionic outer structures on the organism. Consequently, any posttranslational events that change charge or other intrinsic physical properties of the microbial surface will influence subsequent associations with such granule antimicrobial proteins in the phagosome. Such considerations are important to include in a discussion of phagocytic antimicrobial action, as the conventional classification of events as "oxygen-dependent" versus "oxygen-independent" underestimates the complex synergy that occurs in the physiologic setting of an infectious challenge.

As detailed earlier, the assembled oxidase serves as an electron transferase, sequentially moving electrons from cytoplasmic NADPH, across the membrane (either plasma or phagosomal), to molecular oxygen, and thereby generating O_2^- as the proximal product of oxidase activity. Although the magnitude of oxidase activity parallels the intensity of the stimulation, the phagocytosis-triggered oxidase generally generates O_2^- within the phagosome, with very little detected extracellularly unless the number of particles in the challenge is high or closure of the phagosome is incomplete. The turnover number of the phagocyte oxidase is ~150–160 electrons/heme/s, and a soluble agonist such as phorbol myristate acetate can stimulate extracellular generation of 7–10 nmol O_2^-/min/10^6 neutrophils.

As the proximal ROS generated by the phagocyte oxidase, O_2^- has several potential fates. Within the phagosome, O_2^- can undergo protonation to yield the perhydroxy radical OH_2, although the pK_a for dissociation is 4.8. Furthermore, O_2^- can serve either to donate or to accept electrons and thus can react with itself to generate O_2 and H_2O_2. The latter dismutation reaction can occur spontaneously, optimally when the concentrations of the two reactive species are identical (i.e., pH 4.8). In neutrophils, the pH in the phagosome reaches 6–6.5, which would seem to limit spontaneous dismutation of O_2^-. However, the influx of protons during oxidase activation, an obligate response to mediate charge compensation for the shuttling of electrons into the phagosome, likely accounts for conversion of O_2^- to H_2O_2. Catalytic promotion of H_2O_2 generation by superoxide dismutase (SOD) may occur when ingested microbes release their SOD into the phagosome, but

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no experimental data suggest that host-derived SOD contributes in this chemistry. Nearly all of the O_2 consumed by stimulated neutrophils can be recovered as H_2O_2.

Reactive nitrogen species such as nitric oxide (NO) likewise participate in inflammation and in host defense against microbes. Produced from arginine by nitric oxide synthase (NOS), NO has been linked to a wide variety of immune activities, including cell signaling (both within and between cells), tumor killing, and antimicrobial action. Within the context of intracellular events in phagocytes, NO has been most extensively characterized in murine systems. In contrast to the general recognition that NO contributes to murine host defense and participates in both acute and chronic inflammatory states, there is considerable controversy as to the capacity of human phagocytes to produce enzymatically NO in vivo. In the case of monocyte-macrophage production of NO, the story is relatively complicated. In vitro, production of NO by isolated monocytes or monocyte-derived macrophages has not been demonstrated after exposure to a wide array of potential agonists and cytokines. However, macrophages recovered from patients with mycobacterial infection exhibit the capacity to kill mycobacteria in a fashion that is ablated by inhibition of NOS, suggesting that mycobactericidal activity of the macrophages is NO-mediated. It is possible that the failure of the in vitro studies to mirror what is implied in vivo reflects shortcomings in our understanding of human macrophage differentiation in vitro rather than the absence of biologically relevant NOS activity. Elucidation of the role of NO in human macrophage-mediated antimicrobial activity will be anticipated to allow insights into cell models of human macrophage differentiation.

In the case of human neutrophils, evidence supporting the presence of significant NOS activity is likewise controversial. Regardless of the source of NO, there is general agreement that NO participates in influencing the behavior of neutrophils and thus contributes to the biological phenotype during inflammation. NO has been implicated in modulating neutrophil recruitment to sites of inflammation and NO from endothelial cells can interact with neutrophil-derived superoxide anion to generate peroxynitrite, a reactive species with a broad range of biological targets. In this light, one can ascribe to NO a role in influencing neutrophil-dependent antimicrobial activity, even if the inputs are indirect.

Since phagosomes contain large amounts and a wide array of potential substrates derived from both ingested microbes as well as released granule contents, it is impossible to provide a comprehensive list of all the reactive species generated. The complexity of the biochemistry in the phagosome is magnified by recognizing that the microbial contribution will vary as a function of the specific identity as well as the growth phase of the ingested organism, and that the individual phagosome itself is very heterogeneous. Oxidant production is localized to the site on the membrane where the oxidase has assembled, thus providing a point source for oxidant dispersal, the extent of which will reflect the intrinsic diffusion capacity of the specific ROS as well as the probability of its collision with suitable substrates in its environment. Granules likewise fuse at discrete sites on the phagosome and release contents locally. Some granule proteins bind avidly to targets, thus delivering antimicrobial action in close proximity to the potential targets, whereas others may act in solution. Collectively, these factors magnify the inherent challenge of precisely defining the identity of products in a given phagosome or assigning essential antimicrobial activity to any specific constituent.

The direct toxic action of H_2O_2 on microbes is relatively modest at concentrations generated in vivo but is potentiated as much as 100-fold, depending on the organism, in the presence of the azurophilic granule protein myeloperoxidase (MPO). H_2O_2 reacts with the ferric iron in MPO to generate Compound I, a complex in which the heme group of MPO forms a double bond with O_2 and serves as the catalytically active form of MPO (Figure 15.5). Compound I catalyzes a two-electron oxidation of halide ions to produce hypohalous acids. O_2^- can also react directly with MPO and this interaction has been implicated in maintaining optimal activity of the enzyme in the phagosomal environment. In the case of the reaction of Compound I with chloride ion, the predominant halide in the phagosome, Cl^- is oxidized to Cl^+ to yield hypochlorous acid (HOCl), or bleach. Depending on the agonist used for activation and the analytical techniques employed to assess reaction products, ~11%–40% of the O_2 consumed by stimulated human neutrophils can be recovered as a chlorinating species such as HOCl, and stimulated neutrophils can produce ~50 nmoles HOCl/30 minutes/10^6 cells. HOCl is a very potent oxidant and although it can dissociate in a pH-dependent fashion to hypochlorite or generate Cl_2, much of the HOCl generated in the phagosome attacks any potentially oxidizable group to which it has access. Candidate substrates include the sulfur-containing amino acids cysteine and methionine, amino side-groups on all amino acids, carbohydrates, lipids, phospholipids, and nucleic acids. In some cases, the resulting change in the microbe will directly compromise the viability of the organism, whereas in other situations the induced chemical modifications will alter susceptibility to other reactive species in the phagosome. Vulnerable targets are not restricted to the ingested microbe but include as

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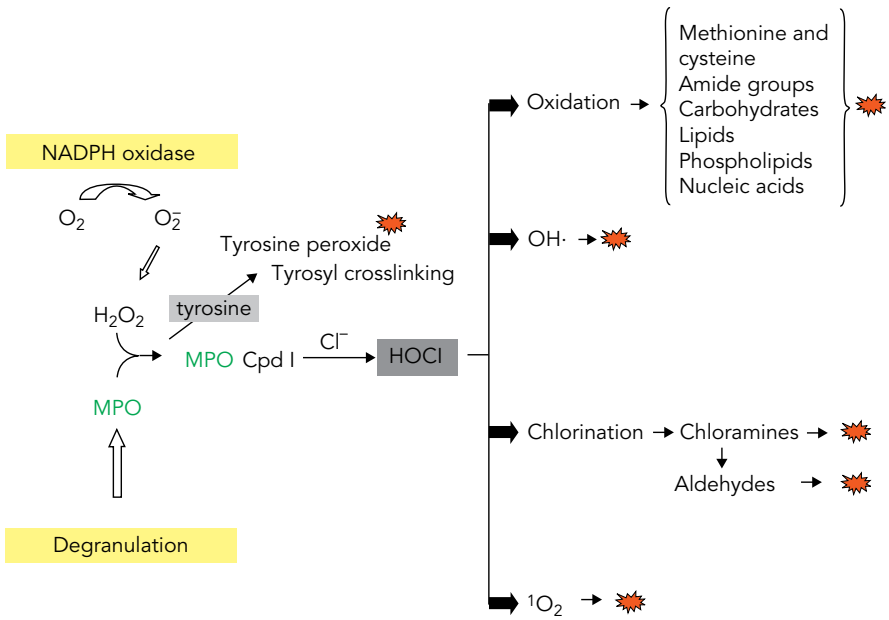


Figure 15.5. Potential MPO-based reactions in the phagosome. Concurrent activation of the NADPH oxidase and release of azurophilic granules into the phagosome delivers reactive oxygen species and granule products, respectively, in close proximity of the ingested microbe. Among the many interactions that occur among oxidants and granule proteins to produce an environment inhospitable to microbes are those dependent directly or indirectly on MPO. MPO and H_2O_2 combine to form Compound I that in turns supports the two-electron oxidation of Cl^- to Cl^+ and thereby generate HOCl, or bleach. MPO and H_2O_2 also directly oxidize tyrosine to generate tyrosyl radicals, a substrate for production of tyrosine peroxide as well as tyrosyl cross-linking of free tyrosines or tyrosyl residues in proteins. HOCl is a very potent oxidant and directly attacks amino acids, especially methionine and cysteine, as well as amide sidechains on peptides and proteins, and other biological substrates. HOCl also supports generation of other reactive species, including hydroxyl radical ($OH^\bullet$), singlet oxygen ($^1O_2$), and chloramines. The latter products include those that are relatively long lived and exert significant antimicrobial activity. Some chloramines decay and release aldehydes, some of which are antimicrobial in their own right. Many of these various derivatives of MPO-dependent HOCl generation mediate antimicrobial activity (indicated by the red starburst).

well host proteins delivered to the phagosome during degranulation.

The HOCl generated by the MPO-H_2O_2-Cl system promotes several other chemical reactions in addition to mediating the direct oxidation of susceptible targets. The downstream products from reactive HOCl create a host of long-lived agents by attacking nitrogen and generating chloramines. Some chloramines retain oxidizing capacity and thus extend the antimicrobial influence of MPO-derived products in the phagosome, whereas others decompose to release aldehydes, some of which possess intrinsic biological activity. The MPO-H_2O_2 system in the absence of chloride can also directly oxidize the aromatic amino acid tyrosine to produce tyrosyl radicals, which in turn promote dityrosine linkages between proteins. Furthermore, tyrosyl radicals and O_2^- can interact to generate tyrosine peroxide, as has been directly demonstrated for stimulated neutrophils.

The MPO-H_2O_2-Cl system supports modification of ROS initially produced by the phagocyte NADPH oxidase. For example, stimulated neutrophils generate low levels of hydroxyl radical ($OH^\bullet$) in an MPO-dependent fashion, via a reaction between HOCl and O_2^-. $OH^\bullet$ reacts readily and rapidly and, in the presence of CO_2, can generate $HCO_3^\bullet$, a radical which would thus be present under physiologic conditions and has demonstrated antimicrobial capacity. In addition, MPO-generated HOCl can interact with H_2O_2 to produce singlet oxygen ($^1O_2$). In both cases, very sensitive analytical systems are required to detect $OH^\bullet$ and $^1O_2$, raising questions about their contribution to the overall toxic environment in the phagosome and physiologic relevance to antimicrobial action. Nonetheless, the species are highly reactive and can be detected in biological settings.

In addition to MPO, other granule proteins exhibit synergistic activity with each other and with oxidants.

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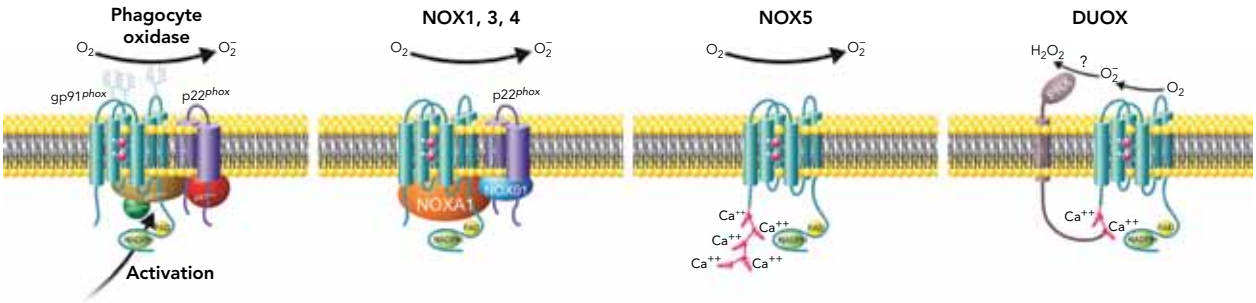


Figure 15.6. Members of the NOX family of NADPH oxidases. The NOX family of NADPH oxidases includes seven proteins (NOX1–5, DUOX1, and DUOX2) that are identified by sequence homology with gp91^{phox} (aka NOX2) and share dependence on NADPH and FAD for activity. Distributed broadly in a nearly all tissues and cell types in both plant and animal kingdoms, the nonphagocyte oxidases serve diverse functions. The organization of the nonphagocyte oxidases differs in several ways from that of the phagocyte oxidase. Only NOX1, 3, and 4 pair with p22^{phox} in the membrane. NOXO1 and NOXA1, homologues of p47^{phox} and p67^{phox} respectively, interact with NOX1, 3, and 4, whereas NOX5 and the DUOX proteins function independently of cytosolic protein cofactors. In contrast to the agonist-dependent assembly and activation seen in the phagocyte oxidase, the nonphagocyte systems function constitutively, with modulation by exposure to various agonists. Both NOX5 and the DUOX proteins possess EF-hands in their cytoplasmic domains and are regulated by changes in intracellular calcium.

The antimicrobial activities of elastase or cathepsin G are potentiated by the MPO-H_2O_2-chloride system by a mechanism that is dependent on the cationic nature of the granule proteins; neither requires enzymatic activity to kill target bacteria. The same synergy probably occurs as well with the other granule antimicrobial proteins, including the defensins and BPI. However, it is important to recognize as well that many of the granule proteins exert potent and effective antimicrobial action even in the absence of a functioning oxidase. Neutrophils from patients with CGD, cells that lack the capacity to generate any ROS, can kill *Escherichia coli*, *Pseudomonas aeruginosa*, and many catalase-negative streptococci normally, thus demonstrating *bona fide* effective antimicrobial action mediated exclusively by the granule proteins.

Given the quantity and expansive functional capacity of granule proteins released into the phagosome, the varied and highly reactive oxidants generated by the phagocyte NADPH oxidase in situ, and the dynamic ways in which these constituents of the phagosome synergize, the phagocyte provides comprehensive and at times redundant biochemical agents to kill and degrade ingested microbes. In addition to supporting antimicrobial activity, neutrophil-derived oxidants contribute to proinflammatory events outside the phagocyte when produced in excess, at inopportune times, or in inappropriate tissue sites. Granule proteins such as MPO and proteinase 3 have been linked to systemic vasculitides and MPO-mediated oxidation of lipoproteins, glycerophospholipids, proteins, and amino acids has been associated with a wide variety of inflammatory diseases. In plasma at physiologic pH, released MPO and H_2O_2 can support the production of HOBr and HOSCN in addition to HOCl, each of which can target the full range of biological molecules. The relationships of MPO and HOCl in modifying lipoproteins and thereby contributing to the development and progression of atherosclerosis are currently being widely studied. MPO can catalyze nitration of targets, most notably tyrosine, either free or as tyrosyl residues in proteins. Although nitration does not occur within the phagosome, MPO-dependent nitration of extracellular targets likely contributes to nitric oxide–mediated tissue injury in inflammation. Thus, much of the potent biochemistry that serves to target microbes captured in the phagosome can induce significant tissue damage as part of the inflammatory response.

NADPH OXIDASE BEYOND PHAGOCYTES

As mentioned earlier, it has been recently appreciated that the phagocyte NADPH oxidase is a member of the NOX protein family, a group defined by homology with gp91^{phox}, aka NOX2 (Figure 15.6). Several excellent reviews of the NOX protein family describe the incredibly wide biological contexts in which they operate, extending through both the plant and animal kingdoms. Although gp91^{phox} serves as the organizing principle for the family, several distinctive features of the nonphagocyte members merit comment.

First, the critical dependence of gp91^{phox} on cofactors is not shared by all NOX proteins. Only NOX 1, 3, and 4 resemble gp91^{phox} in requiring association with p22^{phox} for stability, activity, and proper targeting into membranes. Homologues of p47^{phox} and p67^{phox}, designated as NOXO1 and NOXA1 respectively, serve as cofactors for NOX1 and 3, whereas NOX4, NOX5, and the DUOXes do not depend on cytosolic protein cofactors. Instead, NOX5 and the DUOXes, all EF-hand containing proteins, are regulated by cytosolic calcium and may represent very ancient members of the protein

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family. The extent to which these differences with respect to supporting cofactors provides mechanistic insights into how $p22^{phox}$ precisely influences $gp91^{phox}$ activity or into the intricacies of cytosolic factor modulation of oxidase activity awaits further study. Second, ROS production in the phagocyte system is relatively robust, suitable for damaging microbial targets. In their native setting, the nonphagocyte NOX proteins generally generate lower levels of ROS. With respect to NOX1, 3, and 4, this lower activity seems better suited to mediating signal transduction than rendering irreversible modifications in targets. Third, whereas the phagocyte system is unassembled and inactive in unstimulated cells, some of the NOX proteins are active constitutively but subject to modulation by various factors. For NOX proteins that interact with NOXO1, the constitutive activity may reflect, in part, the absence of an autoinhibitory region in NOXO1. As a consequence, NOXO1 is likely always in an open conformation, supporting continuous interaction with complementary domains at the membrane. Last, studies of the DUOXes detect only H_2O_2 as the proximal product of its activation; no O_2^- is recovered. The biochemical basis for this activity remains undefined, but its eventual elucidation will likely reveal novel biochemistry, either at the level of FAD-supported electron transfer or in the context of novel superoxide dismutase activity in DUOX itself.

Definition and characterization of similarities and contrasts among the NOX proteins with respect to the phagocyte NADPH oxidase promises to expand understanding of the role of ROS in inflammation in its myriad manifestations.

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16 Cell Adhesion Molecules

Lucy V. Norling, Giovanna Leoni, Dianne Cooper, and Mauro Perretti

BIOLOGICAL FUNCTIONS OF ADHESION MOLECULES

Cell adhesion molecules are endowed with many biological functions, often crucial to life. For instance, in gut or lung epithelia they assure cell-to-cell contacts, thus making and maintaining barrier functions, while allowing passage of nutrients and other solutes. Besides maintaining body form and structure, cell adhesion molecules play important roles in tissue development as well as helping to determine its final organization. A common feature of adhesion molecules is their ability to sustain strong bonds between adjacent cells on either a permanent or intermittent basis.

Apart from these *static functions* of cell adhesion molecules, there are many *mobile functions*, whereby they are able to induce intermittent cell-to-cell bonds hence assuring particular cell movements either in the blood vessel (as during inflammation; topic to be investigated in this chapter) or within tissues and organs.

It is therefore evident that, contrary to previous simplistic visions and predictions, cells *are never free floating* but rather they move on surfaces such as the inflamed joint or gut subepithelial strata, penetrate into injured organs or circuit the body (think of T cells traveling the blood lymphatic circulation, and so forth) by activating processes **all** relating to cell adhesion molecule-mediated phenomena.

ADHESION MOLECULES IN INFLAMMATION

Multistep Leukocyte Adhesion Cascade

Leukocyte recruitment in both homeostatic and inflammatory situations follows a generic paradigm characterized by a sequential cascade of interactions between leukocytes and endothelial cells (ECs). Insight into the cellular and molecular processes involved in each step of the cascade have been provided by a range of experimental approaches performed both in vitro and in vivo. These include antibody inhibition studies, static adhesion assays, parallel-plate flow chamber models, as well as the use of intravital microscopy to visualize live interactions of leukocytes with the vessel wall.

Collectively, these studies have helped to elucidate that (i) initial leukocyte–endothelial interactions (capture and rolling) are instigated primarily by a family of molecules called selectins interacting with oligosaccharide residues expressed by counter-ligands, and (ii) firm adhesion and transmigration are mediated by leukocyte integrins interacting with the endothelial immunoglobulin superfamily of adhesion molecules (outlined in Figure 16.1 and described later in detail).

Leukocytes are a heterogeneous population of immune cells with specialized functions. The type of leukocyte recruited to an inflammatory site depends on the stimulus and hence the differential expression of adhesion molecules on the endothelium. In acute inflammatory episodes, the predominant leukocyte influx is neutrophils, with the remaining leukocytes consisting of monocytes or macrophages and, in allergic responses, eosinophils. Comparatively, in chronic inflammatory responses lymphocytes predominate, as well as monocytes and macrophages whereas neutrophils are present to a lesser extent.

Adhesion Molecules Promoting Cell Rolling: The Selectins

Selectins are the main initiators of leukocyte tethering and rolling, although this process can – in some cases – be facilitated by several integrins (especially when selectin deficiency occurs). The selectins are a three-member family (L-, P-, and E-selectin) that share a highly conserved extracellular N-terminal Ca^{2+}-dependent C-type lectin domain. All selectins contain a transmembrane domain and a cytoplasmic

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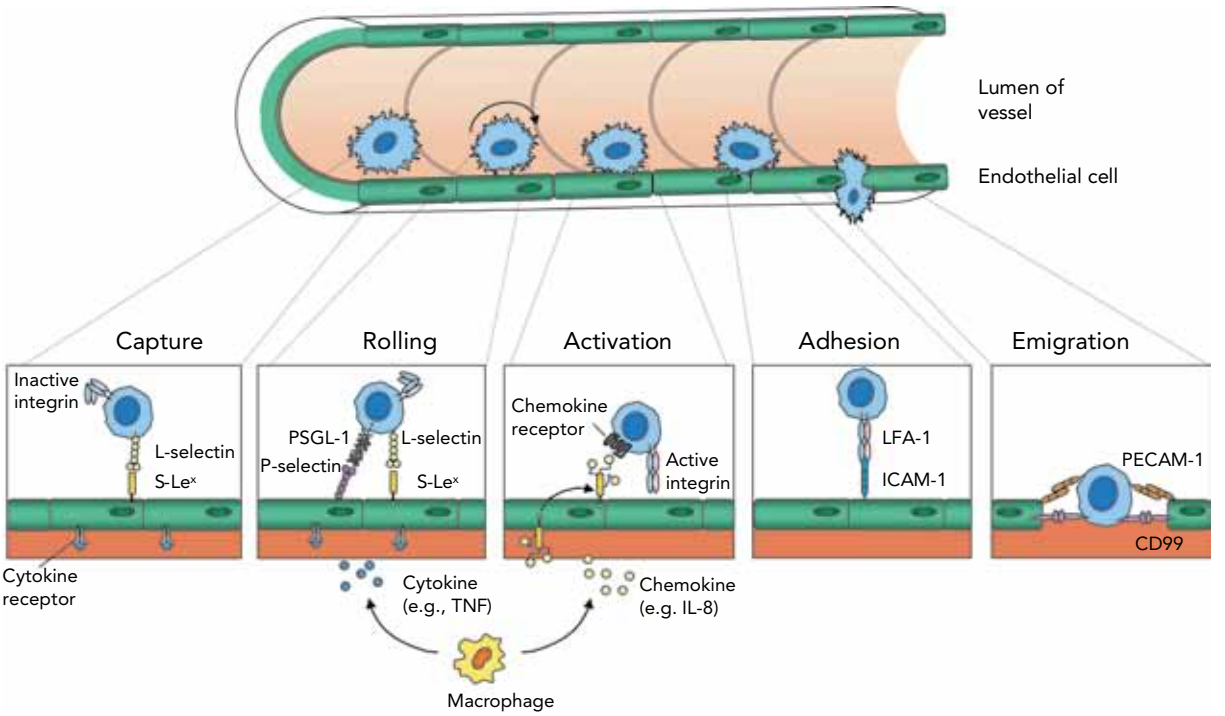


Figure 16.1. Leukocyte recruitment cascade. Following an inflammatory stimulus, tissue-resident macrophages release inflammatory mediators such as the cytokine TNF-α, which induces the rapid expression of preformed P-selectin (and transcription-dependent E-selectin expression) on the endothelium. The interaction between selectins and their glycoprotein ligands initiates leukocyte rolling. Activation by chemokines – and other leukocyte activators (e.g., leukotriene B_4 or platelet-activating factor) – presented on endothelial cells causes leukocyte integrin activation, thus resulting in transition from cell rolling to cell adhesion, in view of the strength of integrin-mediated binding with endothelial immunoglobulin superfamily members. Leukocytes can then transmigrate through the endothelial monolayer and chemotactically move toward the inflammatory stimulus. Examples of adhesion molecules involved in each step are depicted.

tail, the major structural difference between the family members being the number of consensus repeat units. All selectins bind, through their lectin domain, to sialylated forms of oligosaccharides displayed on various mucin-like glycoproteins, for example, sialyl lewis X (sLeX). One of the most extensively characterized selectin ligands is P-selectin glycoprotein ligand-1 (PSGL-1), a highly sialyated mucin found on all leukocytes and a ligand for all three selectins.

Selectin ligands need several carbohydrate modifications before leukocytes will bind; the enzymes involved in these posttranslational modifications include the glycosyltransferases, fucosyltransferases, and the sulfotransferases. Insights into the contributions of these enzymes on ligand structure and hence leukocyte rolling has been demonstrated using enzymatic treatments to cleave certain chains and more specifically in mutant mice with targeted deficiencies.

A rare congenital disease in humans, namely leukocyte adhesion deficiency II (LAD-II), underscores the importance of posttranslational modifications on selectin-ligand biosynthesis. These individuals have a defective Golgi GDP-fucose transporter, resulting in a hypofucosylation of glycoproteins. Patients exhibit persistent leukocytosis, defects in selectin-mediated leukocyte rolling and consequently suffer from recurring bacterial infections because blood-borne leukocytes fail to reach the site of infection/inflammation.

The intrinsic affinity of selectins for their ligands is rather low, and they display extremely rapid association and dissociation rates which assures a transient interaction of the leukocyte with the endothelium, insufficient to permit cell arrest. However, once initial bonds have formed on the endothelium, shear stress from the blood flow *pushes* the leukocyte forward and new bonds occur before the initial bonds break, manifesting in the rolling motion of the leukocyte. Selectins require shear stress to support this interaction, and without flow the cells detach. This counter-intuitive phenomenon is related to the catch bond characteristic of selectins, whereby the bond becomes stronger with applied force.

P-Selectin (CD62P) is preformed in cytoplasmic secretory α-granules of platelets and Weibel-Palade bodies of ECs. Stimulation with vasoactive mediators such as thrombin or histamine results in rapid mobilization of endothelial P-selectin to the external plasma membrane allowing leukocytes to interact. Thus, it is

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not surprising that P-selectin contributes to recruitment early in inflammation and infection. Expression at the cell surface is transient due to internalization by endocytosis for reuse. In addition to this rapid expression, P-selectin can be transcriptionally upregulated in human ECs by cytokines such as IL-4 whereas in murine ECs, TNF-α and LPS can induce its expression. Interactions between P-selectin on activated platelets and its main counterligand PSGL-1 on leukocytes results in leukocyte–platelet interactions that are pertinent for wound healing and contribute to leukocyte recruitment during inflammation.

E-selectin (CD62E) expression is mostly restricted to activated ECs, with the exception of skin microvessels and bone marrow where it is constitutively expressed. E-selectin requires de novo mRNA and protein synthesis and can be induced on cultured EC by cytokines such as IL-1β or TNF-α with maximal expression at 4 hour, reducing toward baseline levels by 24 hours. In comparison to P-selectin, the rate of E-selectin internalization is much slower and it is directed to lysosomes for degradation instead of being recycled.

L-selectin (CD62L) is exclusively and constitutively expressed on leukocytes, particularly neutrophils, monocytes, and naive lymphocytes (but not memory lymphocytes). It serves as a key receptor for lymphocyte homing to secondary lymphoid organs, and at peripheral sites of injury and inflammation. Its location on the tips of microvilli facilitates its interaction with EC, allows high receptor concentrations and also evades the electrostatic repulsion of the negative cell surface. Upon leukocyte activation, L-selectin is rapidly shed; although this mechanism is not fully elucidated, it is known that shedding can be blocked using inhibitors of zinc-based metalloproteases, and that a disintegrin and metalloprotease-17 (ADAM-17) is one of the enzymes responsible for this cleavage.

Essential physiological roles of the selectins have been reinforced by studies using gene-targeted mice and bone marrow–transplanted chimeric mice. In all cases, mice homozygous deficient in selectins are viable and fertile, and display only mild phenotypes suggesting that these molecules are functionally redundant. In sharp contrast, E/P-selectin double null mice exhibit severe defects in leukocyte rolling, with a complete abrogation of trauma-induced rolling in venules 10–120 minutes after exteriorization of the cremaster muscle. These double knockout mice, as well as the triple knockout E/P/L/-selectin triple null mice display an increased susceptibility to spontaneous mucocutaneous infections, which can be lethal.

Creation of mice lacking multiple selectins has helped show overlapping and unique features of the selectins. In addition, generation of quadruple adhesion molecule (E/P/L-selectin and ICAM-1) deficient mice resulted in viable, seemingly healthy mice in the absence of any challenge. Under inflammatory conditions, they exhibited a significantly impaired neutrophil recruitment, but mononuclear cell recruitment was barely affected, because rolling is also facilitated by α4-integrins on these cells. Treating these mice with a blocking monoclonal antibody for the α4-integrin concomitantly with the inflammatory stimulus TNF-α, completely eliminated rolling, and significantly reduced cell adhesion, although a small *residual* level of cell adhesion persisted suggesting again the involvement of additional recruitment mechanisms.

Two α4-integrins, α4β1 (very late antigen-4; VLA-4) and α4β7 (lymphocyte Peyer's patch HEV adhesion molecule-1; LPAM-1), mediate both lymphocyte rolling and firm adhesion, using the same ligand for both processes, notably VCAM-1 and mucosal addressin cell adhesion molecule-1 (MAdCAM-1, a vascular addressin expressed on HEV), respectively.

Adhesion Molecules in Cell Adhesion

Following the initial rolling process mediated by selectins and their ligands, activation of leukocytes coincides with a decrease in their rolling velocity and the transition to firm adhesion. Adherent leukocytes can become elongated and flatten onto the endothelium, decreasing their protrusion into the vessel lumen, and thus their tendency to detach.

Cell adhesion is established by the interaction of integrins with binding partners that belong to the immunoglobulin superfamily, referred to as cellular adhesion molecules (CAMs), including intercellular (I)-CAM-1 (CD54), vascular (V)-CAM-1 (CD106), and platelet-endothelial (PE)-CAM-1 (CD31). These immunoglobulins are expressed basally on the resting endothelium, and can be upregulated during inflammation, as is the case for ICAM-1 and VCAM-1, whilst PECAM-1 is redistributed to the site of leukocyte transmigration and is involved in homophilic PECAM-1 interactions with the leukocyte.

VCAM-1 binds to VLA-4/α4β1 expressed on lymphocytes and monocytes and at very low levels on neutrophils. Furthermore, VCAM-1 can contribute to leukocyte transendothelial migration (TEM) via the paracellular pathway (discussed later in more detail) by causing gap formation between adjacent endothelial cells.

Integrins are a family of type I transmembrane cell surface receptors, each consisting of a noncovalently associated α and β subunit. At present, 18 α-subunits and 8 β-subunits are known, yielding 24 distinct integrins. The overall shape and dimensions of integrins have been determined by electron microscopy, illustrating at least three distinct conformational states. The N-terminal regions of both subunits together form a globular structure that resembles a "head"

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(containing the ligand-binding site), connected to the two "legs." Essential for ligand binding is the metal-ion-dependent adhesion site, where divalent metals normally reside.

Unstimulated leukocytes are nonadherent, yet become adherent in response to an activating stimulus for example, chemokines, cytokines, or antigens. These stimuli activate pathways that act on the cytoplasmic portions of integrins altering their affinity for ligands; key to this process is a conformational change in the integrin structure. In addition, stimuli can cause clustering on the cell surface, altering integrin avidity. GPCR-triggered integrin activation occurs through an "inside-out" signaling process, whereas downstream signaling from ligand binding is referred to as "outside-in" signaling; either process can contribute to the stabilization of adhesion.

Integrins normally exist in a low-affinity conformation in which the leg region is bent, subsequently bringing the ligand-binding head-piece into close proximity to the cell membrane. Activation signals induce a switchblade-like extension of the leg that extends the structure, allowing an intermediate affinity conformation, and an opening of the head-piece resulting in a high-affinity state, and hence increasing the propensity for ligand binding. It is predicted that certain integrins only extend about 5 nm above the plasma membrane in a low-affinity conformation, and can protrude 20–25 nm in an extended higher-affinity form, therefore making them more likely to be topologically accessible to ligands.

Leukocyte integrins are expressed on the cell body rather than microvilli, where L-selectin is expressed. Integrins most relevant to leukocyte arrest include the β1 and β2 subfamilies of integrin receptors. All leukocytes express the β2 integrin (CD18), which pairs with distinct α partners (CD11) on different leukocyte subsets to form different glycoprotein complexes, namely, αLβ2 (LFA-1; CD11a), αMβ2 (Mac-1; CD11b), αXβ2 (p150, 95; CD11c), and αDβ2 (CD11d), binding partners of which are summarized in Table 16.1.

In vivo studies using either CD18-specific monoclonal antibodies or studies in CD18 null mice exhibit a decrease in the percentage of adherent leukocytes in inflamed postcapillary venules. The importance of β2 integrin–mediated adhesion is best exemplified in humans which lack a functional gene for the β2 integrin, referred to as LAD-I genetic deficiency. These individuals have severe infections, and develop spontaneous skin lesions; moreover, the life expectancy of the severely deficient patients is limited.

Adhesion Molecules in Cell Emigration

Following a period of stationary adhesion, a leukocyte may leave the postcapillary venule and pull itself into the subendothelial space. This event called leukocyte extravasation, TEM, or diapadesis is dependent on an array of cellular processes including adhesion molecule expression and activation, cytoskeletal reorganization, and alterations in membrane fluidity. Molecules such as junctional adhesion molecules (JAMs), PECAM-1 (CD31), ICAM-1 (CD54), ICAM-2 (CD102), VCAM-1 (CD106), CD99, and endothelial cell–selective adhesion molecule (ESAM) can be found at endothelial junctions and regulate leukocyte migration into underlying tissues, indeed inflamed endothelial cells reorganize these junctional proteins such that transmigration is favored (Table 16.1).

Two routes of leukocyte diapadesis have been noted so far: a *paracellular* route that dominates most extravasation processes, and a *transcellular* route (Figure 16.2). Recent evidence suggests that the route taken may be cell type specific with neutrophils preferring a paracellular route and lymphocytes a transcellular exit.

Endothelial cells are not merely passive participants in leukocyte transmigration, but are actively involved in this process. Leukocyte adhesion promotes remodeling of the apical endothelial plasma membrane into projections that surround the adherent leukocyte. These transmigratory cups or docking structures are enriched in ICAM-1 and VCAM-1 and may initiate migration by either a transcellular or paracellular route (Figure 16.2).

Transmigration via the paracellular route is a lot more frequent and therefore has been more extensively studied. During this process, leukocytes must cross endothelial junctions that contribute to the barrier function of the blood vessel, and which consist of transmembrane adhesion proteins connected to their intracellular partners. These are composed of two types of junctional structures, tight junctions, which include the claudins, occludins, and junctional adhesion molecules (JAMs) and adherens junctions, the main protein being vascular endothelial (VE)-cadherin. It is thought that ligation of molecules such as ICAM-1 may lead to activation of Rho which then results in opening of endothelial junctions.

Whereas capture, rolling and adhesion all involve heterophilic interactions of adhesion molecules, homophilic interactions between PECAM-1 and CD99 play a major role in transmigration via a paracellular route. Endothelial cell lateral junctions are enriched with these two proteins, which are also expressed on most leukocyte subtypes. Strong in vitro evidence for a role of PECAM-1 in transmigration of leukocytes was originally obtained using anti-PECAM-1 blocking antibodies. In vivo studies using intravital microscopy in PECAM-1 null mice highlighted an additional role for PECAM-1 in the migration of leukocytes through the perivascular basement membrane. PMNs were

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TABLE 16.1. Summary of adhesion molecules involved in leukocyte–endothelial cell interaction

Adhesion molecule	Distribution	Ligands and counter-receptors	Function
L-selectin (CD62L)	All leukocytes except effector and memory effector T cells	PNAd, MAdCAM-1, PSGL-1, E-selectin, P-selectin	Rolling
E-Selectin (CD62E)	Endothelial cells	PSGL-1, ESL-1, CD44*, CD43* (*CLA decorated)	Rolling
P-selectin (CD62P)	Endothelial cells, platelets	PSGL-1, PNAd	Rolling
Selectin ligands			
sLe^x	Myeloid cells, some memory T cells, HEVs	All selectins	Rolling
PSGL-1	All leukocytes	All selectins (essential for P-selectin)	Rolling
PNAd	HEV, some sites of chronic inflammation	L-selectin, P-selectin	Rolling
CLA	Skin-homing T cells, DCs, granulocytes	E-selectin	Rolling
Integrins			
$\alpha_L\beta_{2\ (LFA\text{-}1;\ CD11a/CD18)}$	All leukocytes	ICAM-1, ICAM-2, JAM-A	Adhesion, transmigration
$\alpha_M\beta_{2\ (MAC\text{-}1;\ CD11b/CD18)}$	Granulocytes, monocytes, some activated T cells	ICAM-1, fibrinogen, C3b, JAM-C	Adhesion, transmigration
$\alpha_x\beta_{2\ (p150,95;\ CD11c/CD18)}$	DCs	Fibrinogen, C3b	adhesion
$\alpha_D\beta_{2\ (CD11d/CD18)}$	Monocytes, macrophages, eosinophils	ICAM-1, VCAM-1	adhesion
$\alpha_4\beta_{1\ (VLA\text{-}4)}$	Most leukocytes	VCAM-1, fibrinogen, JAM-B	Rolling, Adhesion
$\alpha_4\beta_{7\ (LPAM\text{-}1)}$	Lymphocytes, NKCs, mast cells, monocytes	MAdCAM-1, fibronectin, VCAM-1	Rolling, Adhesion
Immunoglobulin superfamily			
ICAM-1 (CD54)	Most types of cells	LFA-1 Mac-1, fibrinogen	Adhesion, transmigration
ICAM-1 (CD102)	Endothelial cells, platelets	LFA-1 Mac-1	Adhesion, transmigration
VCAM-1 (CD106)	Endothelial cells	VLA-4, $\alpha_4\beta_7\,\alpha_D\beta_2$	Rolling, adhesion
MAdCAM-1	HEVs in PP and MLN	$\alpha_4\beta_{7,\ L\text{-}selectin}$	Rolling
PECAM-1	Endothelial cells, platelets, leukocytes	PECAM-1	Transmigration
JAM-A	Endothelial cells, epithelial cells, platelets, most leukocytes	JAM-A	Transmigration
JAM-B	Endothelial cells, HEVs	JAM-B, JAM-C	Transmigration
JAM-C	Endothelial cells, HEVs platelets, monocytes, DCs, some T cells	JAM-C, JAM-B	Transmigration
Miscellaneous			
CD99	Endothelial cells, leukocytes	CD99	Transmigration

trapped between the vascular endothelium and the underlying basement membrane, although the number of emigrated leukocytes was unaltered suggesting compensation by other adhesion molecules. Blocking antibodies demonstrated that CD99 is involved in a step in transmigration that is distal to PECAM-1, and blocking both of these adhesion molecules results in an additive inhibition of monocyte transmigration.

The JAM family of adhesion molecules, members of the immunoglobulin superfamily, are targeted to cell–cell contacts and are also implicated in transmigration (Table 16.1 and Figure 16.3). Since JAMs can be expressed on ECs, epithelial cells, platelets, and leukocytes, they function in cell–cell interactions not only between the same types of cells but also among distinct types of cells through homophilic and heterophilic interactions. Furthermore, JAMs exhibit heterophilic interactions via their extracellular domains with the leukocyte integrins LFA-1, VLA-4, and Mac-1 (Figure 16.3). In the context of adhesion and

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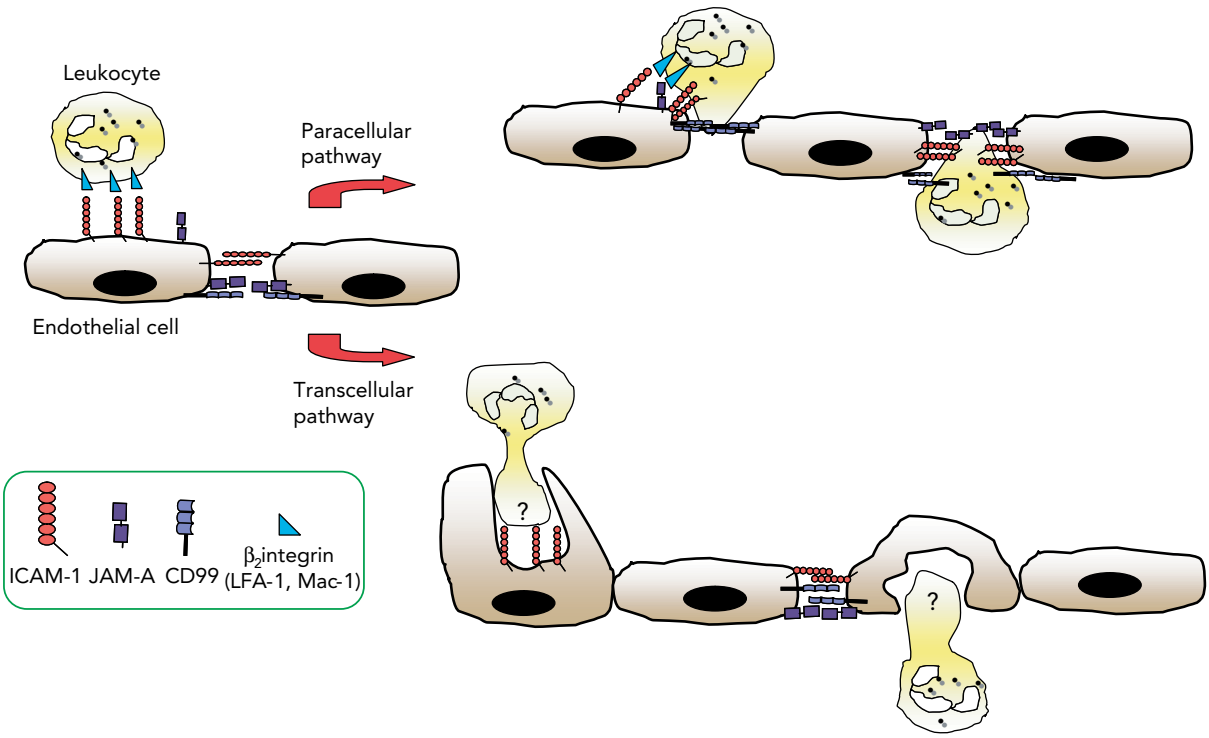


Figure 16.2. Pathways of leukocyte transmigration across the endothelium. To cross the vascular endothelium, leukocytes can take two different pathways. Adherent cells can disengage from the integrin-mediated adhesion and migrate by the paracellular route, passing through interendothelial junctions, squeezing through between adjacent endothelial cells; evidence for leukocytes transmigrating through the transcellular route also exist, whereby the cell will migrate across an individual endothelial cell body.

transmigration, JAM-A and JAM-C have been shown to play a functional role. Blocking JAM-A diminishes monocyte recruitment induced by MCP-3 into subcutaneous air pouches, while JAM-A deficient mice demonstrate reduced neutrophil infiltration in models of peritonitis and ischemia. JAM-C transfected ECs support increased transmigration of lymphocytes in vitro, and in parallel, transmigration could be reduced by 50% using a monoclonal antibody specific for JAM-C or by a soluble form of JAM-C. A prominent role for JAM-C in neutrophil transmigration has also been reported in vitro and in a mouse model of acute inflammatory peritonitis.

ESAM-1 appears to play a particular role in neutrophil transmigration, with diminished or delayed transmigration of these leukocytes in contrast to normal lymphocyte infiltration in ESAM null mice.

Migration via the transcellular route is thought to occur in areas of the endothelial cell that are rich in ICAM-1. Ligation of ICAM-1 by integrins on the surface of the leukocyte results in signaling events leading to the formation of channels in the endothelial cell, through which leukocytes can migrate. Many of the cell adhesion molecules involved in paracellular migration are also thought to play a role in transcellular migration, these being ICAM-1, ICAM-2, PECAM-1, CD99, JAMs, ESAM, VE-cadherin and LFA-1.

Although it appears that leukocyte subtypes may have a preferred route of migration, this may be influenced by the stimulus for migration and may also depend on the vascular bed involved. For example, neutrophil migration in the skin appears to occur largely by a paracellular route in response to high concentrations of FMLP, whereas in the cremaster, chemokine superfusion results in largely transcellular migration.

ASSESSMENT OF ADHESION MOLECULE FUNCTION UNDER FLOW

In Vitro (Flow Chamber)

The parallel-plate flow chamber is an in vitro system widely used to study leukocyte recruitment under defined laminar flow conditions, which more closely resembles the in vivo environment. The flow chamber assay was first performed in late 1980s to examine neutrophil–endothelial interactions, and has since allowed significant advances in the contributions attributed to different adhesion molecules in the process of leukocyte recruitment, demonstrating that

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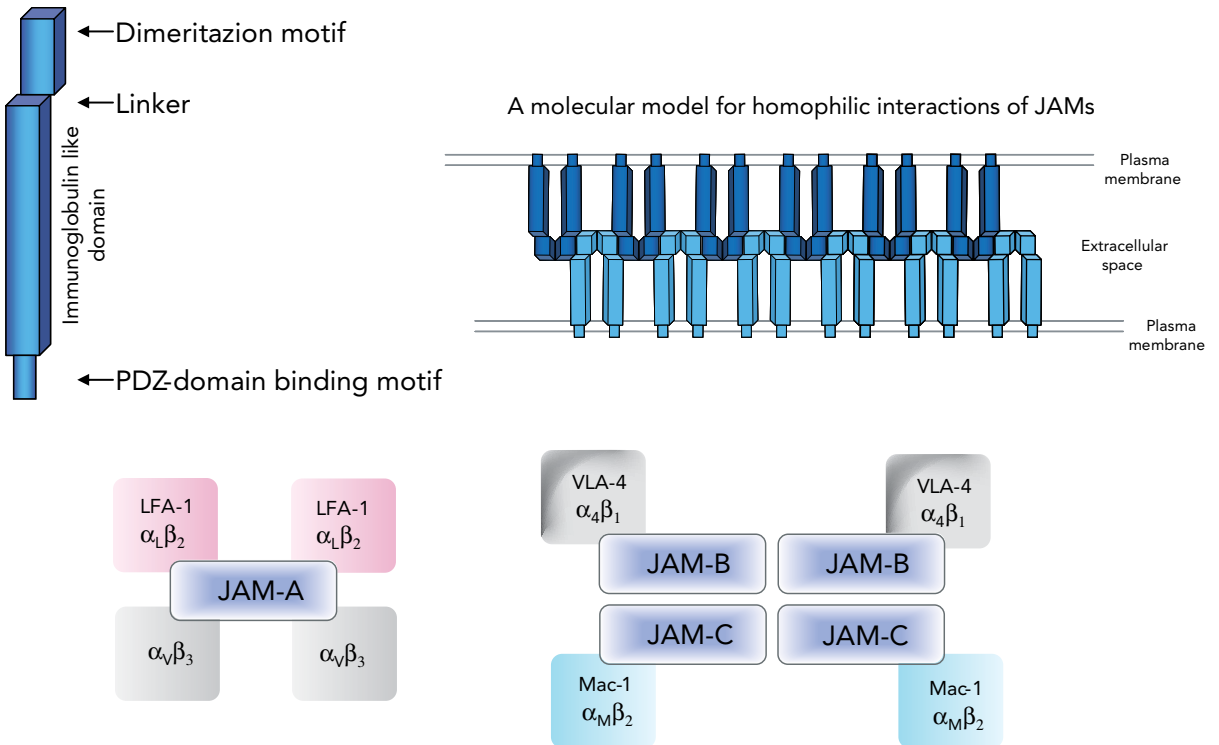


Figure 16.3. Structure of JAM-family members and binding partners. (A) *Upper panel*: JAMs are characterized by two immunoglobulin-like domains in the extracellular portion, a single transmembrane segment and a short cytoplasmatic tail with a PDZ-domain-binding motif (Phe-Leu-Val). A short linker sequence Val-Leu-Val connects the two immunoglobulin domains to impose a bent conformation. A dimerization motif Arg- (Val-Leu-Ile)-Glu in the membrane-distal domain is essential for homodimer formation. *Bottom panel:* Model of JAM homodimerization. (B) JAMs can form homo- and heterodimers, then forming specific *cis*- or *trans*- interactions with other CAMs.

multiple receptor–ligand pairs function in a sequential and orchestrated manner.

In brief, this assay involves perfusing leukocytes (or platelets) in suspension between two flat surfaces. The bottom surface can be coated with a variety of adhesive substrates such as a monolayer of endothelial cells or purified/recombinant adhesion molecules. Confluent endothelial monolayers can be activated with various inflammatory stimuli to delineate the roles of different adhesion molecules, for example, TNF-α (4h) causes an upregulation of E-selectin, ICAM-1, and VCAM-1, whereas histamine (15 minutes stimulation) results in a transient release of P-selectin. Blocking monoclonal antibodies can then be used to confirm the specificity of interactions. Alternatively, stable cell lines can be transfected with specific adhesion molecules to assess their particular function.

Typically, studies using the flow chamber are used to assess the number of interactions with their substrate in terms of quantification of initial capture of cells from the stream of flow, and the proportion of rolling and adherent cells. Once the first few cells start rolling, these cells themselves are also capable of instigating rolling (termed secondary tethering), a process more likely to occur in this assay due to leukocyte gravity sedimentation in the flow chamber. If an endothelial monolayer is used, leukocyte transmigration can also be quantified. Video 16.1 (http://cambridge.org/9780521887298) shows a short clip of neutrophils interacting with a monolayer of human umbilical vein endothelial cells: rolling, adhesion, and transmigration are evident.

A range of defined shear rates and stresses can be produced to facilitate the study of interactions under physiologically relevant conditions. These can be calculated based on a number of known parameters including the chamber geometry, according to the following equation:

$$\tau_w = \frac{6\mu Q}{a^2 b}$$

where τ_w is the wall shear stress (dynes/cm^2); μ is the coefficient of viscosity (Poise); Q is the volumetric flow rate (mL/s); a is the channel height (i.e., gasket thickness, cm); b is the channel width (i.e., gasket width, cm). Certainly, this has proved useful in determining the shear threshold and bond lifetimes of adhesion molecules involved in cell rolling. For instance, E-selectin

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permits the slowest rolling velocities, followed by P-selectin, then L-selectin; this is consistent with the fact that E-selectin demonstrates longer bond lifetimes.

The flow chamber has therefore proved a versatile tool for determining various aspects of the leukocyte recruitment cascade under conditions of shear stress, and the latter point is important and worth remembering.

In Vivo (Intravital Microscopy)

Intravital microscopy (IVM), literally, *microscopy inside life*, is a sophisticated research tool used to visualize and directly observe the microcirculation of animals in vivo. Early pioneers, including Robert Hooke and Antonie van Leeuwenhoek, described microscopic observations of living creatures in the 17th and 18th centuries, and Virchow (1821–1902), Conheim (1839–1884), and Metchnikoff (1845–1916) used microscopy of living blood vessels to advance our understanding of inflammation. Since many immunological disease mechanisms are reflected by primary interactions at the microcirculatory level, IVM can be used to study the underlying trigger and effector mechanisms, their contributions to tissue damage and implications to the immune response of leukocyte trafficking.

Early descriptions of leukocyte rolling and transmigration were provided in the first half of the 19th century and the entire leukocyte recruitment cascade (rolling, firm adhesion, and transmigration) was described well before the 20th century. In the 20th century, photography, film, and video cameras enabled more information to be collected and even more sophisticated and convenient methods are available today. Gus Born was the first (in 1972) to report quantification of rolling velocity of white blood cells on postcapillary venule endothelium.

The basic set-up for intravital fluorescence microscopy requires a high-energy light source (usually a mercury or xenon lamp), a microscope suited for epi- or transillumination, and a camera. Using digital cameras, high-resolution images can be acquired and recorded easily and cheaply for later, off-line analysis. Conventional microscopes illuminate subjects using light from a bulb (halogen, mercury, and xenon) that is either transmitted through or reflected off the subject. All points and focal planes of the subject are illuminated simultaneously and light is collected indiscriminately to form an image. Intravital microscopy allows detailed quantitative analysis of a wide range of microvascular parameters related to leukocyte–endothelium interactions. However, a problem with this method is that the most commonly used variants of intravital microscopy involve surgical preparation and/or exteriorization of the tissue to be examined, for example, mesentery or cremaster. This could induce mild stimulation of the tissue and causes rapid and pronounced upregulation of rolling apparently via partial degranulation of perivascular mast cells and endothelial expression of P-selectin, although this effect has been exploited for studies investigating selectin-dependent recruitment. A further limitation has been the inability to differentiate granulocyte responses from those of mononuclear cells by intravital microscopy. However, this problem has been overcome recently with the use of leukocyte subtype-specific antibodies.

The use of fluorescent dyes such as rhodamine 6G and calcein AM and ever more sophisticated microscopes has broadened the range of vascular beds studied by IVM from thin transparent tissues such as the cremaster to microcirculations of organs such as the brain, lung, and liver.

Table 16.2 depicts the parameters that are quantifiable by IVM pertaining to the leukocyte recruitment cascade. Oedema and arteriolar vasodilation responses are also commonly measured by IVM.

Video 16.2 (http://cambridge.org/9780521887298) shows an inflamed vessel after an ischemic event, with large numbers of leukocytes interacting with a neatly seen postcapillary venule endothelium: cell rolling (including platelets), adhesion, and migration are very distinguishable.

CONCLUSIONS

The fundamental response to tissue damage during an inflammatory response is the recruitment of blood leukocytes to the underlying tissue. Thus pharmacological manipulation of adhesion molecules involved in the cell recruitment cascade represents a promising strategy for therapeutic intervention. Ultimately, the aim is to stop leukocytes in their tracks and break the path of the multistep cascade, insuring rapid inflammatory resolution and restoration to homeostasis. Indeed, compelling data in a multitude of animal models have demonstrated that interfering with key adhesion molecules results in significantly attenuated inflammatory responses. However, despite these findings in mice, the transition to clinical trials in humans, targeting specific adhesion molecules has had a mixed success rate. Generally, endeavors to alleviate chronic relapsing inflammatory diseases (such as multiple sclerosis, psoriasis, and irritable bowel syndrome) have been far more successful than attempts to treat severe acute inflammatory responses.

The major approach of targeting adhesion molecules has been via direct blockade of receptor–ligand interactions through the development of small molecule

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TABLE 16.2. Measurement parameters used in IVM

Parameter	Description
Rolling velocity	Distance traveled/time taken (units are μm/s)
Cell flux	Number of cells passing a fixed point in the vessel per minute
Cell adhesion	Cells that remain stationary for 30 seconds or longer
Cell emigration	Number of cells that had emigrated up to 50 μm from the wall of the 100 μm vessel
Wall shear rate (WSR)	Calculated by the Newtonian definition; WSR = 8000 × (V_{mean}/diameter) (units are s^{-1})

inhibitors or using monoclonal antibodies (either chimaeric or humanized). However, caution must be taken as blocking studies carry a significant risk associated with an impaired immune response. Nevertheless, recent successful trials have been documented using a humanized monoclonal antibody directed against the α_4 integrin chain that blocks binding of $\alpha_4\beta_1$ to VCAM-1 on brain-infiltrating lymphocytes and binding of $\alpha_4\beta_7$ to MadCAM-1 on gut-infiltrating lymphocytes for treatment of multiple sclerosis and Crohn's disease. In addition, a FDA-approved humanized monoclonal antibody (eflizumab) specific for the integrin subunit CD11a is currently being developed for the treatment of plaque-type psoriasis.

Future antiadhesion drug development is aimed at determining which molecular targets offer the best specificity and the least overall risk of reducing host defense mechanisms.

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LIST OF SYNONYMS AND ABBREVIATIONS

Cell adhesion molecules = CAM
$\alpha_4\beta_1$ integrin = VLA-4= CD49d/CD29
$\alpha_5\beta_1$ integrin = VLA-5= CD49e/CD29
$\alpha_6\beta_1$ integrin = VLA-6= CD49f/CD29
$\alpha_L\beta_2$ integrin = LFA-1= CD11a/CD18
$\alpha_M\beta_2$ integrin = MAC-1= CR3= CD11b/CD18
$\alpha_X\beta_2$ integrin = p150, 95= CR14 receptor
$\alpha_4\beta_7$ integrin = LPAM-1= CD49d/CD-
LFA-2 = CD2
LFA-3 = CD58
VCAM = CD106
ICAM-1 = CD54
ICAM-2 = CD102
PE-CAM-1 = CD31
CD36 = "leukocyte differentiation antigen"
MAdCAM-1
E-selectin = CD62E=ELAM-1
P-selectin = CD62P=GMP=140=PADGEM
L-selectin = CD62L=LAM-1=Mel-14

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